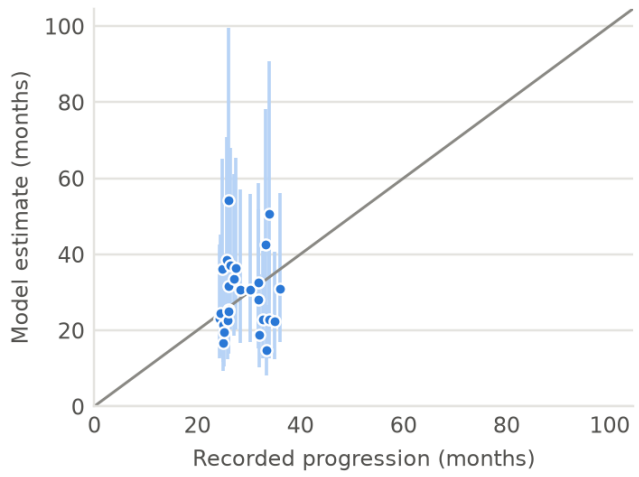
Group C · Progression in year 3 recorded progression-free survival: 24–36 months

Patients	Median model risk percentile	Recorded progression	Model window agreement
27	**71**	**27.1 mo**	**37%**
2.5% of the cohort	cohort-wide risk ranking, out-of-fold	median; range 24.3–36.0 mo	estimate in this window · median |error| 6 mo



Most frequent path end-points in this group	Patients
Platelet degranulation	4
Sperm Motility And Taxes	2
Zinc chloride	2
Synthesis of PA	2
breast adenocarcinoma	2
Striated Muscle Contraction	2
G alpha (q) signalling events	2
Pregnenolone biosynthesis	2

Logic formulas firing most often in this group	Mean weight	Patients
Subtype(p, LumA) ⇒ Progress(p)	-0.011	7
NStage(p, N0) ⇒ Progress(p)	-0.132	7
Stage(p, II) ⇒ Progress(p)	-0.083	6
TStage(p, T4) ⇒ Progress(p)	+0.102	3
Altered(p, g) ∧ InPathway(g, 'Synthesis of PA') ⇒ Progress(p)	-0.010	2
TStage(p, T1) ∧ NStage(p, N1) ⇒ Progress(p)	-0.061	2

Most common driver alterations

PIK3CA mutated (33%), TP53 mutated (26%), KMT2C mutated (22%), FGFR1 amplified (19%), CDH1 mutated (15%), MYC amplified (15%), GATA3 mutated (15%), RB1 mutated (7%), PTEN mutated (7%), PIK3R1 mutated (7%)

Patient index

Patient	Age	Stage	Subtype	Recorded	Model estimate	Window	Risk pct	Report file
TCGA-AR-A0TW	50	III	LumA	24.3 mo	23 mo	✗	84	C_progression_24-36m/TCGA-AR-A0TW.pdf
TCGA-B6-A0I8	46	X	missing	24.5 mo	24 mo	✓	84	C_progression_24-36m/TCGA-B6-A0I8.pdf
TCGA-A2-A0YC	59	II	LumA	24.9 mo	36 mo	✗	18	C_progression_24-36m/TCGA-A2-A0YC.pdf
TCGA-A8-A08O	45	IV	LumA	25.1 mo	17 mo	✗	99	C_progression_24-36m/TCGA-A8-A08O.pdf
TCGA-BH-A1F8	90	III	LumB	25.1 mo	21 mo	✗	82	C_progression_24-36m/TCGA-BH-A1F8.pdf
TCGA-AC-A2FE	62	III	missing	25.2 mo	19 mo	✗	94	C_progression_24-36m/TCGA-AC-A2FE.pdf
TCGA-AC-A2FM	87	II	missing	25.6 mo	39 mo	✗	59	C_progression_24-36m/TCGA-AC-A2FM.pdf
TCGA-BH-A1F0	80	II	Basal	25.8 mo	23 mo	✗	44	C_progression_24-36m/TCGA-BH-A1F0.pdf
TCGA-E9-A226	45	III	LumB	26.0 mo	54 mo	✗	89	C_progression_24-36m/TCGA-E9-A226.pdf
TCGA-A8-A06Y	66	II	LumA	26.0 mo	25 mo	✓	22	C_progression_24-36m/TCGA-A8-A06Y.pdf
TCGA-AO-A0J5	48	IV	LumA	26.0 mo	25 mo	✓	98	C_progression_24-36m/TCGA-AO-A0J5.pdf
TCGA-A7-A13H	61	II	missing	26.1 mo	32 mo	✓	68	C_progression_24-36m/TCGA-A7-A13H.pdf
TCGA-A2-A3XY	49	II	missing	26.4 mo	37 mo	✗	60	C_progression_24-36m/TCGA-A2-A3XY.pdf
TCGA-A2-A0SV	63	IV	LumB	27.1 mo	34 mo	✓	99	C_progression_24-36m/TCGA-A2-A0SV.pdf
TCGA-B6-A0IE	38	III	LumA	27.4 mo	36 mo	✗	71	C_progression_24-36m/TCGA-B6-A0IE.pdf
TCGA-B6-A0I6	49	II	missing	28.2 mo	31 mo	✓	60	C_progression_24-36m/TCGA-B6-A0I6.pdf
TCGA-BH-A18P	60	I	Her2	30.3 mo	31 mo	✓	61	C_progression_24-36m/TCGA-BH-A18P.pdf
TCGA-A1-A0SK	54	II	Basal	31.8 mo	28 mo	✓	54	C_progression_24-36m/TCGA-A1-A0SK.pdf
TCGA-AR-A5QM	62	II	LumA	31.8 mo	33 mo	✓	13	C_progression_24-36m/TCGA-AR-A5QM.pdf
TCGA-E2-A14R	62	II	Basal	31.9 mo	19 mo	✗	44	C_progression_24-36m/TCGA-E2-A14R.pdf
TCGA-B6-A3ZX	50	IV	missing	32.7 mo	23 mo	✗	100	C_progression_24-36m/TCGA-B6-A3ZX.pdf
TCGA-BH-A1FD	68	I	LumB	33.2 mo	43 mo	✗	33	C_progression_24-36m/TCGA-BH-A1FD.pdf
TCGA-EW-A1P1	68	III	Normal	33.3 mo	15 mo	✗	88	C_progression_24-36m/TCGA-EW-A1P1.pdf
TCGA-BH-A1FH	47	IV	LumA	34.0 mo	51 mo	✗	99	C_progression_24-36m/TCGA-BH-A1FH.pdf
TCGA-C8-A3M7	60	III	LumA	34.0 mo	23 mo	✗	75	C_progression_24-36m/TCGA-C8-A3M7.pdf
TCGA-AO-A0J9	61	III	LumA	35.0 mo	22 mo	✗	72	C_progression_24-36m/TCGA-AO-A0J9.pdf
TCGA-EW-A1P0	55	II	LumB	36.0 mo	31 mo	✓	45	C_progression_24-36m/TCGA-EW-A1P0.pdf

■ **Group C** · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**50**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-AR-A0TW-01** · Aneuploidy score: **16** · TMB (nonsyn., /Mb): **1.6** · MSIsensor score: **0.2** · AJCC edition: **—**

2 · Progression assessment

Model risk (cohort percentile)

84th

High risk tier · fused neuro-symbolic model



Probability of progression

15% by 36 mo

12 mo 3% · 24 mo 9% · 60 mo 29%

Model-estimated progression time

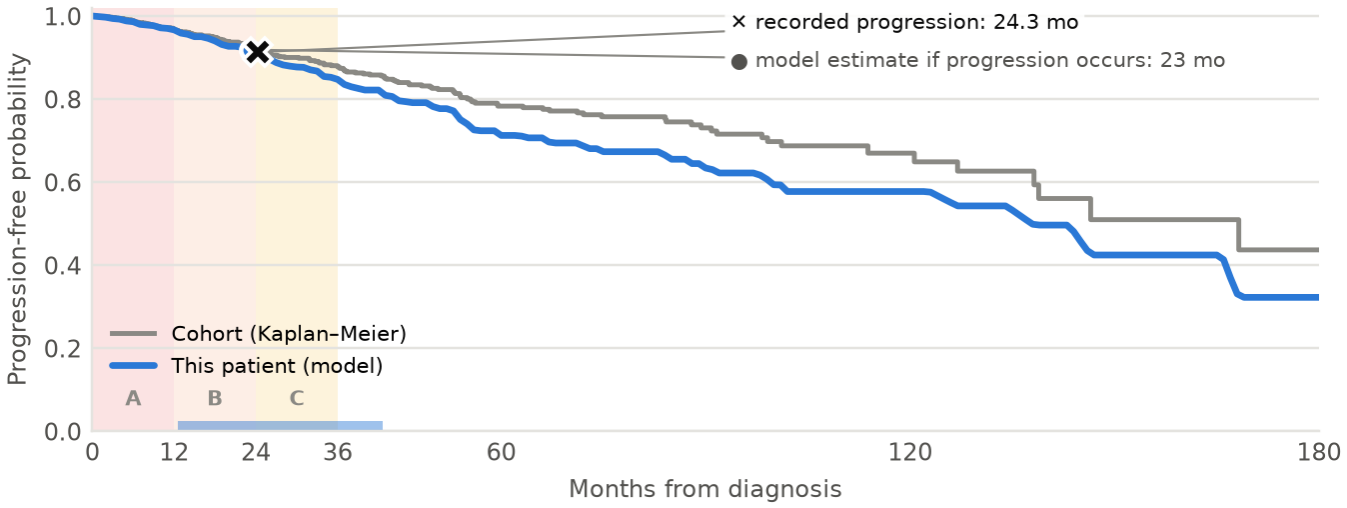
≈ 23 months

Window 12–24 months · IQR 13–43 mo
if progression occurs

Recorded progression

24.3 months

■ **Group C** · Progression in year 3
✗ **recorded window differs** · |error| 1 mo



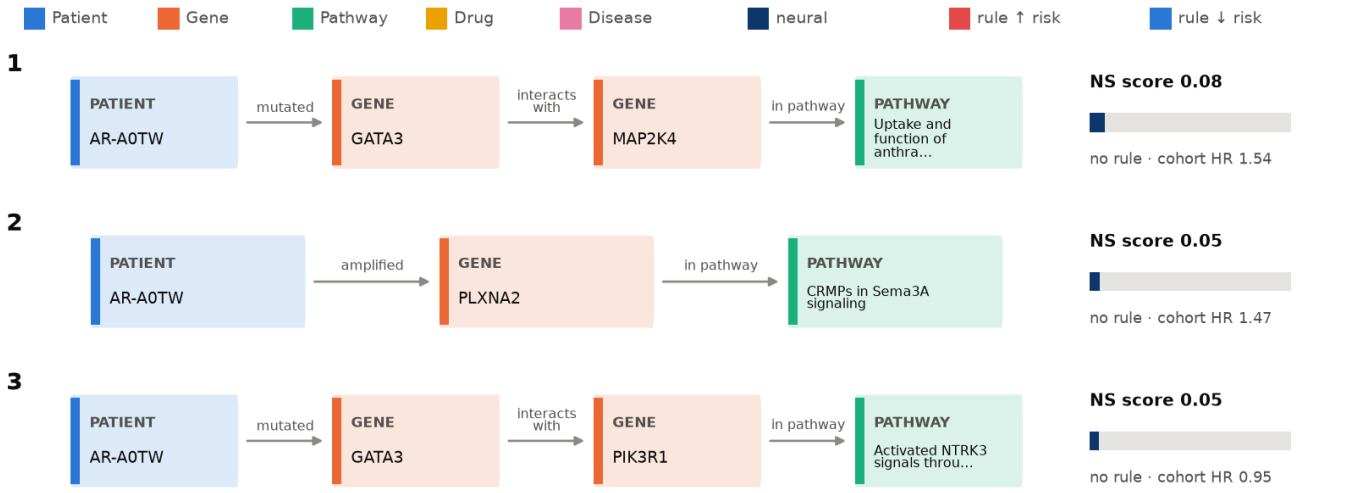
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 50-year-old patient has a LumA tumour, AJCC stage III (T3, N1, M0). The fused neuro-symbolic model places them in the **84th risk percentile** (high tier). The strongest learned formula that holds for them (*Radiation(p, no) ⇒ Progress(p)*) raises the progression hazard (weight +0.023 per SD). The highest-scoring knowledge-graph path runs from **GATA3** to **Uptake and function of anthrax toxins**. Progression was recorded at 24.3 months; the model's estimate given progression was 23 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

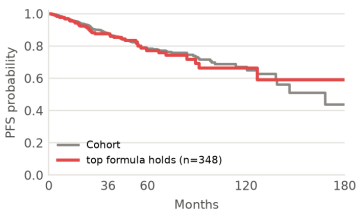


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-AR-A0TW** mutated → **GATA3** interacts with → **MAP2K4** in pathway → **Uptake and function of anthrax toxins**	13	No learned formula fires on this path; its score is neural only. cohort: HR 1.54, n = 59 carriers	**0.08**
2	**TCGA-AR-A0TW** amplified → **PLXNA2** in pathway → **CRMPs in Sema3A signaling**	8	No learned formula fires on this path; its score is neural only. cohort: HR 1.47, n = 90 carriers	**0.05**
3	**TCGA-AR-A0TW** mutated → **GATA3** interacts with → **PIK3R1** in pathway → **Activated NTRK3 signals through PI3K**	8	No learned formula fires on this path; its score is neural only. cohort: HR 0.95, n = 312 carriers	**0.05**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

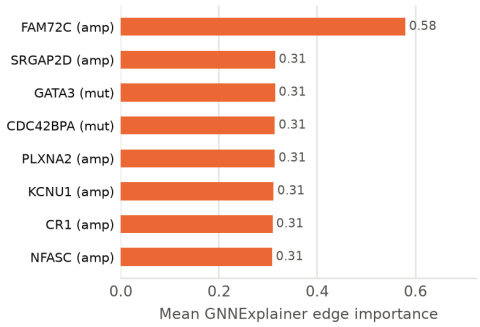
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.023 ↑ risk	Radiation(p, no) ⇒ Progress(p)	single atom	348 pts
-0.016 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	408 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GATA3 · driver	Mutation	p.S401Vfs*106	13	0.31
FAM72C	High-level amplification	GISTIC ±2	0	0.58
SRGAP2D	High-level amplification	GISTIC ±2	0	0.31
CDC42BPA	Mutation	p.S990*	4	0.31
PLXNA2	High-level amplification	GISTIC ±2	4	0.31
KCNU1	High-level amplification	GISTIC ±2	1	0.31
CR1	High-level amplification	GISTIC ±2	3	0.31
NFASC	High-level amplification	GISTIC ±2	3	0.31



8 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A0AZ	47	III	LumA	0.87	no progression, followed 63.1 mo
TCGA-A1-A0SJ	39	III	LumA	0.75	no progression, followed 13.7 mo
TCGA-BH-A0BS	55	III	LumA	0.73	no progression, followed 85.9 mo
TCGA-EW-A3U0	61	III	missing	0.71	no progression, followed 17.5 mo
TCGA-BH-A18M	39	III	LumA	0.69	no progression, followed 72.6 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group C** · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**46**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE X**
Genetic ancestry	**EUR**	Pathologic T	**T1**
Prior diagnosis	**Yes**	Pathologic N	**NX**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-B6-A0I8-01** · Aneuploidy score: **—** · TMB (nonsyn., /Mb): **3.73** · MSIsensor score: **0.57** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

84th

High risk tier · fused neuro-symbolic model



Probability of progression

17% by 36 mo

12 mo 5% · 24 mo 9% · 60 mo 27%

Model-estimated progression time

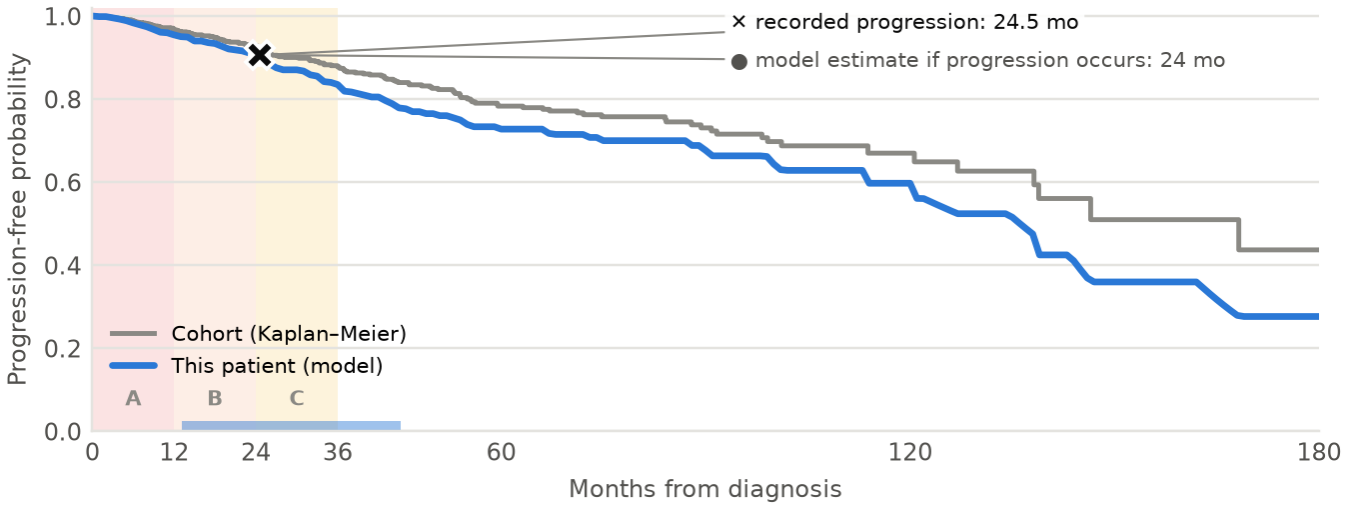
≈ 24 months

Window 24–36 months · IQR 13–45 mo
if progression occurs

Recorded progression

24.5 months

■ **Group C** · Progression in year 3
✓ **estimate in recorded window** · |error| 0 mo



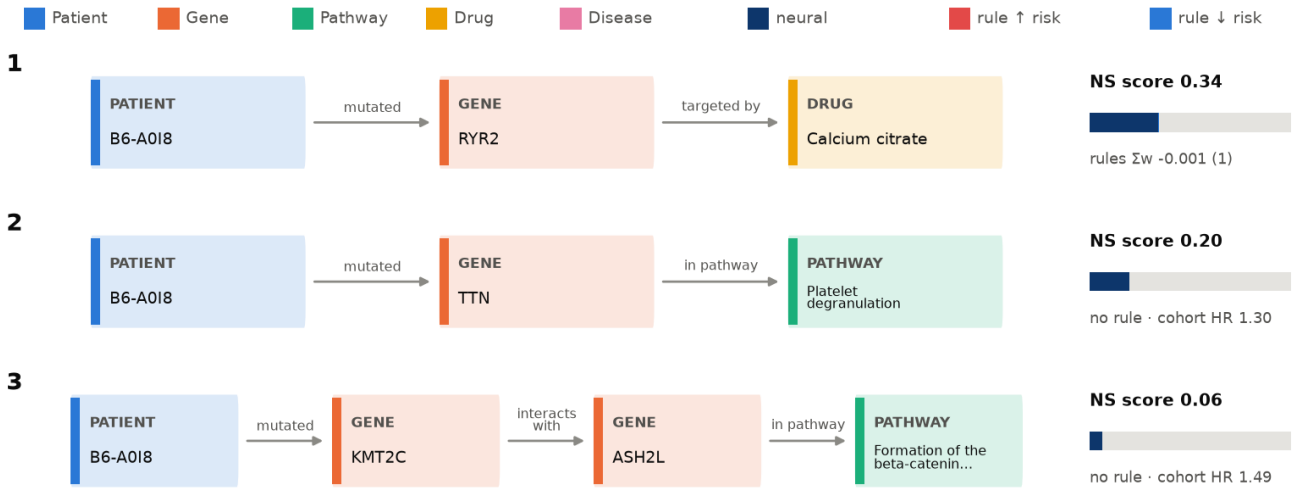
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 46-year-old patient has a subtype-unassigned tumour, AJCC stage X (T1, missing, M0). The fused neuro-symbolic model places them in the **84th risk percentile** (high tier). The strongest learned formula that holds for them (*TStage(p, T1)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.042 per SD). The highest-scoring knowledge-graph path runs from **RYR2** to **Calcium citrate**; the learned formulas it instantiates point to slower progression (Σw -0.001). Progression was recorded at 24.5 months; the model's estimate given progression was 24 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

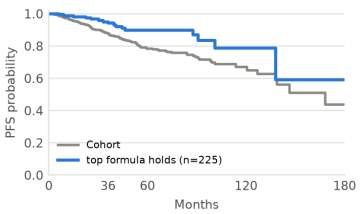


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-B6-A0I8** mutated → **RYR2** targeted by → **Calcium citrate**	57	↓ **slower progression** · Σw -0.001 over 1 formula(s) Altered(p, g) ∧ TargetedBy(g, 'Flecainide') ⇒ Progress(p) cohort: HR 0.86, n = 193 carriers	**0.34**
2	**TCGA-B6-A0I8** mutated → **TTN** in pathway → **Platelet degranulation**	33	No learned formula fires on this path; its score is neural only. cohort: HR 1.30, n = 311 carriers	**0.20**
3	**TCGA-B6-A0I8** mutated → **KMT2C** interacts with → **ASH2L** in pathway → **Formation of the beta-catenin:TCF transactivating complex**	10	No learned formula fires on this path; its score is neural only. cohort: HR 1.49, n = 189 carriers	**0.06**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

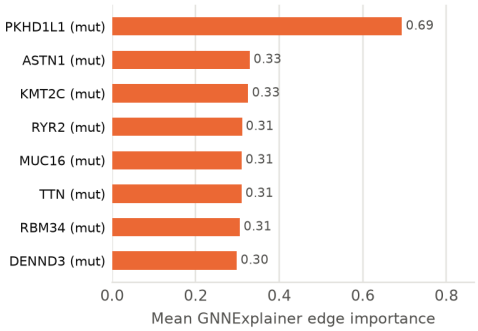
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.042 ↓ risk	TStage(p, T1) ⇒ Progress(p)	single atom	225 pts
-0.001 ↓ risk	Altered(p, g) ∧ TargetedBy(g, 'Flecainide') ⇒ Progress(p) [KG]	single atom	147 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
KMT2C · driver	Mutation	p.T1171S	4	0.33
PKHD1L1	Mutation	p.L211Q	0	0.69
ASTN1	Mutation	p.F92=	0	0.33
RYR2	Mutation	p.S3959=	7	0.31
MUC16	Mutation	p.P7152=	8	0.31
TTN	Mutation	p.A5218Lfs*2, p.R20026P	2	0.31
RBM34	Mutation	p.*38*	0	0.31
DENND3	Mutation	p.Q704E	1	0.30
SPTA1	Mutation	p.R1757C	4	0.30



9 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

RYR2 → Flecainide (DB01195); RYR2 → Tetracaine (DB09085); RYR2 → Calcium citrate (DB11093); RYR2 → Calcium Phosphate (DB11348); RYR2 → Calcium phosphate dihydrate (DB14481); MUC16 → Oregovomab (DB04964). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-B6-A0I6	49	II	missing	0.55	progressed at 28.2 mo (group C)
TCGA-B6-A402	47	I	missing	0.54	progressed at 49.8 mo (group D)
TCGA-BH-A0HN	67	I	missing	0.53	no progression, followed 17.0 mo
TCGA-A8-A09A	40	X	LumA	0.52	no progression, followed 10.0 mo
TCGA-D8-A1XD	36	III	missing	0.51	no progression, followed 17.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group C · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**59**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1MI**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

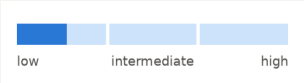
Sample: **TCGA-A2-A0YC-01** · Aneuploidy score: **4** · TMB (nonsyn., /Mb): **0.77** · MSIsensor score: **0.02** · AJCC edition: —

2 · Progression assessment

Model risk (cohort percentile)

18th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

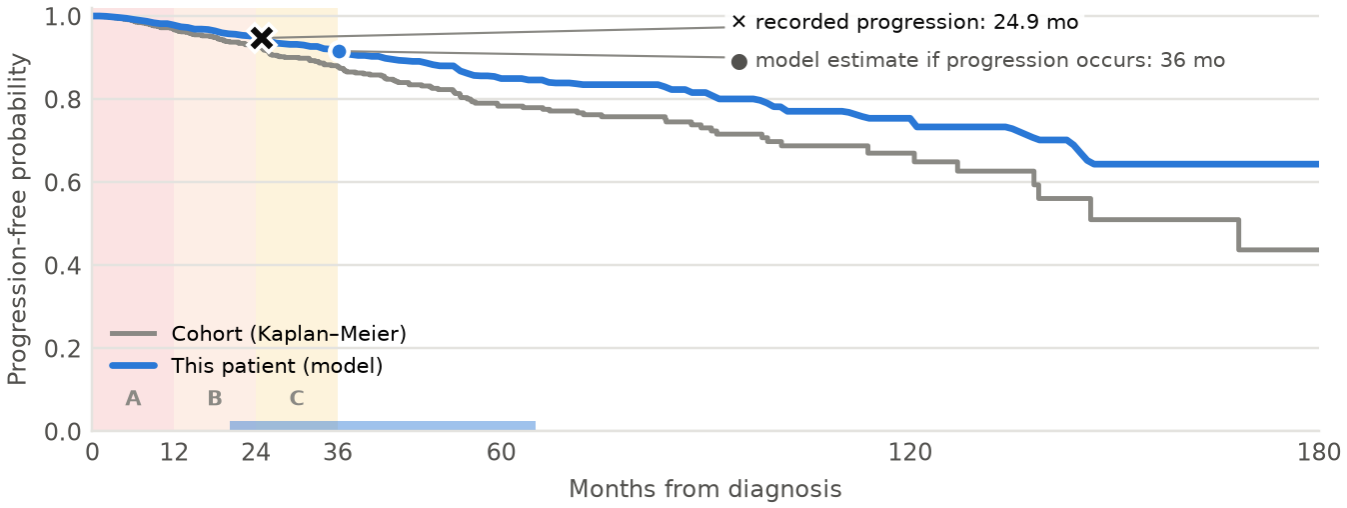
≈ 36 months

Window > 36 months · IQR 20–65 mo if progression occurs

Recorded progression

24.9 months

Group C · Progression in year 3
✗ recorded window differs · |error| 11 mo



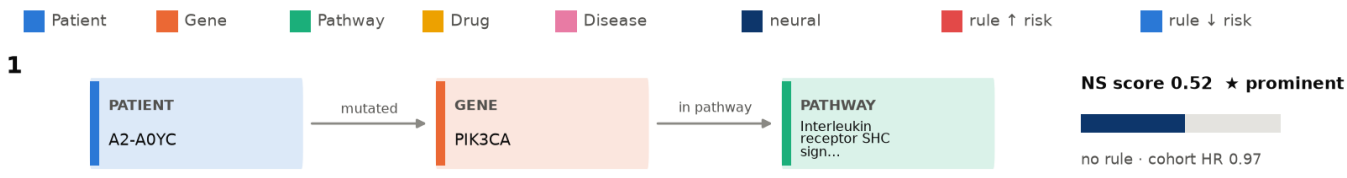
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 59-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **18th risk percentile** (low tier). The strongest learned formula that holds for them (*Subtype(p, LumA) ⇒ Progress(p)*) lowers the progression hazard (weight -0.010 per SD). The highest-scoring knowledge-graph path runs from **PIK3CA** to **Interleukin receptor SHC signaling**. Progression was recorded at 24.9 months; the model's estimate given progression was 36 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

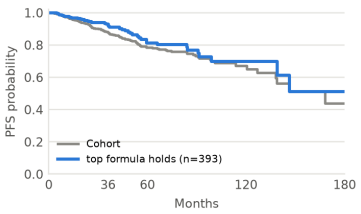


1 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-A2-A0YC** mutated → **PIK3CA** in pathway → **Interleukin receptor SHC signaling**	87	No learned formula fires on this path; its score is neural only. cohort: HR 0.97, n = 324 carriers	**0.52**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

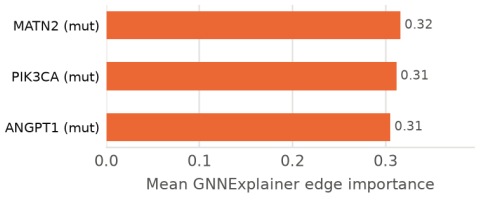
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.010 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	390 pts
-0.006 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	487 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3CA · driver	Mutation	p.E545K	70	0.31
MATN2	Mutation	p.E22K	0	0.32
ANGPT1	Mutation	p.G347V	2	0.31



3 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A10B	67	II	LumA	0.99	progressed at 6.4 mo (group A)
TCGA-AR-A0TR	68	II	LumA	0.98	no progression, followed 5.3 mo
TCGA-E9-A227	42	II	LumA	0.96	no progression, followed 32.1 mo
TCGA-BH-A0BJ	41	II	LumA	0.96	no progression, followed 21.7 mo
TCGA-EW-A1P5	77	II	LumA	0.94	no progression, followed 23.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group C** · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**45**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**—**	PAM50 subtype	**LumA**
Ethnicity	**—**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N3A**
Radiation therapy	**No**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-A8-A08O-01** · Aneuploidy score: **7** · TMB (nonsyn., /Mb): **0.7** · MSIsensor score: **0.08** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

99th

High risk tier · fused neuro-symbolic model



Probability of progression

63% by 36 mo

12 mo 22% · 24 mo 45% · 60 mo 84%

Model-estimated progression time

≈ 17 months

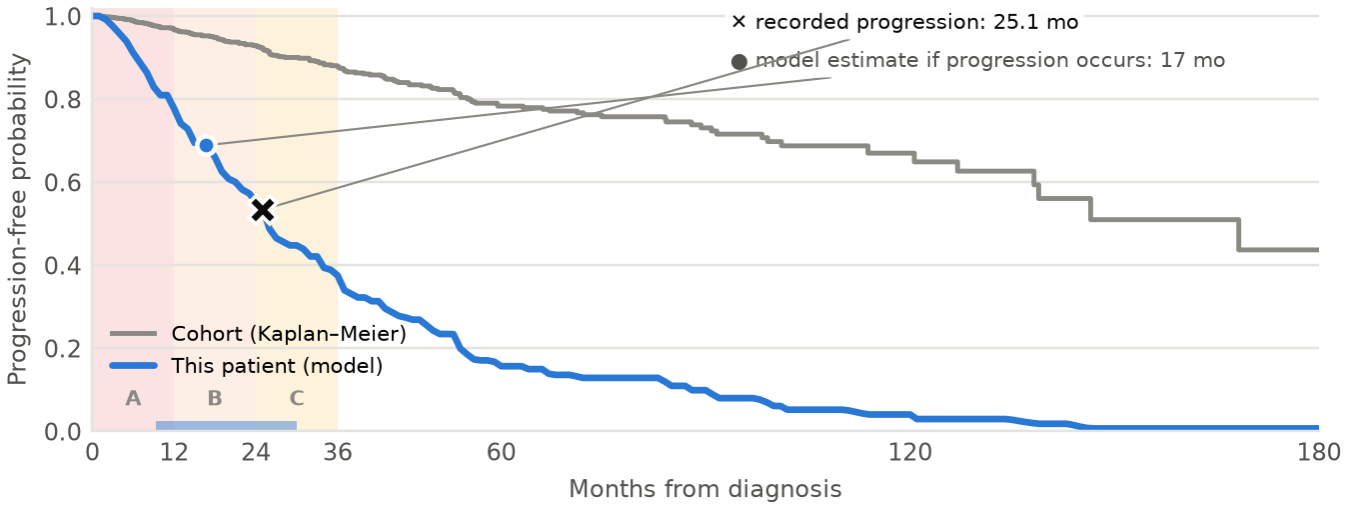
Window 12–24 months · IQR 9–30 mo if progression occurs

Recorded progression

25.1 months

■ **Group C** · Progression in year 3

✗ **recorded window differs** · |error| 8 mo



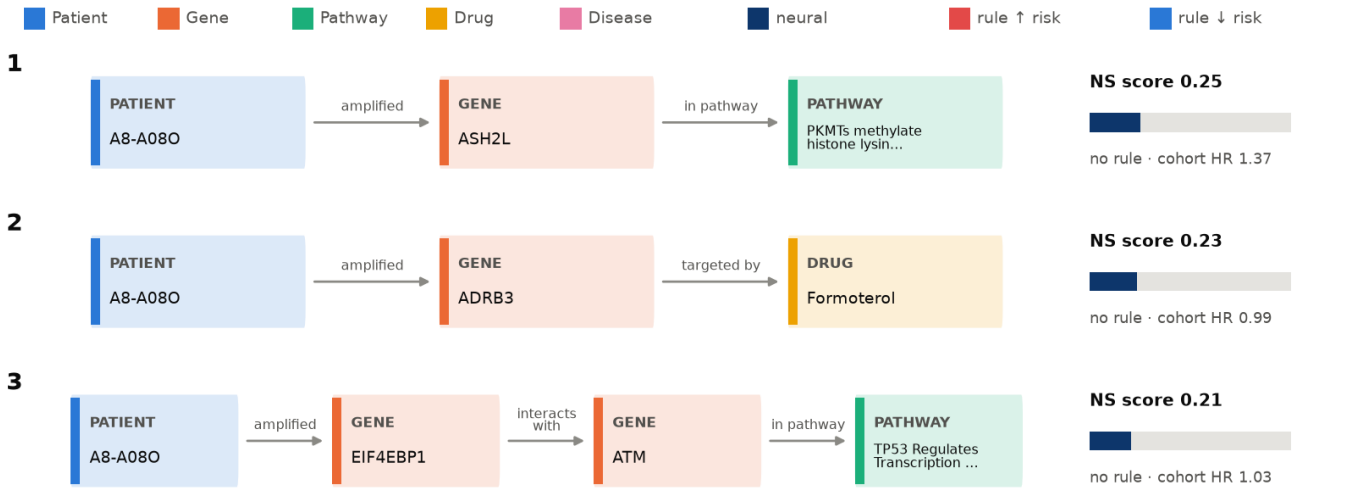
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 45-year-old patient has a LumA tumour, AJCC stage IV (T2, N3, M1). The fused neuro-symbolic model places them in the **99th risk percentile** (high tier). The strongest learned formula that holds for them (*NStage(p, N3) ⇒ Progress(p)*) raises the progression hazard (weight +0.098 per SD). The highest-scoring knowledge-graph path runs from **ASH2L** to **PKMTs methylate histone lysines**. Progression was recorded at 25.1 months; the model's estimate given progression was 17 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

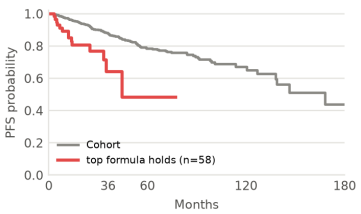


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-A8-A08O** amplified → **ASH2L** in pathway → **PKMTs methylate histone lysines**	42	No learned formula fires on this path; its score is neural only. cohort: HR 1.37, n = 323 carriers	**0.25**
2	**TCGA-A8-A08O** amplified → **ADRB3** targeted by → **Formoterol**	39	No learned formula fires on this path; its score is neural only. cohort: HR 0.99, n = 111 carriers	**0.23**
3	**TCGA-A8-A08O** amplified → **EIF4EBP1** interacts with → **ATM** in pathway → **TP53 Regulates Transcription of Caspase Activators and Caspases**	34	No learned formula fires on this path; its score is neural only. cohort: HR 1.03, n = 306 carriers	**0.21**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

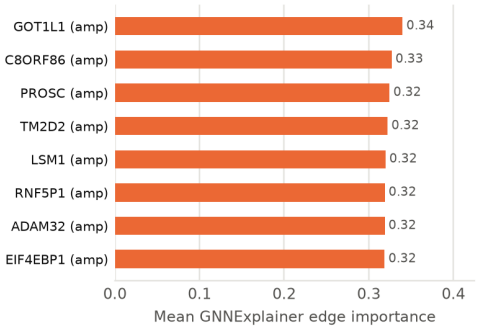
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.098 ↑ risk	NStage(p, N3) ⇒ Progress(p)	single atom	57 pts
-0.010 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	390 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.31
PIK3CA · driver	Mutation	p.H1047R	70	0.31
MAP2K4 · driver	Homozygous deletion	GISTIC ±2	12	0.31
FOXA1 · driver	High-level amplification	GISTIC ±2	9	0.31
GOT1L1	High-level amplification	GISTIC ±2	0	0.34
C8ORF86	High-level amplification	GISTIC ±2	0	0.33
PROSC	High-level amplification	GISTIC ±2	0	0.32
TM2D2	High-level amplification	GISTIC ±2	0	0.32
LSM1	High-level amplification	GISTIC ±2	1	0.32
RNF5P1	High-level amplification	GISTIC ±2	0	0.32
ADAM32	High-level amplification	GISTIC ±2	0	0.32
EIF4EBP1	High-level amplification	GISTIC ±2	2	0.32



31 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896); FGFR1 → Ponatinib (DB08901) … and 52 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A1R2	51	III	LumA	0.73	no progression, followed 34.9 mo
TCGA-A2-A0SV	63	IV	LumB	0.68	progressed at 27.1 mo (group C)
TCGA-D8-A27R	41	III	LumB	0.66	no progression, followed 10.1 mo
TCGA-WT-AB41	55	II	LumA	0.66	no progression, followed 53.0 mo
TCGA-HN-A2OB	45	II	LumA	0.65	progressed at 50.7 mo (group D)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-BH-A1F8 · Group C

Report date 31 Oct 2026

■ **Group C** · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**90**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIB**
Genetic ancestry	**EUR**	Pathologic T	**T4B**
Prior diagnosis	**No**	Pathologic N	**N2**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-BH-A1F8-01** · Aneuploidy score: **16** · TMB (nonsyn., /Mb): **4.33** · MSIsensor score: **0.62** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

82nd

High risk tier · fused neuro-symbolic model



Probability of progression

15% by 36 mo

12 mo 4% · 24 mo 9% · 60 mo 27%

Model-estimated progression time

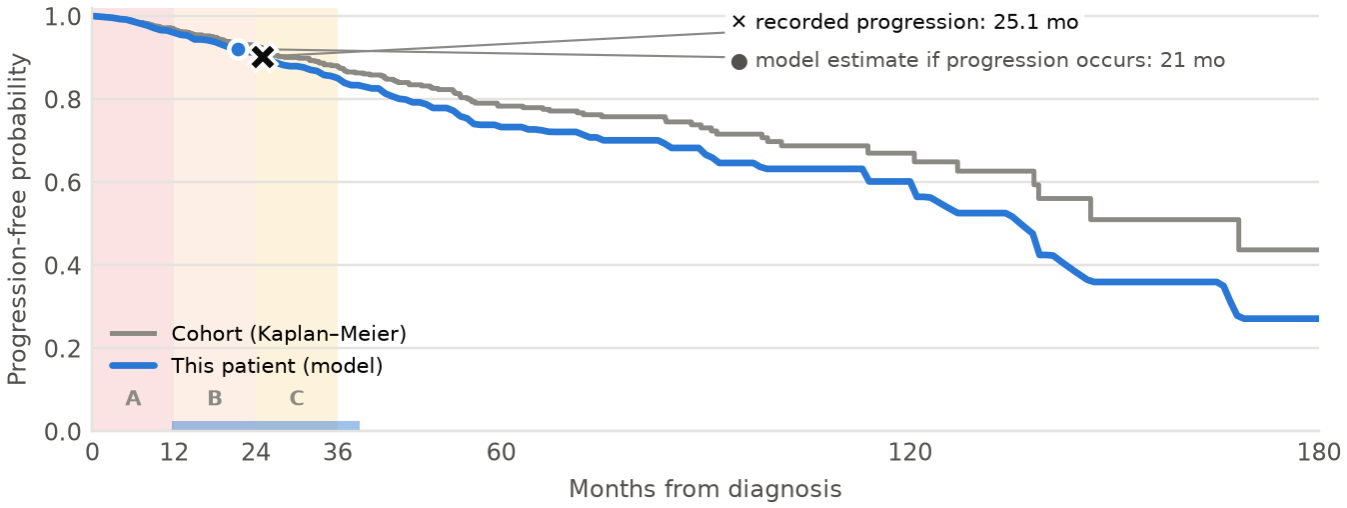
≈ 21 months

Window 12–24 months · IQR 12–39 mo
if progression occurs

Recorded progression

25.1 months

■ **Group C** · Progression in year 3
✗ **recorded window differs** · |error| 4 mo



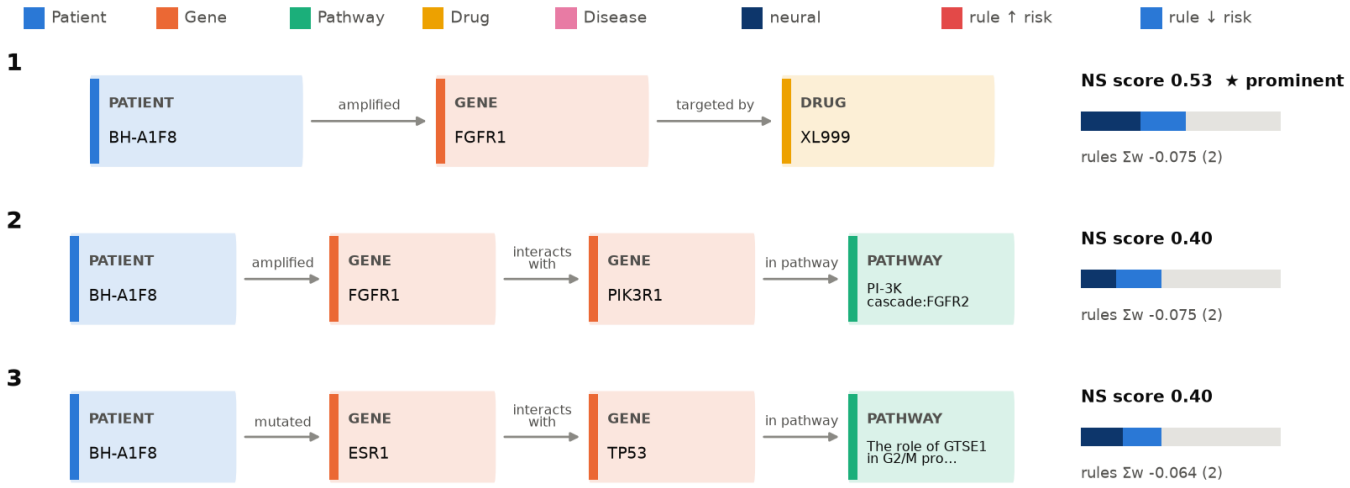
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 90-year-old patient has a LumB tumour, AJCC stage III (T4, N2, M0). The fused neuro-symbolic model places them in the **82nd risk percentile** (high tier). The strongest learned formula that holds for them (*TStage(p, T4)* ⇒ *Progress(p)*) raises the progression hazard (weight +0.113 per SD). The highest-scoring knowledge-graph path runs from **FGFR1** to **XL999**; the learned formulas it instantiates point to slower progression (Σw -0.075). Progression was recorded at 25.1 months; the model's estimate given progression was 21 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

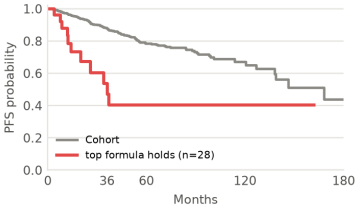


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-BH-A1F8** amplified → **FGFR1** targeted by → **XL999**	50	↓ **slower progression** · Σw -0.075 over 2 formula(s) Altered(p, g) ∧ InPathway(g, 'Vitamin B2 (riboflavin) metabolism') ∧ Altered(p, g) ∧ InPathway(g, 'Signal transduction by L1') ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) cohort: HR 0.63, n = 108 carriers	**0.53**
2	**TCGA-BH-A1F8** amplified → **FGFR1** interacts with → **PIK3R1** in pathway → **PI-3K cascade:FGFR2**	30	↓ **slower progression** · Σw -0.075 over 2 formula(s) Altered(p, g) ∧ InPathway(g, 'Vitamin B2 (riboflavin) metabolism') ∧ Altered(p, g) ∧ InPathway(g, 'Signal transduction by L1') ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) cohort: HR 0.76, n = 386 carriers	**0.40**
3	**TCGA-BH-A1F8** mutated → **ESR1** interacts with → **TP53** in pathway → **The role of GTSE1 in G2/M progression after G2 checkpoint**	35	↓ **slower progression** · Σw -0.064 over 2 formula(s) Altered(p, g) ∧ InPathway(g, 'Vitamin B2 (riboflavin) metabolism') ∧ Altered(p, g) ∧ InPathway(g, 'Signal transduction by L1') ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) cohort: HR 0.95, n = 283 carriers	**0.40**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

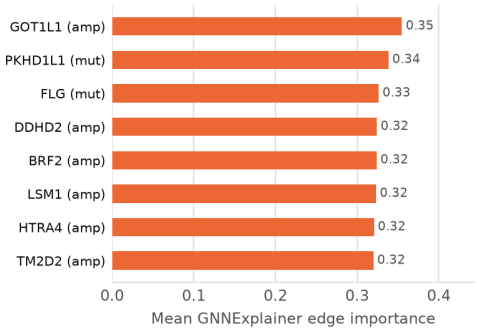
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.113 ↑ risk	TStage(p, T4) ⇒ Progress(p)	single atom	26 pts
-0.067 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Vitamin B2 (riboflavin) metabolism') ∧ Altered(p, g) ∧ InPathway(g, 'Signal transduction by L1') ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) [KG]	RL-proposed	39 pts
-0.010 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Synthesis of PA') ⇒ Progress(p) [KG]	single atom	187 pts
-0.008 ↓ risk	Stage(p, III) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	134 pts
+0.004 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Nuclear signaling by ERBB4') ⇒ Progress(p) [KG]	single atom	30 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3CA · driver	Mutation	p.H1047R	70	0.32
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.32
ESR1 · driver	Mutation	p.L370F	141	0.31
PTEN · driver	Mutation	p.N48I	22	0.31
GOT1L1	High-level amplification	GISTIC ±2	0	0.35
PKHD1L1	Mutation	p.S2910L	0	0.34
FLG	Mutation	p.D2784H	1	0.33
DDHD2	High-level amplification	GISTIC ±2	1	0.32
BRF2	High-level amplification	GISTIC ±2	2	0.32
LSM1	High-level amplification	GISTIC ±2	1	0.32
HTRA4	High-level amplification	GISTIC ±2	0	0.32
TM2D2	High-level amplification	GISTIC ±2	0	0.32



42 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109) … and 172 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A1JC	59	III	LumB	0.67	no progression, followed 15.8 mo
TCGA-A2-A0SV	63	IV	LumB	0.62	progressed at 27.1 mo (group C)
TCGA-A8-A092	48	III	LumB	0.62	no progression, followed 31.0 mo
TCGA-BH-A1FM	44	III	LumB	0.62	no progression, followed 45.6 mo
TCGA-E2-A9RU	90	III	LumB	0.61	no progression, followed 17.7 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group C** · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**62**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N3A**
Radiation therapy	**Yes**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-AC-A2FE-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **0** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

94th

High risk tier · fused neuro-symbolic model



Probability of progression

20% by 36 mo

12 mo 6% · 24 mo 12% · 60 mo 36%

Model-estimated progression time

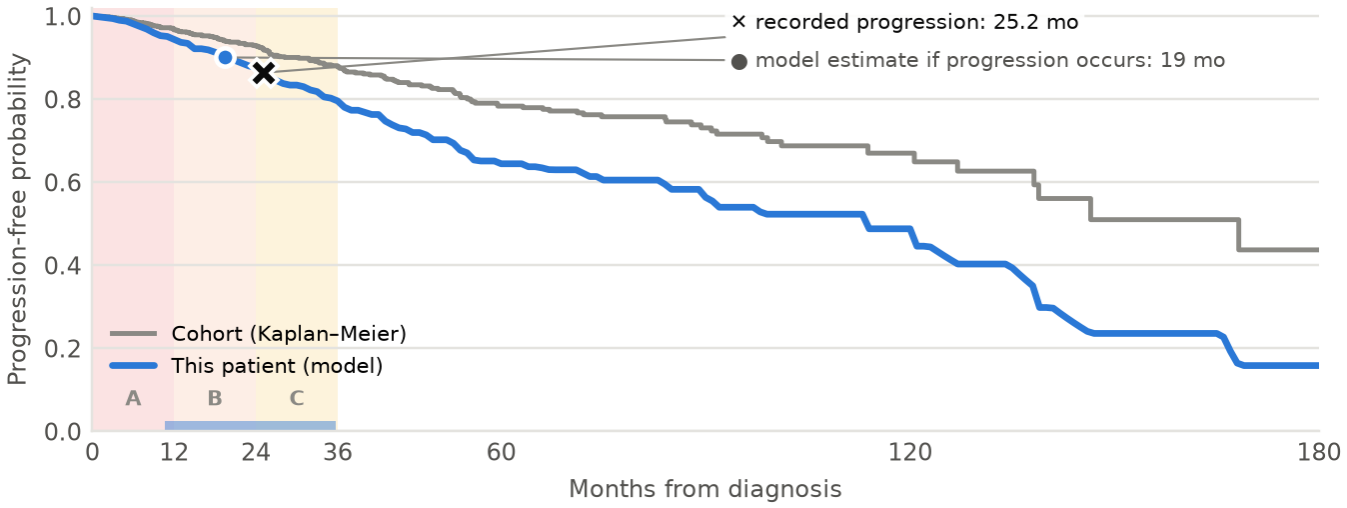
≈ 19 months

Window 12–24 months · IQR 11–36 mo
if progression occurs

Recorded progression

25.2 months

■ **Group C** · Progression in year 3
✗ **recorded window differs** · |error| 6 mo



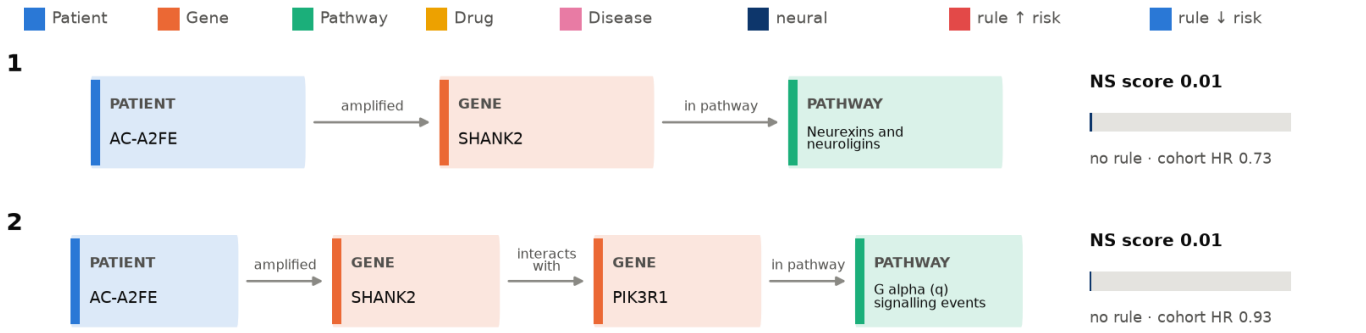
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 62-year-old patient has a subtype-unassigned tumour, AJCC stage III (T3, N3, MX). The fused neuro-symbolic model places them in the **94th risk percentile** (high tier). The strongest learned formula that holds for them (*Histology(p, lobular) ⇒ Progress(p)*) lowers the progression hazard (weight -0.017 per SD). The highest-scoring knowledge-graph path runs from **SHANK2** to **Neurexins and neuroligins**. Progression was recorded at 25.2 months; the model's estimate given progression was 19 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

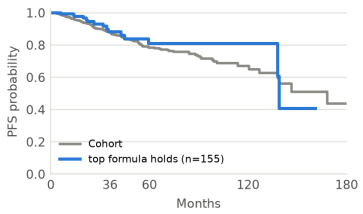


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 2 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-AC-A2FE** amplified → **SHANK2** in pathway → **Neurexins and neuroligins**	2	No learned formula fires on this path; its score is neural only. cohort: HR 0.73, n = 213 carriers	**0.01**
2	**TCGA-AC-A2FE** amplified → **SHANK2** interacts with → **PIK3R1** in pathway → **G alpha (q) signalling events**	1	No learned formula fires on this path; its score is neural only. cohort: HR 0.93, n = 370 carriers	**0.01**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

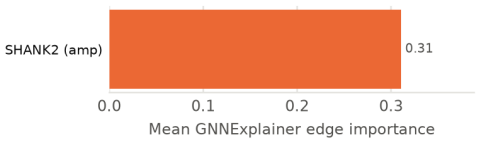
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.017 ↓ risk	Histology(p, lobular) ⇒ Progress(p)	single atom	154 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
SHANK2	High-level amplification	GISTIC ±2	1	0.31



1 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A7-A426	50	III	missing	0.94	no progression, followed 12.0 mo
TCGA-EW-A424	51	III	missing	0.79	no progression, followed 23.5 mo
TCGA-AC-A2QI	76	III	missing	0.79	no progression, followed 19.3 mo
TCGA-E9-A3Q9	78	III	missing	0.78	no progression, followed 32.9 mo
TCGA-AC-A3BB	46	III	missing	0.78	no progression, followed 32.4 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time

estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group C** · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**87**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1MI**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

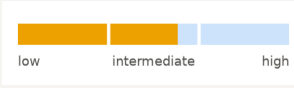
Sample: **TCGA-AC-A2FM-01** · Aneuploidy score: **11** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **0.07** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

59th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

11% by 36 mo

12 mo 3% · 24 mo 6% · 60 mo 19%

Model-estimated progression time

≈ 39 months

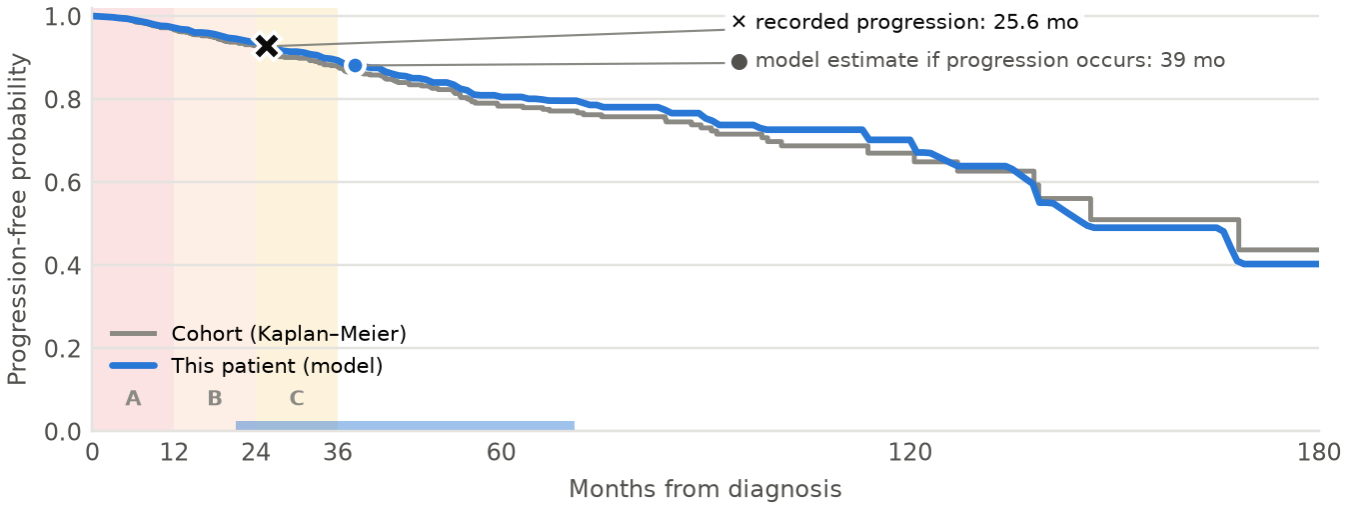
Window > 36 months · IQR 21–71 mo if progression occurs

Recorded progression

25.6 months

■ **Group C** · Progression in year 3

✗ **recorded window differs** · |error| 13 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 87-year-old patient has a subtype-unassigned tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **59th risk percentile** (intermediate tier). The strongest learned formula that holds for them (*Stage(p, II)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.151 per SD). Progression was recorded at 25.6 months; the model's estimate given progression was 39 months.

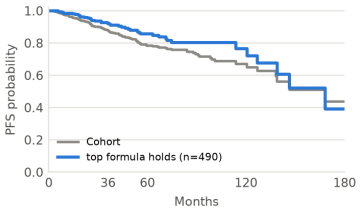
3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

No explanation path is available: none of this patient's altered panel genes has a stored PrimeKG pathway, drug, or breast-disease link (or the tumour carries no qualifying alteration). The risk estimate therefore rests on the clinical block and the symbolic rules below.

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.151 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	482 pts
-0.017 ↓ risk	Histology(p, lobular) ⇒ Progress(p)	single atom	154 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

No mutation or high-level CNA on the 500-gene panel was recorded for this tumour.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-GM-A3NY	72	II	missing	0.98	no progression, followed 38.2 mo
TCGA-MS-A51U	44	II	missing	0.81	no progression, followed 22.4 mo
TCGA-AC-A3W7	66	II	missing	0.80	no progression, followed 15.5 mo
TCGA-JL-A3YW	49	II	missing	0.80	no progression, followed 11.8 mo
TCGA-A7-A3IZ	62	II	missing	0.78	no progression, followed 10.6 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group C** · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**80**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T1A**
Prior diagnosis	**No**	Pathologic N	**N1C**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-BH-A1F0-01** · Aneuploidy score: **7** · TMB (nonsyn., /Mb): **1.97** · MSIsensor score: **0.01** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

44th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 17%

Model-estimated progression time

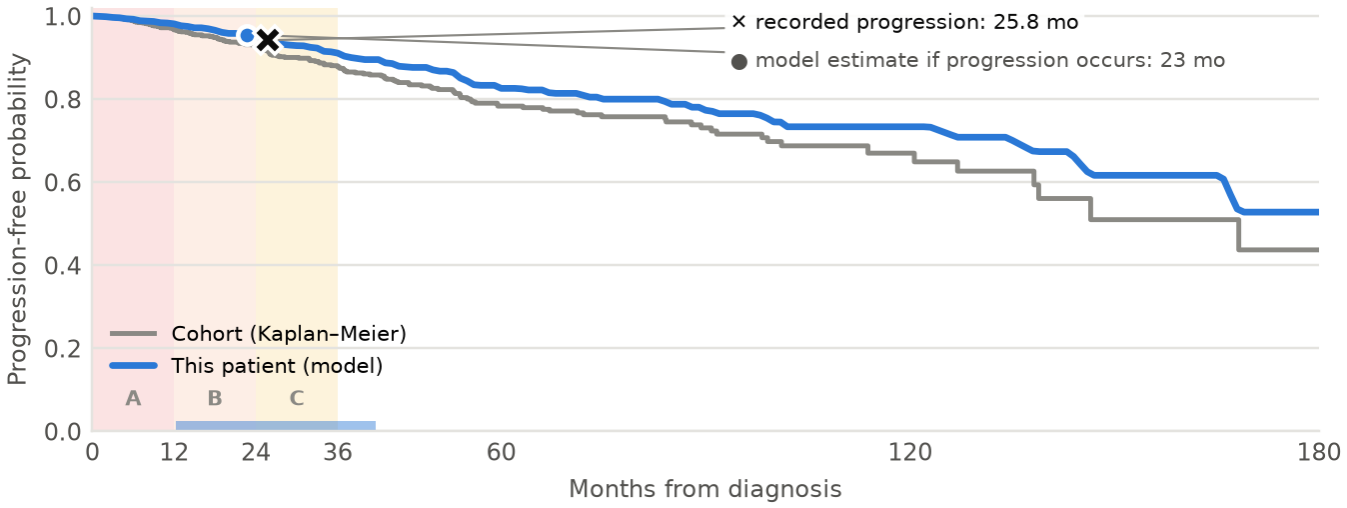
≈ 23 months

Window 12–24 months · IQR 12–42 mo
if progression occurs

Recorded progression

25.8 months

■ **Group C** · Progression in year 3
✗ **recorded window differs** · |error| 3 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 80-year-old patient has a Basal tumour, AJCC stage II (T1, N1, M0). The fused neuro-symbolic model places them in the **44th risk percentile** (intermediate tier). The strongest learned formula that holds for them (*TStage(p, T1)* ∧ *NStage(p, N1)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.090 per SD). The highest-scoring knowledge-graph path runs from **COL22A1** to **Collagen biosynthesis and modifying enzymes**. Progression was recorded at 25.8 months; the model's estimate given progression was 23 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

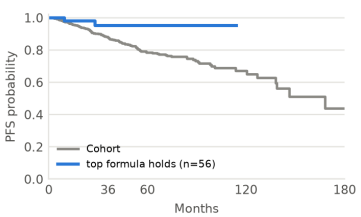


5 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-BH-A1F0** mutated → **COL22A1** in pathway → **Collagen biosynthesis and modifying enzymes**	100	No learned formula fires on this path; its score is neural only. cohort: HR 1.20, n = 151 carriers	**0.60**
2 ★	**TCGA-BH-A1F0** mutated → **ESRP1** in pathway → **FGFR2 alternative splicing**	100	No learned formula fires on this path; its score is neural only. cohort: HR 1.06, n = 113 carriers	**0.60**
3 ★	**TCGA-BH-A1F0** mutated → **ESRP1** interacts with → **ZNF703** in pathway → **Generic Transcription Pathway**	98	No learned formula fires on this path; its score is neural only. cohort: HR 1.13, n = 315 carriers	**0.59**
4 ★	**TCGA-BH-A1F0** mutated → **RB1** in pathway → **Phosphorylation of proteins involved in G1/S transition by active Cyclin E:Cdk2 complexes**	98	No learned formula fires on this path; its score is neural only. cohort: HR 0.82, n = 51 carriers	**0.59**
5 ★	**TCGA-BH-A1F0** mutated → **CR1** in pathway → **RUNX1 and FOXP3 control the development of regulatory T lymphocytes (Tregs)**	97	No learned formula fires on this path; its score is neural only. cohort: HR 1.17, n = 166 carriers	**0.58**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

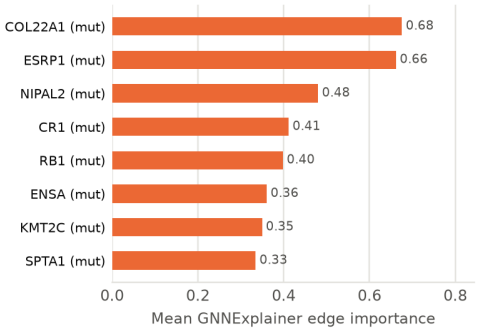
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.090 ↓ risk	TStage(p, T1) ∧ NStage(p, N1) ⇒ Progress(p)	mined pair	56 pts
-0.049 ↓ risk	Stage(p, II) ∧ TStage(p, T1) ⇒ Progress(p)	mined pair	51 pts
-0.034 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	481 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
RB1 · driver	Mutation	p.K548=	25	0.40
KMT2C · driver	Mutation	p.E1096V	4	0.35
TP53 · driver	Mutation	p.V147Lfs*23	57	0.31
COL22A1	Mutation	p.L1467=	2	0.68
ESRP1	Mutation	3'UTR	2	0.66
NIPAL2	Mutation	p.V209=	1	0.48
CR1	Mutation	p.S1296F, p.S1992Y	3	0.41
ENSA	Mutation	p.A11=	1	0.36
SPTA1	Mutation	p.G14=	4	0.33
ATP1A4	Mutation	p.A632=, p.K202=	4	0.33



10 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

RB1 → Insulin human (DB00030); RB1 → Insulin pork (DB00071); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548) … and 1 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0ST	62	II	Basal	0.79	no progression, followed 99.2 mo
TCGA-EW-A1PH	52	II	Basal	0.78	no progression, followed 20.0 mo
TCGA-BH-A18Q	56	II	Basal	0.75	no progression, followed 55.6 mo
TCGA-E2-A1AZ	63	II	Basal	0.74	no progression, followed 76.6 mo
TCGA-S3-AA0Z	63	II	Basal	0.73	no progression, followed 20.7 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group C** · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**45**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N2**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

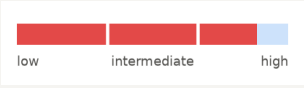
Sample: **TCGA-E9-A226-01** · Aneuploidy score: **7** · TMB (nonsyn., /Mb): **2.03** · MSIsensor score: **0.31** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

89th

High risk tier · fused neuro-symbolic model



Probability of progression

17% by 36 mo

12 mo 4% · 24 mo 9% · 60 mo 31%

Model-estimated progression time

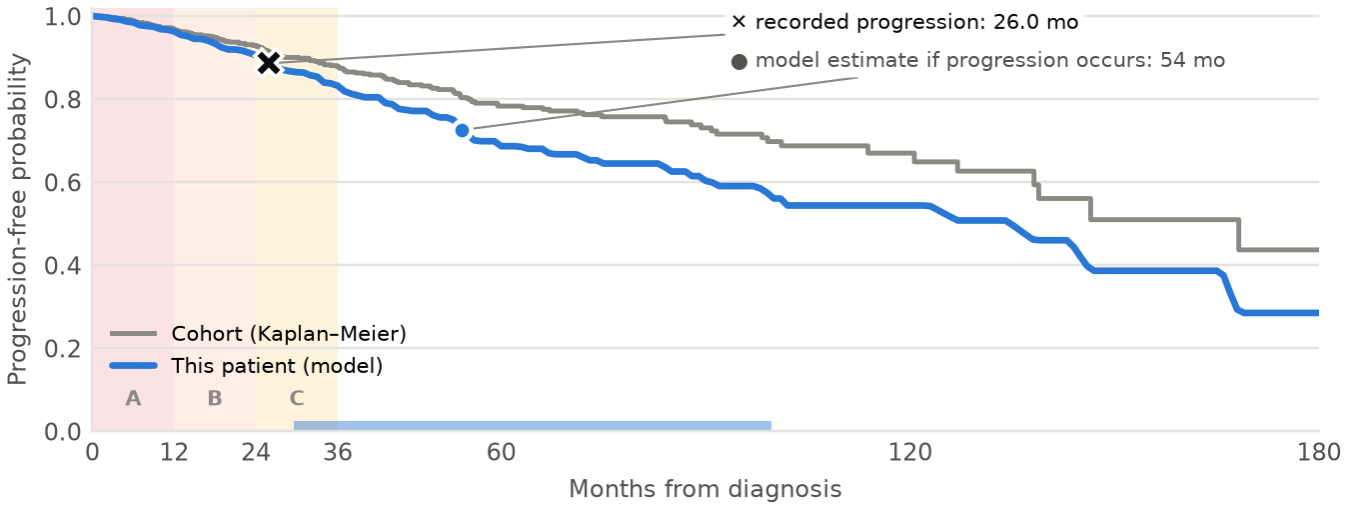
≈ 54 months

Window > 36 months · IQR 30–100 mo
if progression occurs

Recorded progression

26.0 months

■ **Group C** · Progression in year 3
✗ **recorded window differs** · |error| 28 mo



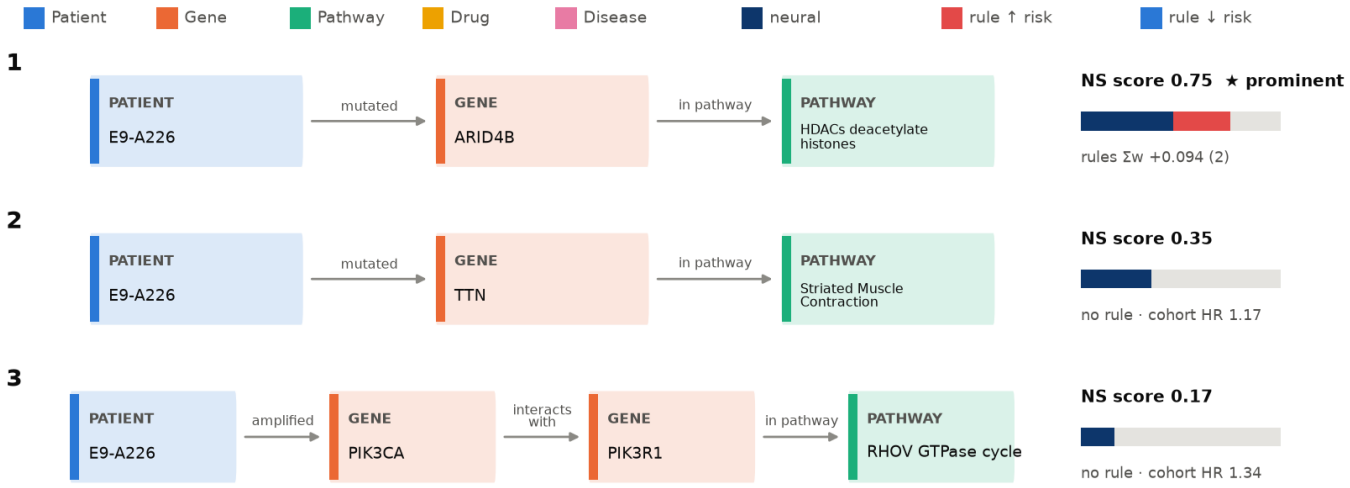
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 45-year-old patient has a LumB tumour, AJCC stage III (T2, N2, M0). The fused neuro-symbolic model places them in the **89th risk percentile** (high tier). The strongest learned formula that holds for them (*Altered(p, g) ∧ InPathway(g, 'HDACs deacetylate histones') ⇒ Progress(p)*) raises the progression hazard (weight +0.071 per SD). The highest-scoring knowledge-graph path runs from **ARID4B** to **HDACs deacetylate histones**; the learned formulas it instantiates point to faster progression (Σw +0.094). Progression was recorded at 26.0 months; the model's estimate given progression was 54 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

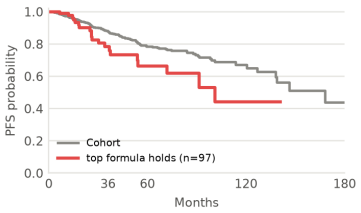


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-E9-A226** mutated → **ARID4B** in pathway → **HDACs deacetylate histones**	77	↑ **faster progression** · Σw +0.094 over 2 formula(s) Altered(p, g) ∧ InPathway(g, 'HDACs deacetylate histones') ⇒ Progress(p) cohort: HR 1.84, n = 97 carriers	**0.75**
2	**TCGA-E9-A226** mutated → **TTN** in pathway → **Striated Muscle Contraction**	59	No learned formula fires on this path; its score is neural only. cohort: HR 1.17, n = 329 carriers	**0.35**
3	**TCGA-E9-A226** amplified → **PIK3CA** interacts with → **PIK3R1** in pathway → **RHOV GTPase cycle**	28	No learned formula fires on this path; its score is neural only. cohort: HR 1.34, n = 31 carriers	**0.17**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

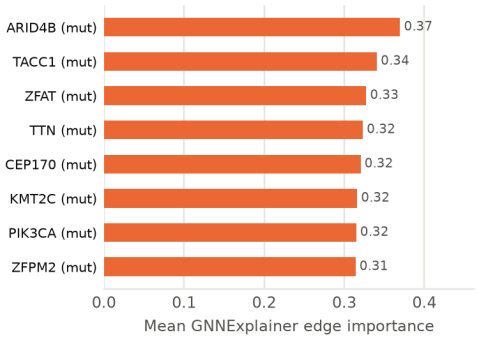
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.071 ↑ risk	Altered(p, g) ∧ InPathway(g, 'HDACs deacetylate histones') ⇒ Progress(p) [KG]	single atom	97 pts
+0.023 ↑ risk	Altered(p, g) ∧ InPathway(g, 'HDACs deacetylate histones') ∧ Altered(p, g) ∧ InPathway(g, 'Ovarian tumor domain proteases') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of TP53 Activity through Acetylation') ⇒ Progress(p) [KG]	RL-proposed	46 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
KMT2C · driver	Mutation	p.E3679Dfs*2	4	0.32
PIK3CA · driver	High-level amplification	GISTIC ±2	70	0.32
PIK3CA · driver	Mutation	p.N345K	70	0.32
TP53 · driver	Mutation	p.P152Afs*14	57	0.31
ARID4B	Mutation	p.E250G	3	0.37
TACC1	Mutation	p.A605=	0	0.34
ZFAT	Mutation	p.V560=	0	0.33
TTN	Mutation	p.I2124=	2	0.32
CEP170	Mutation	p.R462T	0	0.32
ZFPM2	Mutation	p.E1061D	2	0.31



10 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241) … and 12 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A27N	36	III	LumB	0.83	no progression, followed 17.1 mo
TCGA-BH-A0BF	56	II	LumB	0.77	no progression, followed 43.5 mo
TCGA-A8-A092	48	III	LumB	0.76	no progression, followed 31.0 mo
TCGA-AR-A24K	46	II	LumB	0.74	no progression, followed 50.9 mo
TCGA-A2-A0YT	56	III	LumB	0.74	no progression, followed 23.8 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group C** · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**66**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**—**	PAM50 subtype	**LumA**
Ethnicity	**—**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

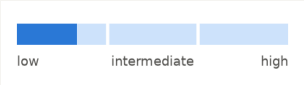
Sample: **TCGA-A8-A06Y-01** · Aneuploidy score: **12** · TMB (nonsyn., /Mb): **0.97** · MSIsensor score: **0.87** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

22nd

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

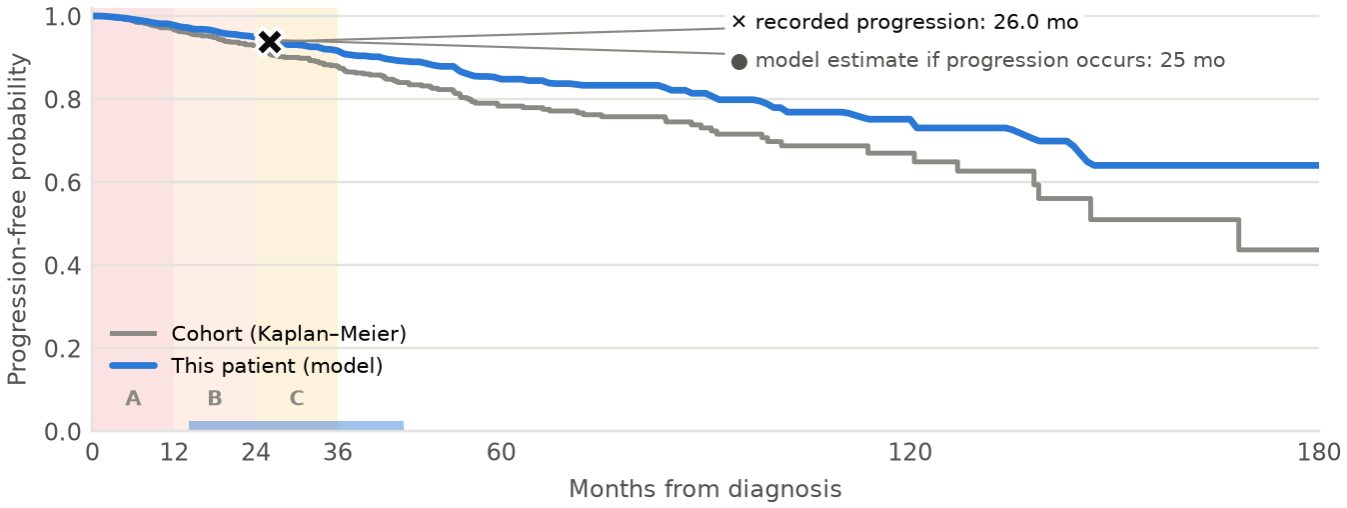
≈ 25 months

Window 24–36 months · IQR 14–46 mo
if progression occurs

Recorded progression

26.0 months

■ **Group C** · Progression in year 3
✓ **estimate in recorded window** · |error| 1 mo



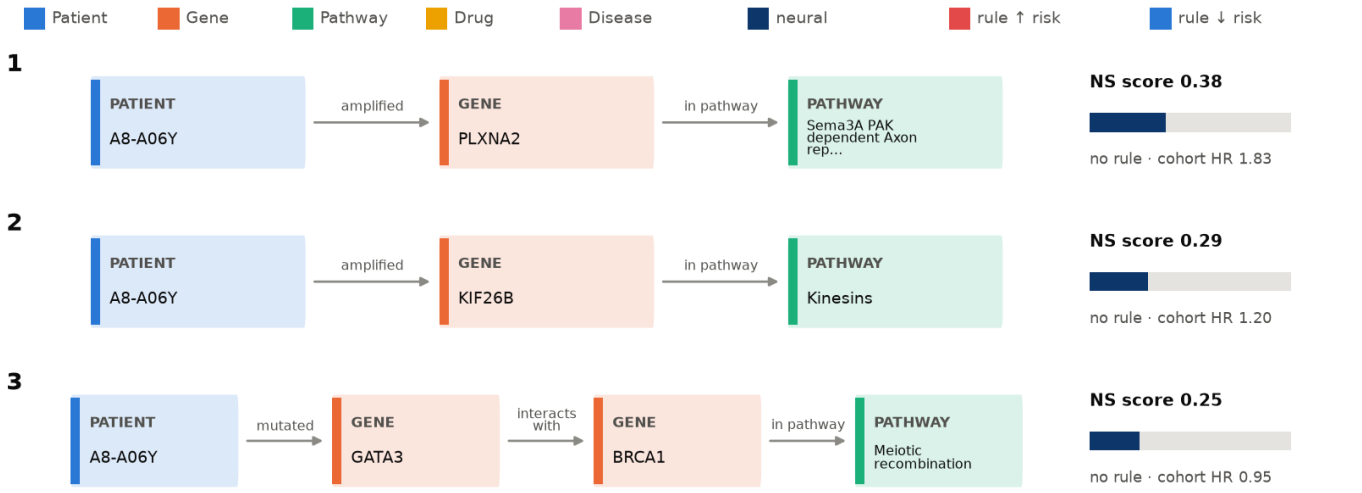
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 66-year-old patient has a LumA tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **22nd risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0) ⇒ Progress(p)*) lowers the progression hazard (weight -0.095 per SD). The highest-scoring knowledge-graph path runs from **PLXNA2** to **Sema3A PAK dependent Axon repulsion**. Progression was recorded at 26.0 months; the model's estimate given progression was 25 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

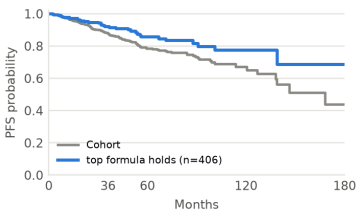


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-A8-A06Y** amplified → **PLXNA2** in pathway → **Sema3A PAK dependent Axon repulsion**	63	No learned formula fires on this path; its score is neural only. cohort: HR 1.83, n = 92 carriers	**0.38**
2	**TCGA-A8-A06Y** amplified → **KIF26B** in pathway → **Kinesins**	49	No learned formula fires on this path; its score is neural only. cohort: HR 1.20, n = 191 carriers	**0.29**
3	**TCGA-A8-A06Y** mutated → **GATA3** interacts with → **BRCA1** in pathway → **Meiotic recombination**	41	No learned formula fires on this path; its score is neural only. cohort: HR 0.95, n = 107 carriers	**0.25**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

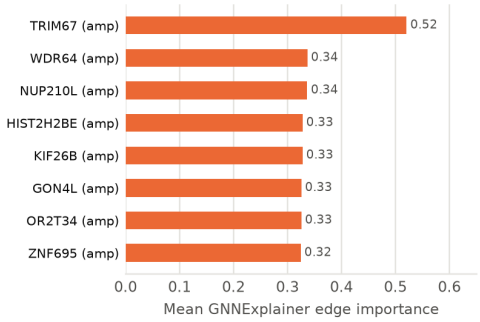
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.095 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	397 pts
+0.058 ↑ risk	Amplified(p, GOLPH3L) ∧ Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ⇒ Progress(p) [KG]	mined pair	55 pts
-0.010 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	390 pts
-0.006 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	487 pts
+0.005 ↑ risk	Amplified(p, GOLPH3L) ∧ Altered(p, g) ∧ InPathway(g, 'RUNX1 and FOXP3 control the development of regulatory T lymphocytes (Tregs)') ⇒ Progress(p) [KG]	mined pair	48 pts
+0.004 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Defective B3GALTL causes Peters-plus syndrome (PpS)') ⇒ Progress(p) [KG]	single atom	94 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GATA3 · driver	Mutation	p.L354Pfs*17	13	0.31
TRIM67	High-level amplification	GISTIC ±2	0	0.52
WDR64	High-level amplification	GISTIC ±2	0	0.34
NUP210L	High-level amplification	GISTIC ±2	0	0.34
HIST2H2BE	High-level amplification	GISTIC ±2	0	0.33
KIF26B	High-level amplification	GISTIC ±2	2	0.33
GON4L	High-level amplification	GISTIC ±2	0	0.33
OR2T34	High-level amplification	GISTIC ±2	1	0.33
ZNF695	High-level amplification	GISTIC ±2	0	0.32
OR2L13	High-level amplification	GISTIC ±2	1	0.32
FLG2	High-level amplification	GISTIC ±2	1	0.32
RBM34	High-level amplification	GISTIC ±2	0	0.32



108 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

NID1 → Urokinase (DB00013); SV2A → Levetiracetam (DB01202); SV2A → Brivaracetam (DB05541); SV2A → Seletracetam (DB05885); ATP1A4 → Rubidium Rb-82 (DB09479); CA14 → Benzthiazide (DB00562); CA14 → Cyclothiazide (DB00606); CA14 → Acetazolamide (DB00819); CA14 → Zonisamide (DB00909); CA14 → Ellagic acid (DB08846) … and 47 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AO-A12A	47	II	LumA	0.96	no progression, followed 102.3 mo
TCGA-EW-A1P3	48	II	LumA	0.92	no progression, followed 53.0 mo
TCGA-BH-A18F	50	II	LumA	0.92	no progression, followed 32.9 mo
TCGA-E9-A249	45	II	LumB	0.91	no progression, followed 7.1 mo
TCGA-A8-A07F	65	II	LumA	0.90	no progression, followed 19.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group C** · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**48**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**EUR**	Pathologic T	**T4**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**Yes**	Pathologic M	**M1**
Neoadjuvant therapy	**Yes**	ICD-O-3 histology	**8523/3**

Sample: **TCGA-AO-A0J5-01** · Aneuploidy score: **3** · TMB (nonsyn., /Mb): **1.77** · MSIsensor score: **0.12** · AJCC edition: **—**

2 · Progression assessment

Model risk (cohort percentile)

98th

High risk tier · fused neuro-symbolic model



Probability of progression

51% by 36 mo

12 mo 17% · 24 mo 34% · 60 mo 78%

Model-estimated progression time

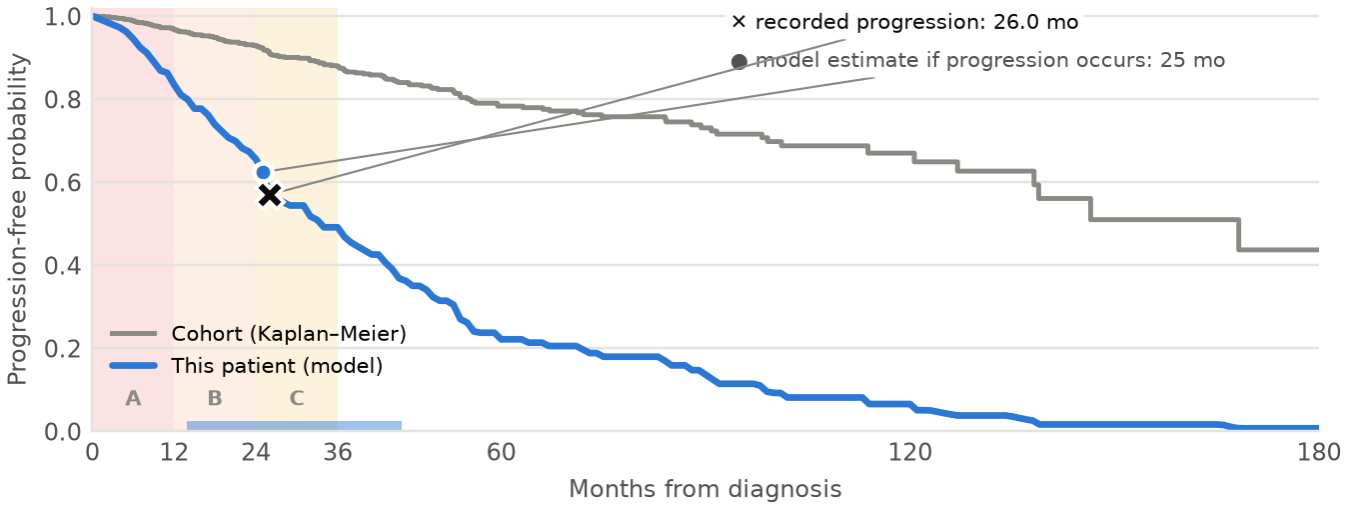
≈ 25 months

Window 24–36 months · IQR 14–45 mo
if progression occurs

Recorded progression

26.0 months

■ **Group C** · Progression in year 3
✓ **estimate in recorded window** · |error| 1 mo



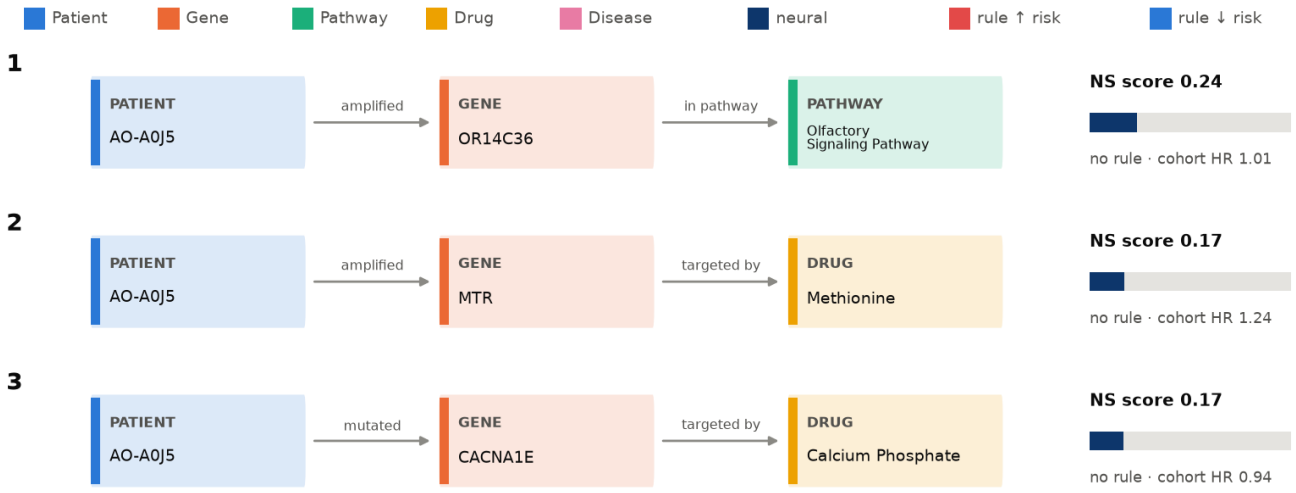
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 48-year-old patient has a LumA tumour, AJCC stage IV (T4, N1, M1). The fused neuro-symbolic model places them in the **98th risk percentile** (high tier). The strongest learned formula that holds for them (*TStage(p, T4)* ⇒ *Progress(p)*) raises the progression hazard (weight +0.096 per SD). The highest-scoring knowledge-graph path runs from **OR14C36** to **Olfactory Signaling Pathway**. Progression was recorded at 26.0 months; the model's estimate given progression was 25 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

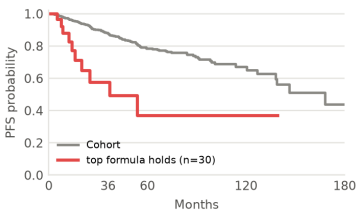


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-AO-A0J5** amplified → **OR14C36** in pathway → **Olfactory Signaling Pathway**	39	No learned formula fires on this path; its score is neural only. cohort: HR 1.01, n = 158 carriers	**0.24**
2	**TCGA-AO-A0J5** amplified → **MTR** targeted by → **Methionine**	29	No learned formula fires on this path; its score is neural only. cohort: HR 1.24, n = 86 carriers	**0.17**
3	**TCGA-AO-A0J5** mutated → **CACNA1E** targeted by → **Calcium Phosphate**	28	No learned formula fires on this path; its score is neural only. cohort: HR 0.94, n = 188 carriers	**0.17**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

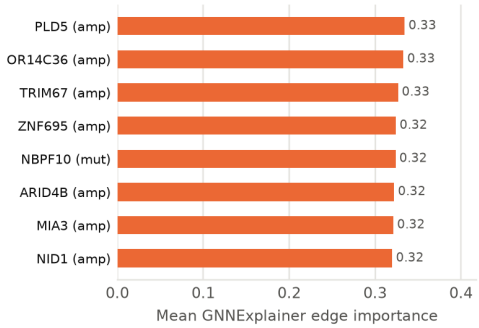
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.096 ↑ risk	TStage(p, T4) ⇒ Progress(p)	single atom	28 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3CA · driver	Mutation	p.H1047R	70	0.31
GATA3 · driver	Mutation	p.R330Lfs*24	13	0.31
PLD5	High-level amplification	GISTIC ±2	0	0.33
OR14C36	High-level amplification	GISTIC ±2	1	0.33
TRIM67	High-level amplification	GISTIC ±2	0	0.33
ZNF695	High-level amplification	GISTIC ±2	0	0.32
NBPF10	Mutation	p.G1285=	0	0.32
ARID4B	High-level amplification	GISTIC ±2	3	0.32
MIA3	High-level amplification	GISTIC ±2	3	0.32
NID1	High-level amplification	GISTIC ±2	3	0.32
CHML	High-level amplification	GISTIC ±2	2	0.32
OXR1	Mutation	p.E653V	0	0.32



79 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); NID1 → Urokinase (DB00013); RYR2 → Flecainide (DB01195); RYR2 → Tetracaine (DB09085) … and 47 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AR-A5QP	54	II	LumA	0.72	no progression, followed 39.0 mo
TCGA-E9-A1NI	51	II	LumB	0.70	no progression, followed 9.9 mo
TCGA-BH-A28Q	46	II	LumA	0.69	no progression, followed 36.8 mo
TCGA-D8-A1JB	54	II	LumA	0.69	no progression, followed 55.5 mo
TCGA-E9-A229	37	I	LumA	0.69	no progression, followed 37.7 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group C** · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**61**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**No**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-A7-A13H-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **—** · MSIsensor score: **0** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

68th

High risk tier · fused neuro-symbolic model



Probability of progression

11% by 36 mo

12 mo 3% · 24 mo 7% · 60 mo 23%

Model-estimated progression time

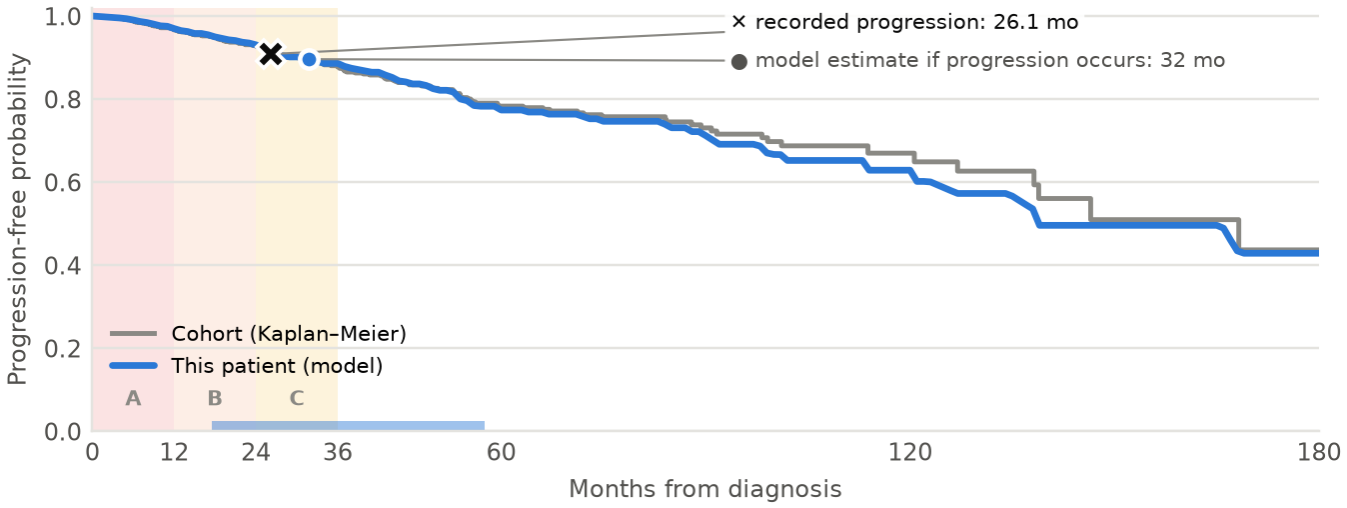
≈ 32 months

Window 24–36 months · IQR 17–58 mo
if progression occurs

Recorded progression

26.1 months

■ **Group C** · Progression in year 3
✓ **estimate in recorded window** · |error| 6 mo



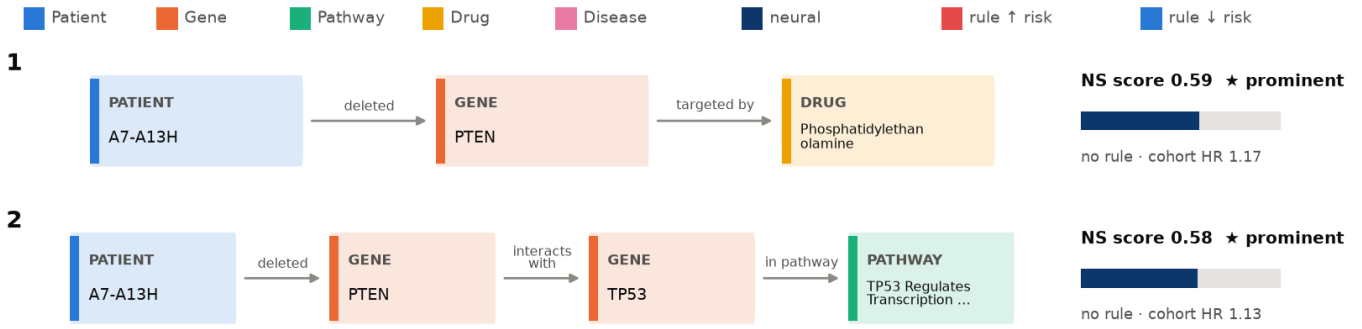
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 61-year-old patient has a subtype-unassigned tumour, AJCC stage II (T2, N1, MX). The fused neuro-symbolic model places them in the **68th risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **PTEN** to **Phosphatidylethanolamine**. Progression was recorded at 26.1 months; the model's estimate given progression was 32 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



2 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

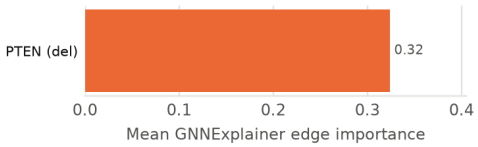
#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-A7-A13H** deleted → **PTEN** targeted by → **Phosphatidylethanolamine**	99	No learned formula fires on this path; its score is neural only. cohort: HR 1.17, n = 90 carriers	**0.59**
2 ★	**TCGA-A7-A13H** deleted → **PTEN** interacts with → **TP53** in pathway → **TP53 Regulates Transcription of Death Receptors and Ligands**	97	No learned formula fires on this path; its score is neural only. cohort: HR 1.13, n = 344 carriers	**0.58**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

No learned formula with a non-zero weight fires for this patient; the symbolic risk is the baseline of the logic model.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PTEN · driver	Homozygous deletion	GISTIC ±2	22	0.32



1 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PTEN → Phosphatidylethanolamine (DB04327). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AC-A3W7	66	II	missing	0.86	no progression, followed 15.5 mo
TCGA-A7-A4SA	40	II	missing	0.84	no progression, followed 14.9 mo
TCGA-AC-A62Y	79	II	LumA	0.81	no progression, followed 17.4 mo
TCGA-BH-A8G0	54	II	LumA	0.71	no progression, followed 21.8 mo
TCGA-GM-A3NY	72	II	missing	0.71	no progression, followed 38.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group C · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**49**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**AFR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

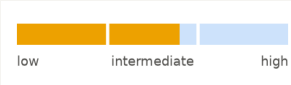
Sample: **TCGA-A2-A3XY-01** · Aneuploidy score: **8** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **0.02** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

60th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

11% by 36 mo

12 mo 3% · 24 mo 6% · 60 mo 20%

Model-estimated progression time

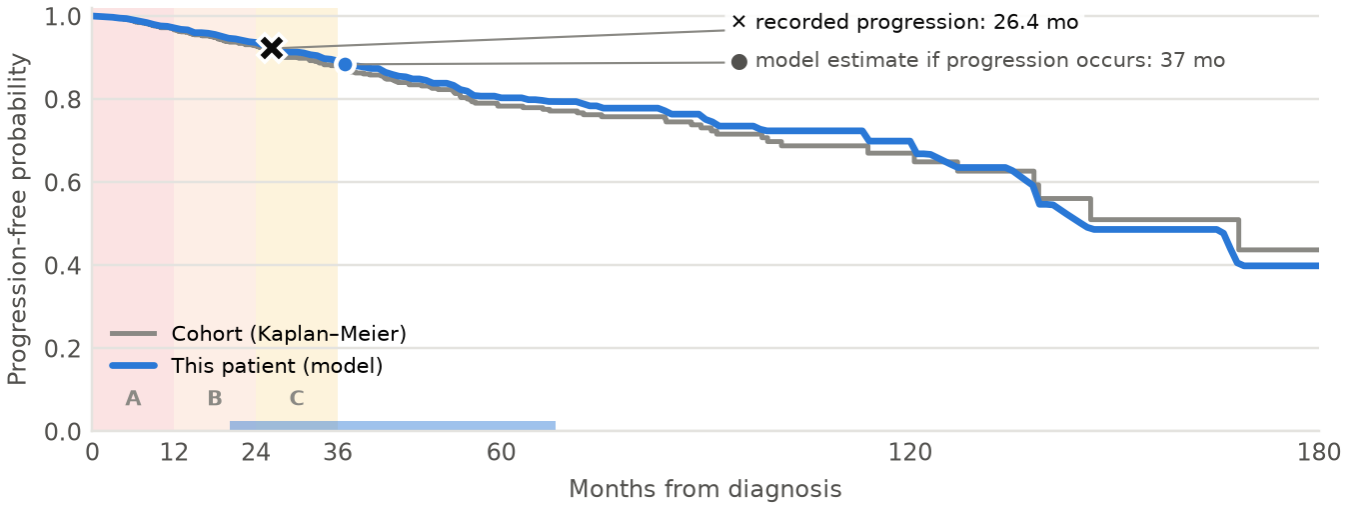
≈ 37 months

Window > 36 months · IQR 20–68 mo if progression occurs

Recorded progression

26.4 months

Group C · Progression in year 3
✗ recorded window differs · |error| 11 mo



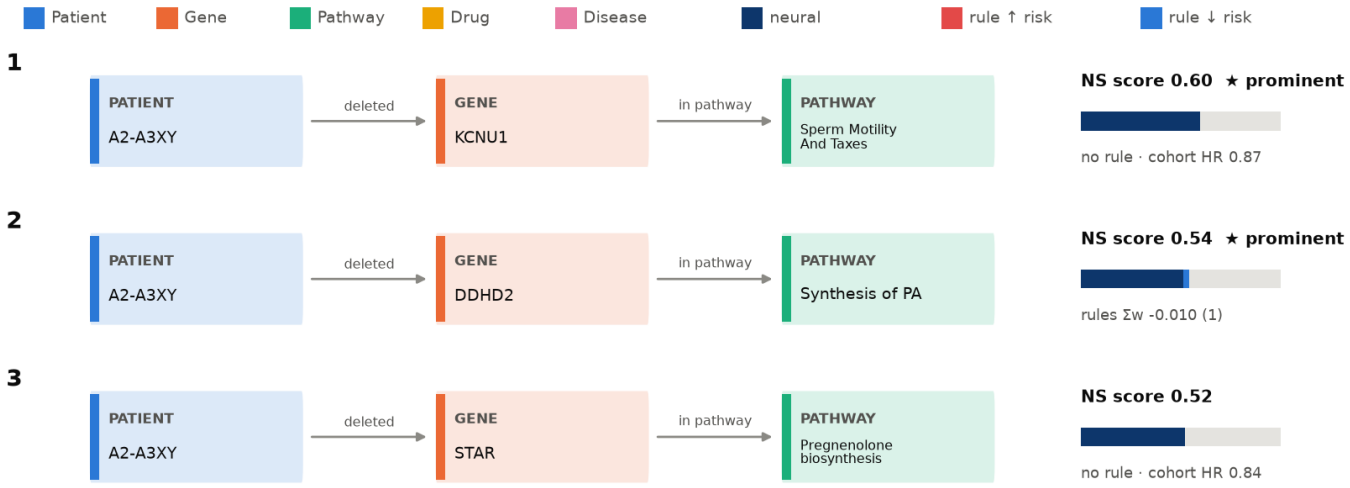
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 49-year-old patient has a subtype-unassigned tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **60th risk percentile** (intermediate tier). The strongest learned formula that holds for them (*Stage(p, II)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.151 per SD). The highest-scoring knowledge-graph path runs from **KCNU1** to **Sperm Motility And Taxes**. Progression was recorded at 26.4 months; the model's estimate given progression was 37 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

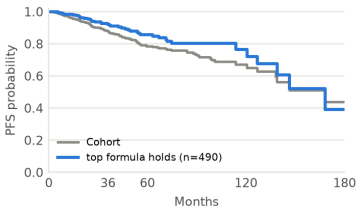


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-A2-A3XY** deleted → **KCNU1** in pathway → **Sperm Motility And Taxes**	99	No learned formula fires on this path; its score is neural only. cohort: HR 0.87, n = 97 carriers	**0.60**
2 ★	**TCGA-A2-A3XY** deleted → **DDHD2** in pathway → **Synthesis of PA**	85	↓ **slower progression** · Σw -0.010 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Synthesis of PA') ⇒ Progress(p) cohort: HR 0.80, n = 189 carriers	**0.54**
3	**TCGA-A2-A3XY** deleted → **STAR** in pathway → **Pregnenolone biosynthesis**	87	No learned formula fires on this path; its score is neural only. cohort: HR 0.84, n = 196 carriers	**0.52**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

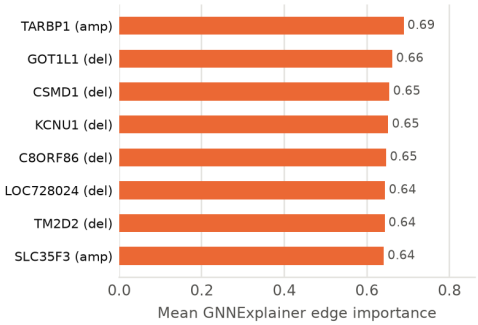
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.151 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	482 pts
-0.041 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	337 pts
-0.010 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Synthesis of PA') ⇒ Progress(p) [KG]	single atom	187 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FGFR1 · driver	Homozygous deletion	GISTIC ±2	43	0.32
PIK3CA · driver	High-level amplification	GISTIC ±2	70	0.31
TARBP1	High-level amplification	GISTIC ±2	0	0.69
GOT1L1	Homozygous deletion	GISTIC ±2	0	0.66
CSMD1	Homozygous deletion	GISTIC ±2	0	0.65
KCNU1	Homozygous deletion	GISTIC ±2	1	0.65
C8ORF86	Homozygous deletion	GISTIC ±2	0	0.65
LOC728024	Homozygous deletion	GISTIC ±2	0	0.64
TM2D2	Homozygous deletion	GISTIC ±2	0	0.64
SLC35F3	High-level amplification	GISTIC ±2	0	0.64
LETM2	Homozygous deletion	GISTIC ±2	0	0.63
PROSC	Homozygous deletion	GISTIC ±2	0	0.63



39 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896); FGFR1 → Ponatinib (DB08901) … and 52 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A1ES	35	II	missing	0.80	progressed at 113.8 mo (group D)
TCGA-E2-A1IE	61	II	LumA	0.75	progressed at 59.5 mo (group D)
TCGA-WT-AB41	55	II	LumA	0.74	no progression, followed 53.0 mo
TCGA-B6-A0WT	61	II	LumA	0.73	progressed at 127.0 mo (group D)
TCGA-BH-A204	80	II	LumB	0.73	no progression, followed 83.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group C** · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**63**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N2A**
Radiation therapy	**Yes**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-A2-A0SV-01** · Aneuploidy score: **13** · TMB (nonsyn., /Mb): **1.1** · MSIsensor score: **0.59** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

99th

High risk tier · fused neuro-symbolic model



Probability of progression

56% by 36 mo

12 mo 19% · 24 mo 38% · 60 mo 82%

Model-estimated progression time

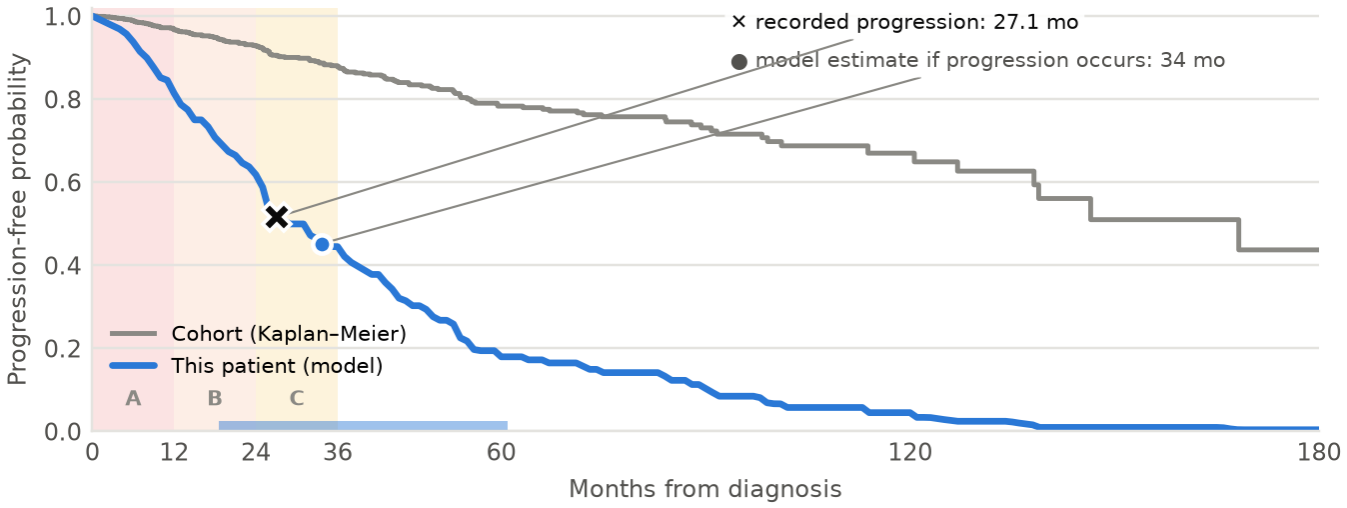
≈ 34 months

Window 24–36 months · IQR 19–61 mo
if progression occurs

Recorded progression

27.1 months

■ **Group C** · Progression in year 3
✓ **estimate in recorded window** · |error| 6 mo



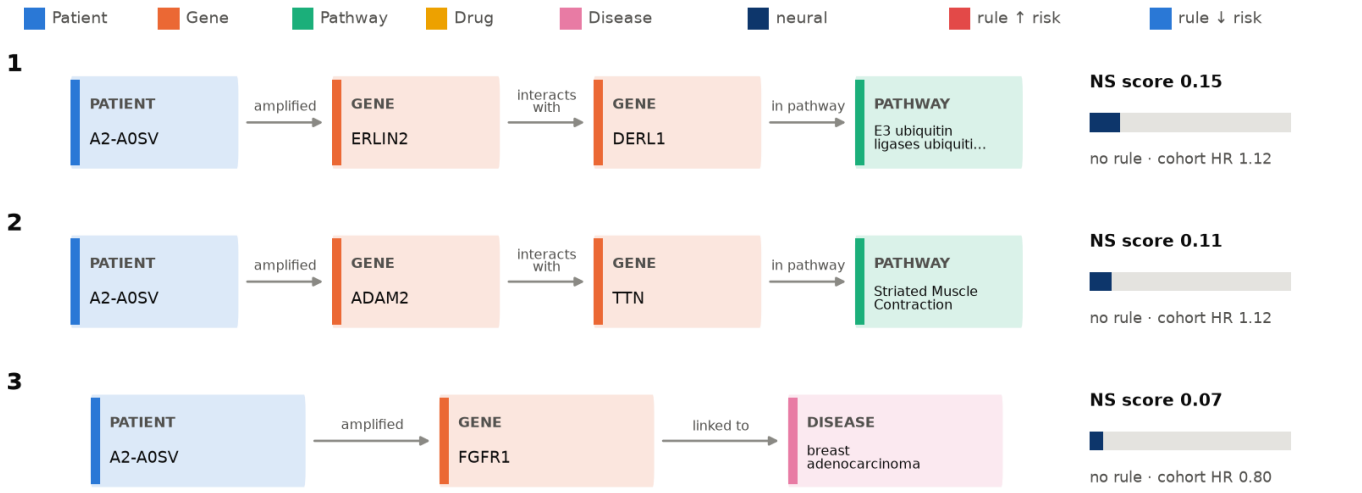
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 63-year-old patient has a LumB tumour, AJCC stage IV (T2, N2, M1). The fused neuro-symbolic model places them in the **99th risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **ERLIN2** to **E3 ubiquitin ligases ubiquitinate target proteins**. Progression was recorded at 27.1 months; the model's estimate given progression was 34 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

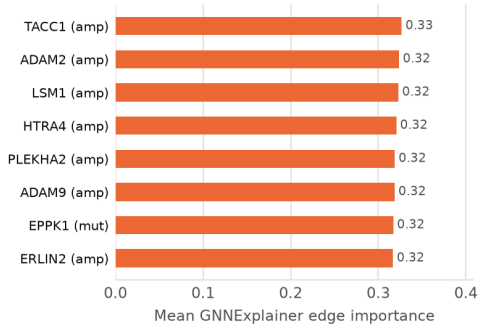
#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-A2-A0SV** amplified → **ERLIN2** interacts with → **DERL1** in pathway → **E3 ubiquitin ligases ubiquitinate target proteins**	25	No learned formula fires on this path; its score is neural only. cohort: HR 1.12, n = 107 carriers	**0.15**
2	**TCGA-A2-A0SV** amplified → **ADAM2** interacts with → **TTN** in pathway → **Striated Muscle Contraction**	18	No learned formula fires on this path; its score is neural only. cohort: HR 1.12, n = 328 carriers	**0.11**
3	**TCGA-A2-A0SV** amplified → **FGFR1** linked to → **breast adenocarcinoma**	11	No learned formula fires on this path; its score is neural only. cohort: HR 0.80, n = 743 carriers	**0.07**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

No learned formula with a non-zero weight fires for this patient; the symbolic risk is the baseline of the logic model.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PTEN · driver	Mutation	p.H64_Y65del	22	0.31
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.31
KMT2C · driver	Mutation	p.I1381V	4	0.31
TACC1	High-level amplification	GISTIC ±2	0	0.33
ADAM2	High-level amplification	GISTIC ±2	1	0.32
LSM1	High-level amplification	GISTIC ±2	1	0.32
HTRA4	High-level amplification	GISTIC ±2	0	0.32
PLEKHA2	High-level amplification	GISTIC ±2	1	0.32
ADAM9	High-level amplification	GISTIC ±2	2	0.32
EPPK1	Mutation	p.P2313L, p.T2311A	0	0.32
ERLIN2	High-level amplification	GISTIC ±2	4	0.32
LETM2	High-level amplification	GISTIC ±2	0	0.32



34 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PTEN → Phosphatidylethanolamine (DB04327); FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896) … and 46 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A8-A08O	45	IV	LumA	0.68	progressed at 25.1 mo (group C)
TCGA-D8-A1JC	59	III	LumB	0.66	no progression, followed 15.8 mo
TCGA-BH-A0E2	49	III	LumA	0.65	no progression, followed 14.3 mo
TCGA-AN-A0XN	68	III	LumA	0.64	no progression, followed 0.3 mo
TCGA-B6-A0WT	61	II	LumA	0.64	progressed at 127.0 mo (group D)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group C** · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**38**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**Black or African American**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**AFR_ADMIX**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-B6-A0IE-01** · Aneuploidy score: **12** · TMB (nonsyn., /Mb): **1** · MSIsensor score: **0.22** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

71st

High risk tier · fused neuro-symbolic model



Probability of progression

13% by 36 mo

12 mo 4% · 24 mo 8% · 60 mo 24%

Model-estimated progression time

≈ 36 months

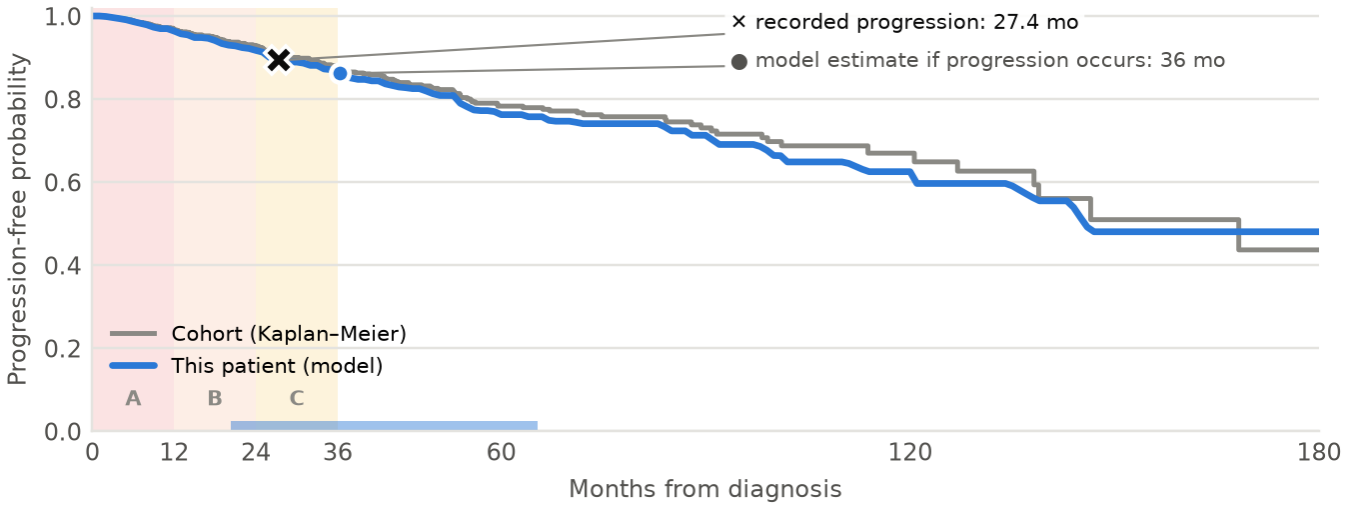
Window > 36 months · IQR 20–65 mo if progression occurs

Recorded progression

27.4 months

■ **Group C** · Progression in year 3

✗ **recorded window differs** · |error| 9 mo



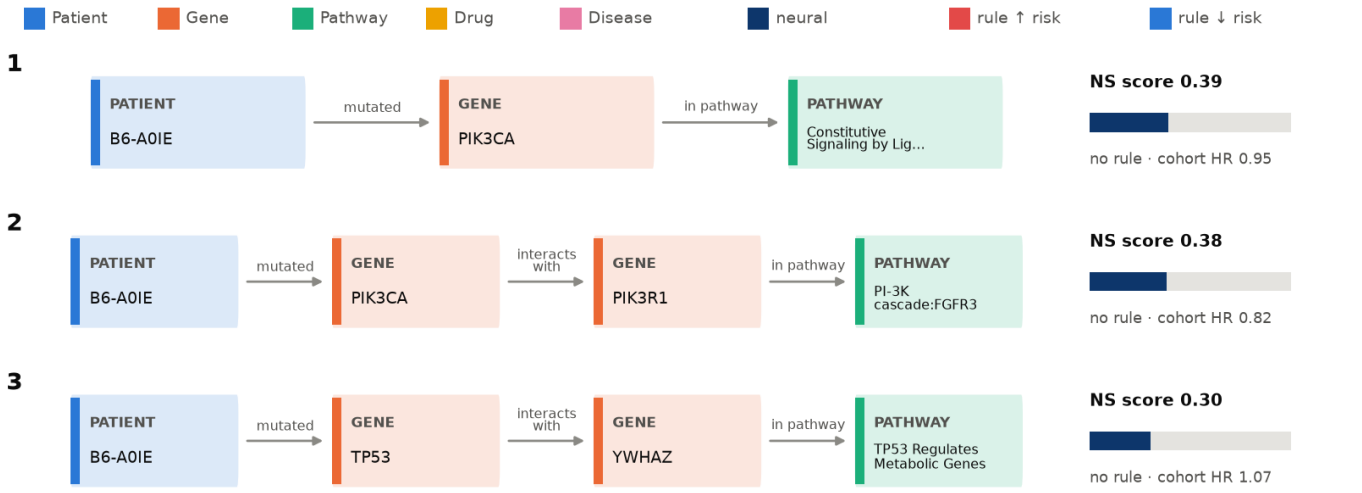
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 38-year-old patient has a LumA tumour, AJCC stage III (T3, N1, M0). The fused neuro-symbolic model places them in the **71st risk percentile** (high tier). The strongest learned formula that holds for them (*Age(p) < 40 ⇒ Progress(p)*) raises the progression hazard (weight +0.025 per SD). The highest-scoring knowledge-graph path runs from **PIK3CA** to **Constitutive Signaling by Ligand-Responsive EGFR Cancer Variants**. Progression was recorded at 27.4 months; the model's estimate given progression was 36 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

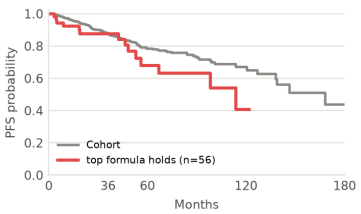


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-B6-A0IE** mutated → **PIK3CA** in pathway → **Constitutive Signaling by Ligand-Responsive EGFR Cancer Variants**	65	No learned formula fires on this path; its score is neural only. cohort: HR 0.95, n = 341 carriers	**0.39**
2	**TCGA-B6-A0IE** mutated → **PIK3CA** interacts with → **PIK3R1** in pathway → **PI-3K cascade:FGFR3**	64	No learned formula fires on this path; its score is neural only. cohort: HR 0.82, n = 394 carriers	**0.38**
3	**TCGA-B6-A0IE** mutated → **TP53** interacts with → **YWHAZ** in pathway → **TP53 Regulates Metabolic Genes**	50	No learned formula fires on this path; its score is neural only. cohort: HR 1.07, n = 449 carriers	**0.30**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

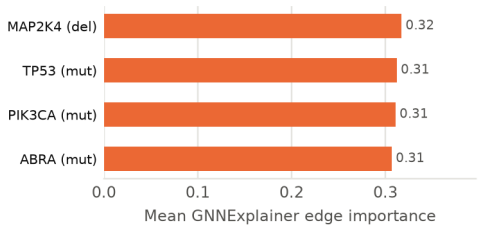
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.025 ↑ risk	Age(p) < 40 ⇒ Progress(p)	single atom	56 pts
-0.010 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	390 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MAP2K4 · driver	Homozygous deletion	GISTIC ±2	12	0.32
TP53 · driver	Mutation	p.X307_splice	57	0.31
PIK3CA · driver	Mutation	p.H1047R	70	0.31
ABRA	Mutation	p.S77I	0	0.31



4 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201) … and 5 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AR-A5QQ	68	III	Basal	0.82	progressed at 9.9 mo (group A)
TCGA-C8-A133	65	III	LumA	0.75	no progression, followed 0.0 mo
TCGA-A2-A3KD	47	III	LumA	0.74	no progression, followed 39.6 mo
TCGA-AR-A24O	43	III	LumA	0.72	progressed at 84.2 mo (group D)
TCGA-E9-A1N5	45	II	LumA	0.68	no progression, followed 36.8 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group C · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**49**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**AFR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

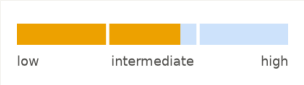
Sample: **TCGA-B6-A0I6-01** · Aneuploidy score: **—** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **2.69** · AJCC edition: **3RD**

2 · Progression assessment

Model risk (cohort percentile)

60th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

12% by 36 mo

12 mo 3% · 24 mo 6% · 60 mo 20%

Model-estimated progression time

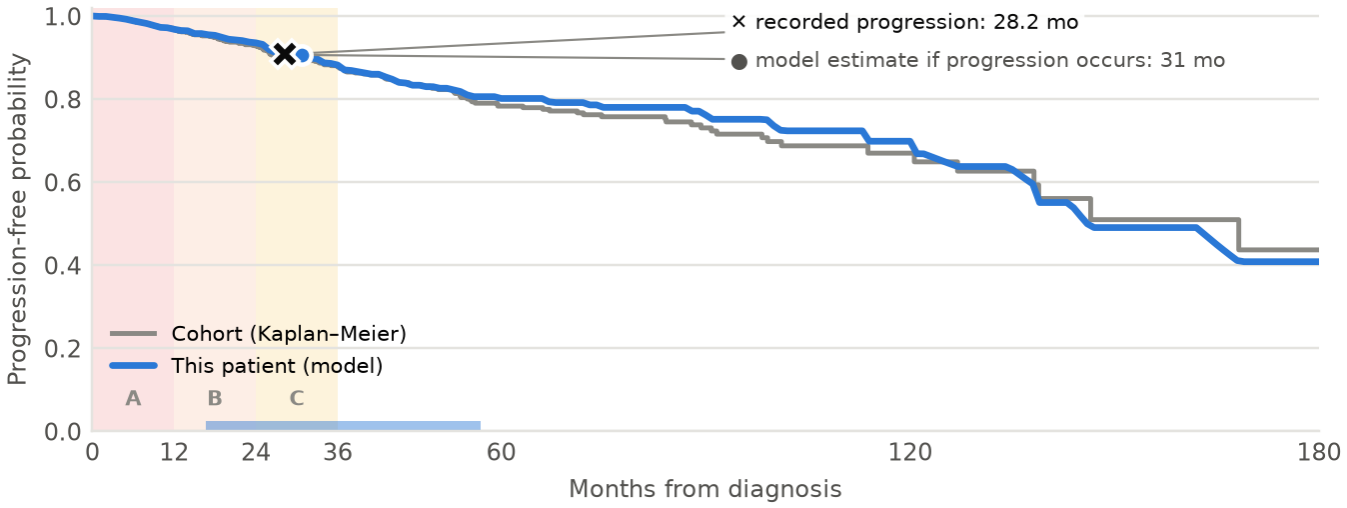
≈ 31 months

Window 24–36 months · IQR 17–57 mo
if progression occurs

Recorded progression

28.2 months

Group C · Progression in year 3
✓ **estimate in recorded window** · |error| 3 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 49-year-old patient has a subtype-unassigned tumour, AJCC stage II (T1, N1, M0). The fused neuro-symbolic model places them in the **60th risk percentile** (intermediate tier). The strongest learned formula that holds for them (*TStage(p, T1) ⇒ Progress(p)*) lowers the progression hazard (weight -0.042 per SD). Progression was recorded at 28.2 months; the model's estimate given progression was 31 months.

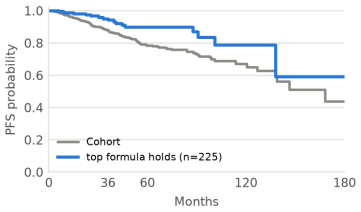
3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

No explanation path is available: none of this patient's altered panel genes has a stored PrimeKG pathway, drug, or breast-disease link (or the tumour carries no qualifying alteration). The risk estimate therefore rests on the clinical block and the symbolic rules below.

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.042 ↓ risk	TStage(p, T1) ⇒ Progress(p)	single atom	225 pts
-0.032 ↓ risk	TStage(p, T1) ∧ NStage(p, N1) ⇒ Progress(p)	mined pair	61 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

No mutation or high-level CNA on the 500-gene panel was recorded for this tumour.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A10E	64	II	missing	0.95	no progression, followed 28.4 mo
TCGA-BH-A0HO	48	II	LumA	0.82	no progression, followed 2.5 mo
TCGA-A8-A086	59	II	LumA	0.79	no progression, followed 13.0 mo
TCGA-A8-A085	44	II	missing	0.79	no progression, followed 37.0 mo
TCGA-AO-A0JI	56	II	LumA	0.78	no progression, followed 50.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-BH-A18P · Group C

Report date 31 Oct 2026

Group C · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**60**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Her2**
Ethnicity	**—**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**EUR**	Pathologic T	**T1**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-BH-A18P-01** · Aneuploidy score: **6** · TMB (nonsyn., /Mb): **9.2** · MSIsensor score: **0.06** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

61st

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

10% by 36 mo

12 mo 3% · 24 mo 6% · 60 mo 21%

Model-estimated progression time

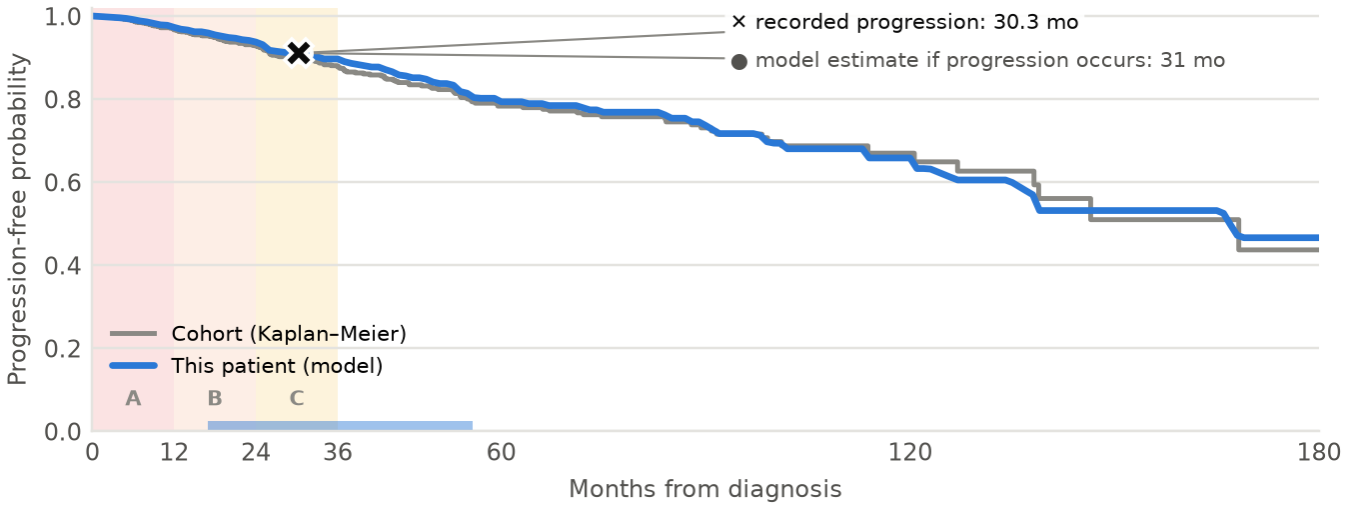
≈ 31 months

Window 24–36 months · IQR 17–56 mo
if progression occurs

Recorded progression

30.3 months

Group C · Progression in year 3
✓ **estimate in recorded window** · |error| 0 mo



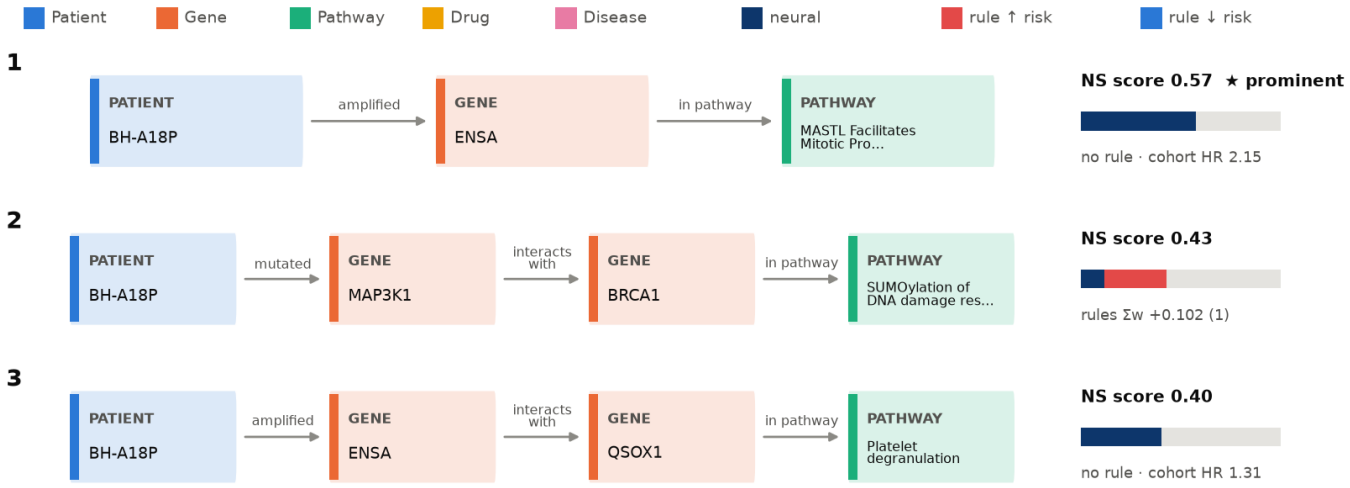
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 60-year-old patient has a Her2 tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **61st risk percentile** (intermediate tier). The strongest learned formula that holds for them (*Altered(p, g)* ∧ *InPathway(g, 'Neurofascin interactions')* ∧ *Altered(p, g)* ∧ *InPathway(g, 'Metalloprotease DUBs')* ∧ *Altered(p, g)* ∧ *InPathway(g, 'Estrogen-dependent gene expression')* ⇒ *Progress(p)*) raises the progression hazard (weight +0.102 per SD). The highest-scoring knowledge-graph path runs from **ENSA** to **MASTL Facilitates Mitotic Progression**. Progression was recorded at 30.3 months; the model's estimate given progression was 31 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

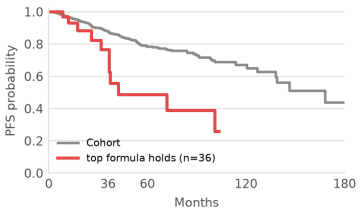


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-BH-A18P** amplified → **ENSA** in pathway → **MASTL Facilitates Mitotic Progression**	96	No learned formula fires on this path; its score is neural only. cohort: HR 2.15, n = 91 carriers	**0.57**
2	**TCGA-BH-A18P** mutated → **MAP3K1** interacts with → **BRCA1** in pathway → **SUMOylation of DNA damage response and repair proteins**	20	↑ **faster progression** · Σw +0.102 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ∧ Altered(p, g) ∧ InPathway(g, 'Metalloprotease DUBs') ∧ Altered(p, g) ∧ InPathway(g, 'Estrogen-dependent gene expression') ⇒ Progress(p) cohort: HR 1.55, n = 233 carriers	**0.43**
3	**TCGA-BH-A18P** amplified → **ENSA** interacts with → **QSOX1** in pathway → **Platelet degranulation**	67	No learned formula fires on this path; its score is neural only. cohort: HR 1.31, n = 312 carriers	**0.40**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

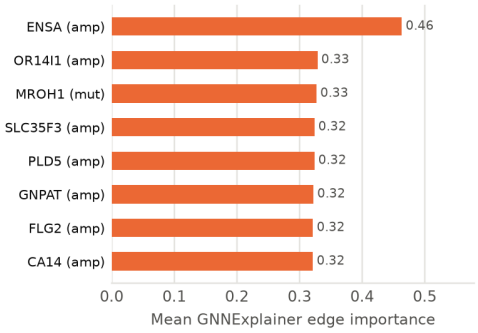
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.102 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ∧ Altered(p, g) ∧ InPathway(g, 'Metalloprotease DUBs') ∧ Altered(p, g) ∧ InPathway(g, 'Estrogen-dependent gene expression') ⇒ Progress(p) [KG]	RL-proposed	36 pts
-0.099 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	398 pts
+0.058 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Defective B3GALTL causes Peters-plus syndrome (PpS)') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of Complement cascade') ⇒ Progress(p) [KG]	RL-proposed	44 pts
-0.009 ↓ risk	NStage(p, N0) ∧ Altered(p, g) ∧ InPathway(g, 'Synthesis of PIPs at the plasma membrane') ⇒ Progress(p) [KG]	mined pair	212 pts
+0.007 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ∧ Altered(p, g) ∧ InPathway(g, 'Estrogen-dependent gene expression') ∧ Altered(p, g) ∧ TargetedBy(g, 'Zinc') ⇒ Progress(p) [KG]	RL-proposed	65 pts
-0.004 ↓ risk	NStage(p, N0) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) [KG]	mined pair	252 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
KMT2C · driver	Mutation	p.E1992Q	4	0.32
FOXA1 · driver	Mutation	p.I176M	9	0.32
CDH1 · driver	Mutation	p.S36Afs*20	17	0.32
RB1 · driver	Mutation	p.R908*	25	0.31
PIK3R1 · driver	Mutation	p.E635K	79	0.31
MAP3K1 · driver	Mutation	p.P233Qfs*30	13	0.31
ERBB2 · driver	Mutation	p.L755S	55	0.31
ENSA	High-level amplification	GISTIC ±2	1	0.46
OR14I1	High-level amplification	GISTIC ±2	1	0.33
MROH1	Mutation	p.L267=	0	0.33
SLC35F3	High-level amplification	GISTIC ±2	0	0.32
PLD5	High-level amplification	GISTIC ±2	0	0.32



124 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

RB1 → Insulin human (DB00030); RB1 → Insulin pork (DB00071); PIK3R1 → Isoprenaline (DB01064); PIK3R1 → SF1126 (DB05210); PIK3R1 → Enzastaurin (DB06486); PIK3R1 → Wortmannin (DB08059); MAP3K1 → AZD-8330 (DB06061); MAP3K1 → Binimetinib (DB11967); MAP3K1 → Fostamatinib (DB12010); ERBB2 → Trastuzumab (DB00072) … and 71 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-B6-A1KI	63	I	LumA	0.88	no progression, followed 73.5 mo
TCGA-AN-A0FN	61	I	LumA	0.84	no progression, followed 7.2 mo
TCGA-AN-A0FF	32	I	LumB	0.83	no progression, followed 5.7 mo
TCGA-BH-A18F	50	II	LumA	0.81	no progression, followed 32.9 mo
TCGA-B6-A40B	76	I	LumA	0.80	no progression, followed 103.6 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group C** · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**54**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**Asian**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EAS**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8013/3**

Sample: **TCGA-A1-A0SK-01** · Aneuploidy score: **16** · TMB (nonsyn., /Mb): **2.03** · MSIsensor score: **1.02** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

54th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

10% by 36 mo

12 mo 3% · 24 mo 6% · 60 mo 19%

Model-estimated progression time

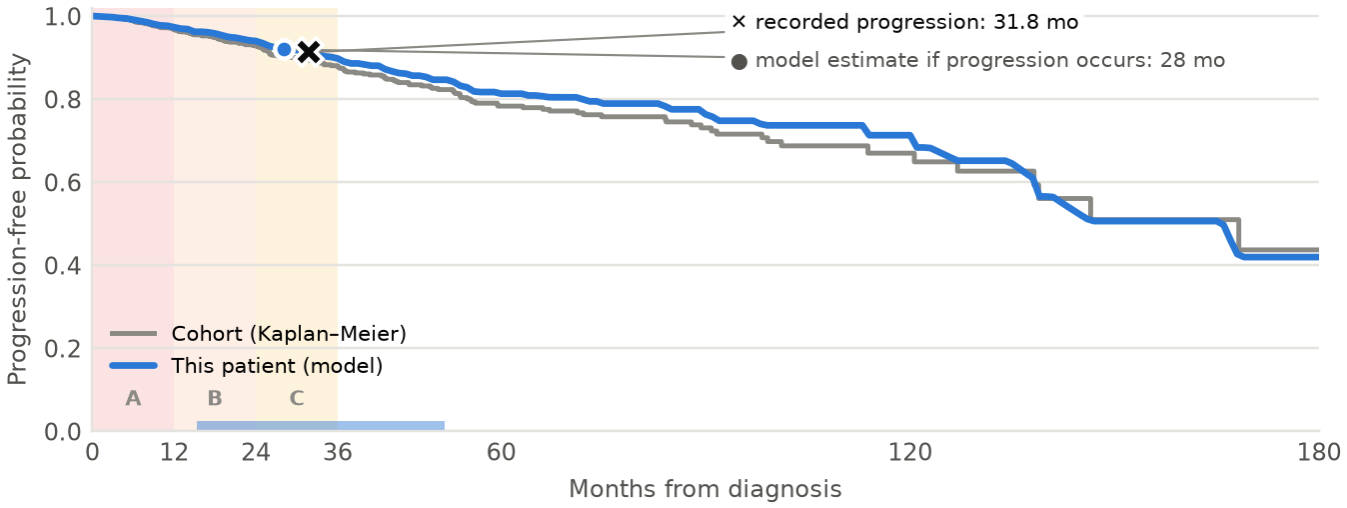
≈ 28 months

Window 24–36 months · IQR 15–52 mo
if progression occurs

Recorded progression

31.8 months

■ **Group C** · Progression in year 3
✓ **estimate in recorded window** · |error| 4 mo



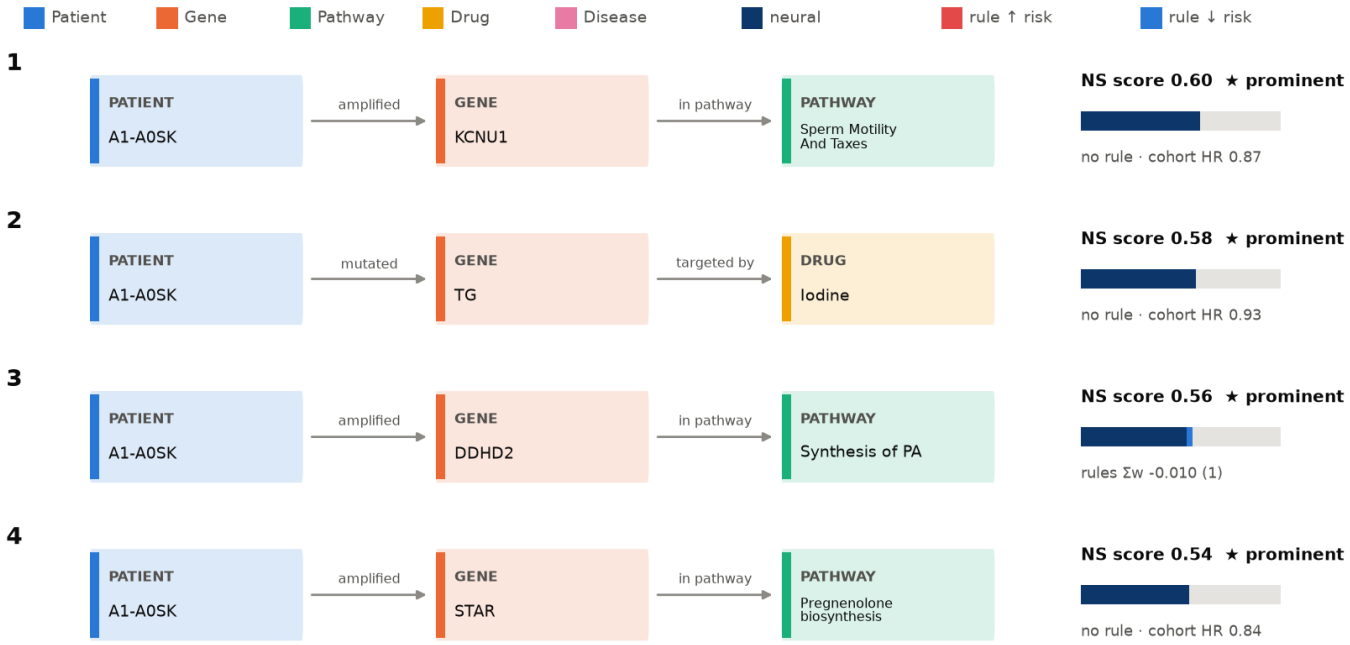
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 54-year-old patient has a Basal tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **54th risk percentile** (intermediate tier). The strongest learned formula that holds for them (*Stage(p, II) ⇒ Progress(p)*) lowers the progression hazard (weight -0.151 per SD). The highest-scoring knowledge-graph path runs from **KCNU1** to **Sperm Motility And Taxes**. Progression was recorded at 31.8 months; the model's estimate given progression was 28 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

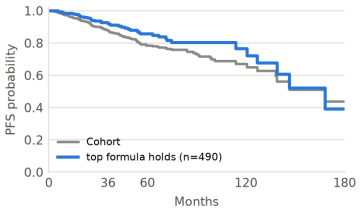


4 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-A1-A0SK** amplified → **KCNU1** in pathway → **Sperm Motility And Taxes**	100	No learned formula fires on this path; its score is neural only. cohort: HR 0.87, n = 97 carriers	**0.60**
2 ★	**TCGA-A1-A0SK** mutated → **TG** targeted by → **Iodine**	96	No learned formula fires on this path; its score is neural only. cohort: HR 0.93, n = 130 carriers	**0.58**
3 ★	**TCGA-A1-A0SK** amplified → **DDHD2** in pathway → **Synthesis of PA**	88	↓ **slower progression** · Σw -0.010 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Synthesis of PA') ⇒ Progress(p) cohort: HR 0.80, n = 189 carriers	**0.56**
4 ★	**TCGA-A1-A0SK** amplified → **STAR** in pathway → **Pregnenolone biosynthesis**	91	No learned formula fires on this path; its score is neural only. cohort: HR 0.84, n = 196 carriers	**0.54**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

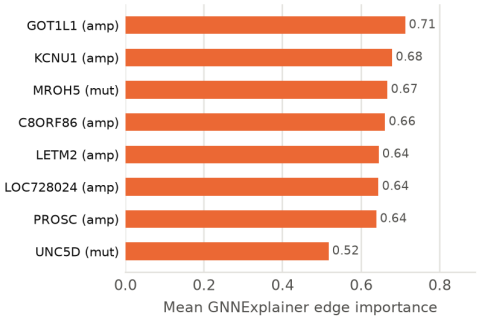
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.151 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	482 pts
-0.117 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	388 pts
-0.041 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	337 pts
+0.019 ↑ risk	Histology(p, other) ⇒ Progress(p)	single atom	81 pts
-0.010 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Synthesis of PA') ⇒ Progress(p) [KG]	single atom	187 pts
-0.007 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) [KG]	mined pair	306 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
BRCA1 · driver	Mutation	p.A521=	29	0.32
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.31
TP53 · driver	Mutation	p.M133K	57	0.31
GOT1L1	High-level amplification	GISTIC ±2	0	0.71
KCNU1	High-level amplification	GISTIC ±2	1	0.68
MROH5	Mutation	p.L110R	0	0.67
C8ORF86	High-level amplification	GISTIC ±2	0	0.66
LETM2	High-level amplification	GISTIC ±2	0	0.64
LOC728024	High-level amplification	GISTIC ±2	0	0.64
PROSC	High-level amplification	GISTIC ±2	0	0.64
UNC5D	High-level amplification	GISTIC ±2	1	0.52
UNC5D	Mutation	p.V547F	1	0.52



29 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896); FGFR1 → Ponatinib (DB08901) … and 53 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0YM	67	II	Basal	0.77	no progression, followed 31.7 mo
TCGA-AO-A03R	57	II	Normal	0.75	no progression, followed 68.7 mo
TCGA-E9-A1RB	40	II	LumB	0.75	no progression, followed 32.1 mo
TCGA-A2-A4S1	66	II	Basal	0.73	no progression, followed 27.0 mo
TCGA-E9-A1NC	61	II	Basal	0.72	no progression, followed 39.6 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-AR-A5QM · Group C

Report date 31 Oct 2026

■ **Group C** · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**62**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**Yes**	Pathologic N	**N0**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

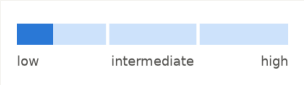
Sample: **TCGA-AR-A5QM-01** · Aneuploidy score: **19** · TMB (nonsyn., /Mb): **0.87** · MSIsensor score: **0.04** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

13th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

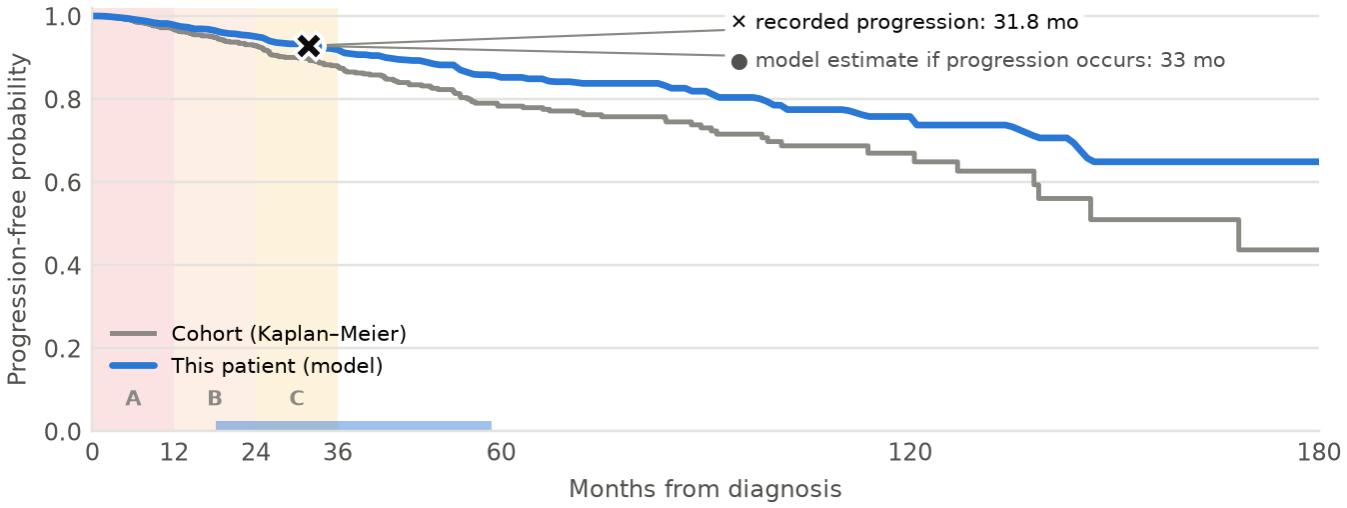
≈ 33 months

Window 24–36 months · IQR 18–59 mo
if progression occurs

Recorded progression

31.8 months

■ **Group C** · Progression in year 3
✓ **estimate in recorded window** · |error| 1 mo



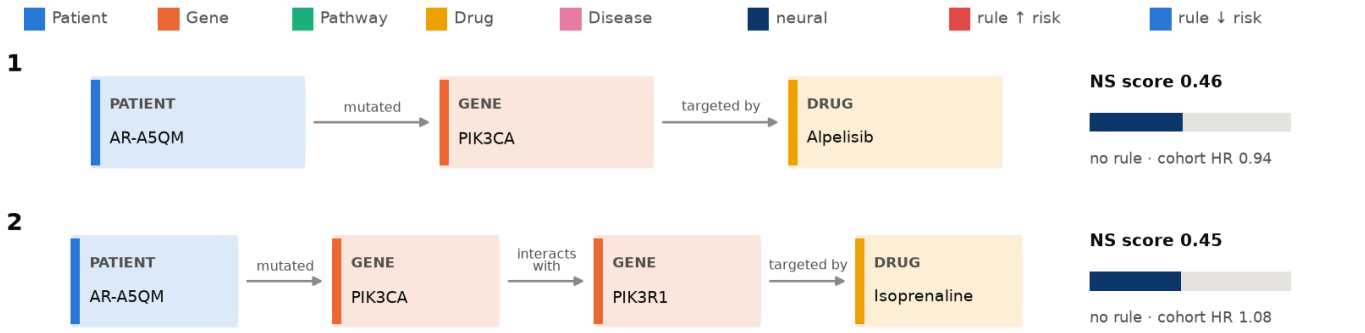
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 62-year-old patient has a LumA tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **13th risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0) ⇒ Progress(p)*) lowers the progression hazard (weight -0.095 per SD). The highest-scoring knowledge-graph path runs from **PIK3CA** to **Alpelisib**. Progression was recorded at 31.8 months; the model's estimate given progression was 33 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

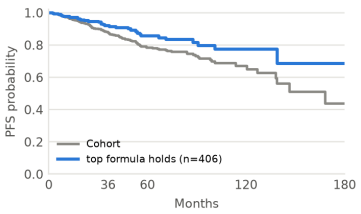


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 2 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-AR-A5QM** mutated → **PIK3CA** targeted by → **Alpelisib**	77	No learned formula fires on this path; its score is neural only. cohort: HR 0.94, n = 301 carriers	**0.46**
2	**TCGA-AR-A5QM** mutated → **PIK3CA** interacts with → **PIK3R1** targeted by → **Isoprenaline**	76	No learned formula fires on this path; its score is neural only. cohort: HR 1.08, n = 141 carriers	**0.45**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

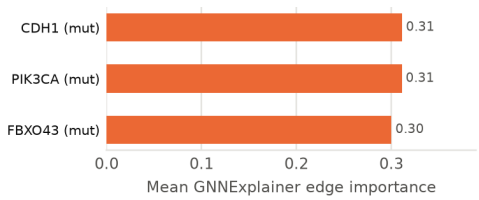
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.095 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	397 pts
-0.010 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	390 pts
-0.006 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	487 pts
-0.005 ↓ risk	HighAneuploidy(p) ⇒ Progress(p)	single atom	214 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.R74*	17	0.31
PIK3CA · driver	Mutation	p.H1047R	70	0.31
FBXO43	Mutation	p.C673Y	0	0.30



3 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-PE-A5DD	64	II	LumA	1.00	no progression, followed 64.2 mo
TCGA-E2-A10F	47	II	LumA	0.97	no progression, followed 28.9 mo
TCGA-B6-A2IU	62	II	LumA	0.89	progressed at 138.9 mo (group D)
TCGA-E2-A1L8	52	II	LumA	0.89	no progression, followed 73.6 mo

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A1XO	56	II	LumA	0.84	no progression, followed 55.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group C** · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**62**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

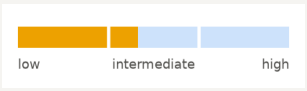
Sample: **TCGA-E2-A14R-01** · Aneuploidy score: **11** · TMB (nonsyn., /Mb): **3.7** · MSIsensor score: **1.89** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

44th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 18%

Model-estimated progression time

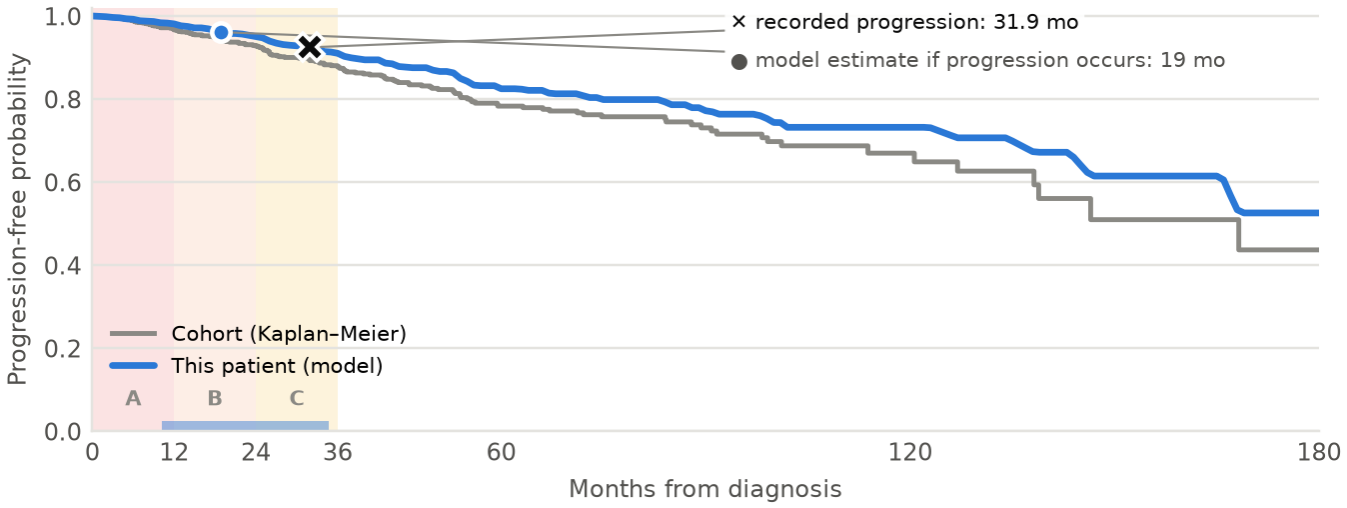
≈ 19 months

Window 12–24 months · IQR 10–35 mo
if progression occurs

Recorded progression

31.9 months

■ **Group C** · Progression in year 3
✗ **recorded window differs** · |error| 13 mo



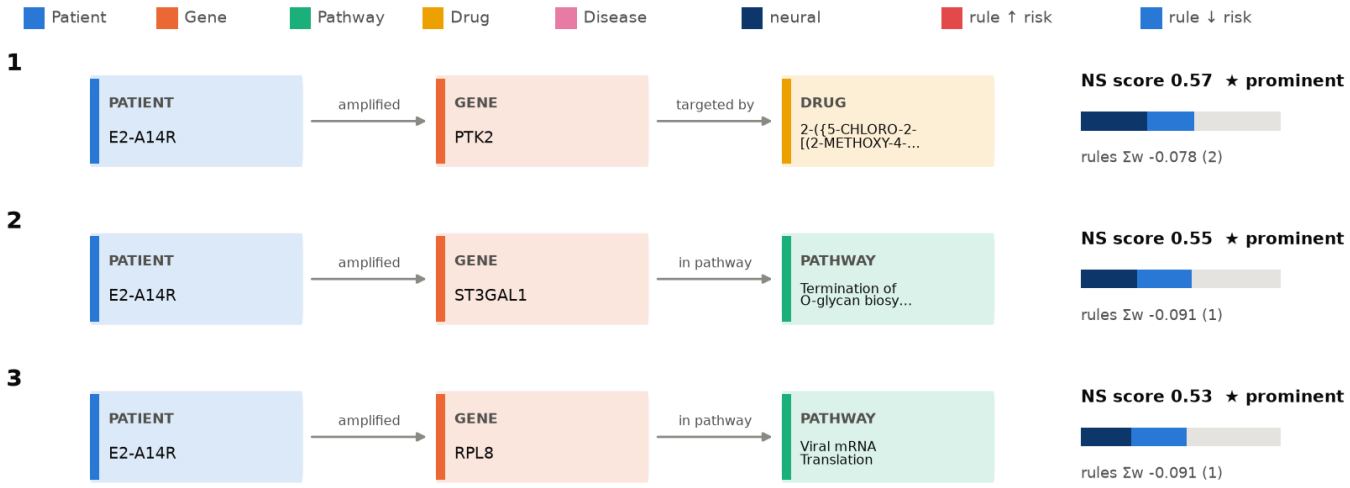
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 62-year-old patient has a Basal tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **44th risk percentile** (intermediate tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.304 per SD). The highest-scoring knowledge-graph path runs from **PTK2** to **2-({5-CHLORO-2-[(2-METHOXY-4-MORPHOLIN-4-YLPHENYL)AMINO]PYRIMIDIN-4-YL}AMINO)-N-METHYLBENZAMIDE**; the learned formulas it instantiates point to slower progression (Σw -0.078). Progression was recorded at 31.9 months; the model's estimate given progression was 19 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

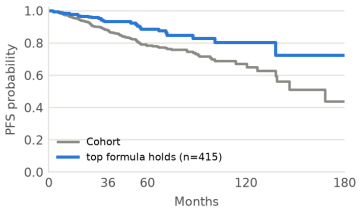


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-E2-A14R** amplified → **PTK2** targeted by → **2-({5-CHLORO-2-[(2-METHOXY-4-MORPHOLIN-4-YLPHENYL)AMINO]PYRIMIDIN-4-YL}AMINO)-N-METHYLBENZAMIDE**	56	↓ **slower progression** · Σw -0.078 over 2 formula(s) Altered(p, g) ∧ InPathway(g, 'Integrin signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Cytoprotection by HMOX1') ∧ Altered(p, g) ∧ InPathway(g, 'Factors involved in megakaryocyte development and platelet production') ⇒ Progress(p) cohort: HR 0.78, n = 100 carriers	**0.57**
2 ★	**TCGA-E2-A14R** amplified → **ST3GAL1** in pathway → **Termination of O-glycan biosynthesis**	47	↓ **slower progression** · Σw -0.091 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Keratan sulfate biosynthesis') ∧ Altered(p, g) ∧ InPathway(g, 'PKMTs methylate histone lysines') ∧ Altered(p, g) ∧ InPathway(g, 'Major pathway of rRNA processing in the nucleolus and cytosol') ⇒ Progress(p) cohort: HR 1.13, n = 199 carriers	**0.55**
3 ★	**TCGA-E2-A14R** amplified → **RPL8** in pathway → **Viral mRNA Translation**	42	↓ **slower progression** · Σw -0.091 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Keratan sulfate biosynthesis') ∧ Altered(p, g) ∧ InPathway(g, 'PKMTs methylate histone lysines') ∧ Altered(p, g) ∧ InPathway(g, 'Major pathway of rRNA processing in the nucleolus and cytosol') ⇒ Progress(p) cohort: HR 0.82, n = 91 carriers	**0.53**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

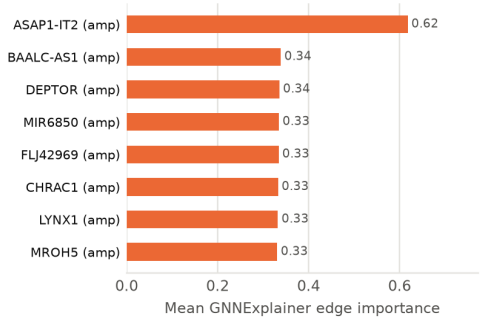
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.304 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	408 pts
+0.101 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Mitochondrial translation initiation') ⇒ Progress(p) [KG]	single atom	104 pts
-0.091 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Keratan sulfate biosynthesis') ∧ Altered(p, g) ∧ InPathway(g, 'PKMTs methylate histone lysines') ∧ Altered(p, g) ∧ InPathway(g, 'Major pathway of rRNA processing in the nucleolus and cytosol') ⇒ Progress(p) [KG]	RL-proposed	45 pts
+0.082 ↑ risk	Altered(p, g) ∧ TargetedBy(g, 'Ellagic acid') ⇒ Progress(p) [KG]	single atom	185 pts
-0.057 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Integrin signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Cytoprotection by HMOX1') ∧ Altered(p, g) ∧ InPathway(g, 'Factors involved in megakaryocyte development and platelet production') ⇒ Progress(p) [KG]	RL-proposed	65 pts
+0.048 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ∧ Altered(p, g) ∧ InPathway(g, 'Transcriptional Regulation by VENTX') ∧ Altered(p, g) ∧ InPathway(g, 'Factors involved in megakaryocyte development and platelet production') ⇒ Progress(p) [KG]	RL-proposed	64 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MYC · driver	High-level amplification	GISTIC ±2	20	0.32
TP53 · driver	Mutation	p.I195T	57	0.31
ASAP1-IT2	High-level amplification	GISTIC ±2	0	0.62
BAALC-AS1	High-level amplification	GISTIC ±2	0	0.34
DEPTOR	High-level amplification	GISTIC ±2	0	0.34
MIR6850	High-level amplification	GISTIC ±2	0	0.33
FLJ42969	High-level amplification	GISTIC ±2	0	0.33
CHRAC1	High-level amplification	GISTIC ±2	0	0.33
LYNX1	High-level amplification	GISTIC ±2	0	0.33
MROH5	High-level amplification	GISTIC ±2	0	0.33
LRRC14	High-level amplification	GISTIC ±2	0	0.33
NRBP2	High-level amplification	GISTIC ±2	0	0.33



333 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548) … and 101 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A1XQ	69	II	Basal	0.93	no progression, followed 16.4 mo
TCGA-B6-A0I1	73	II	Basal	0.91	progressed at 36.8 mo (group D)
TCGA-AN-A0FX	52	II	Basal	0.89	no progression, followed 0.3 mo
TCGA-E9-A3QA	33	II	Basal	0.89	no progression, followed 30.2 mo
TCGA-E2-A14Y	35	II	Basal	0.88	no progression, followed 69.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month

is always shown beside it.

■ **Group C** · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**50**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**Black or African American**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**AFR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**Yes**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8522/3**

Sample: **TCGA-B6-A3ZX-01** · Aneuploidy score: **19** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **0** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

100th

High risk tier · fused neuro-symbolic model



Probability of progression

77% by 36 mo

12 mo 31% · 24 mo 58% · 60 mo 94%

Model-estimated progression time

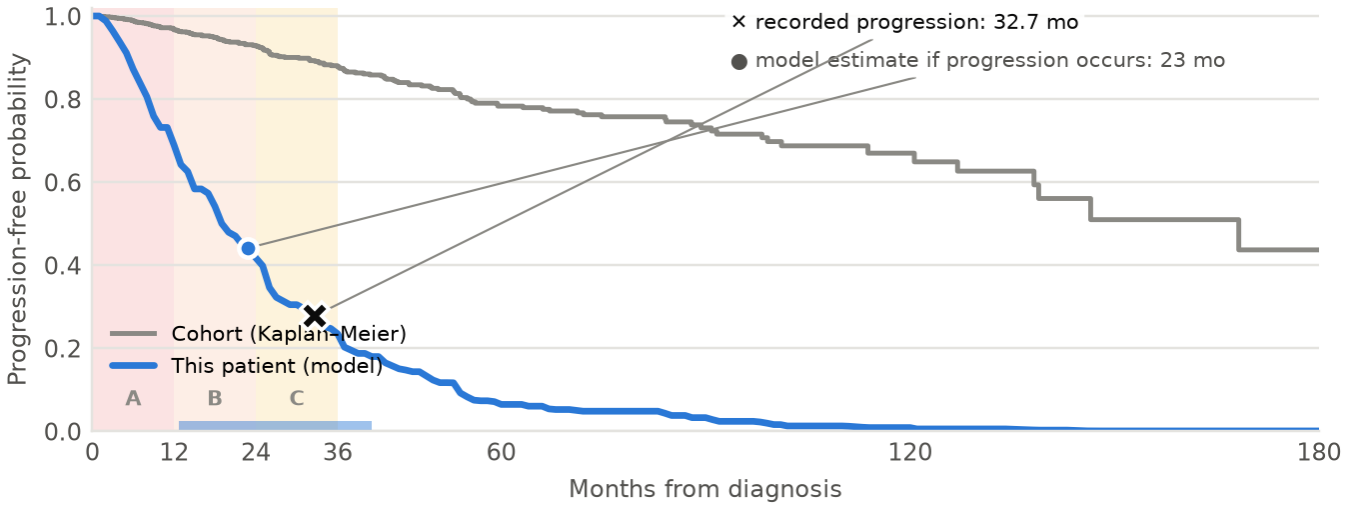
≈ 23 months

Window 12–24 months · IQR 13–41 mo
if progression occurs

Recorded progression

32.7 months

■ **Group C** · Progression in year 3
✗ **recorded window differs** · |error| 10 mo



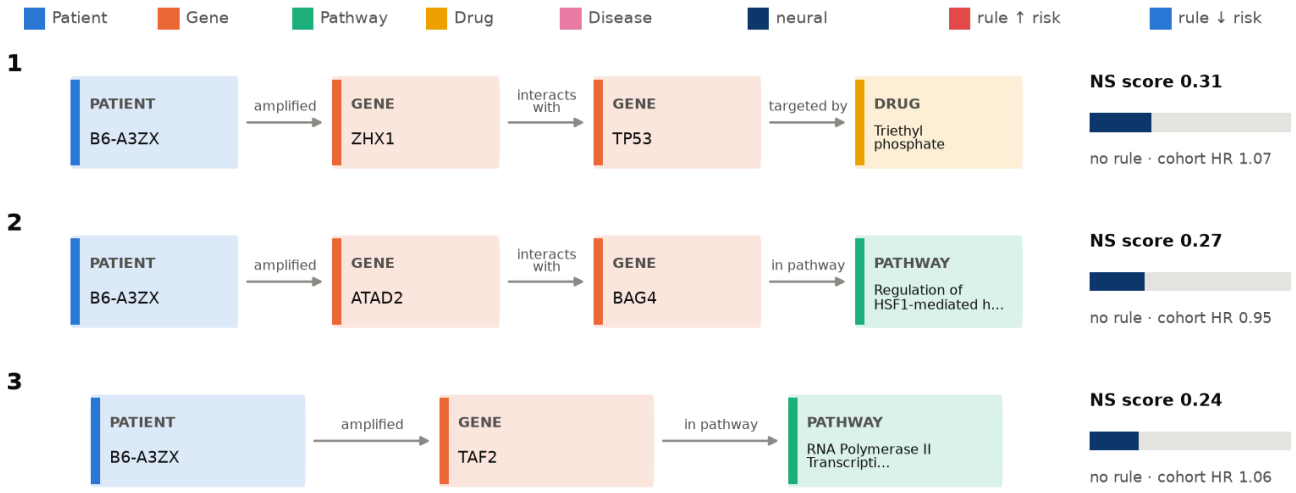
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 50-year-old patient has a subtype-unassigned tumour, AJCC stage IV (T3, N1, M1). The fused neuro-symbolic model places them in the **100th risk percentile** (high tier). The strongest learned formula that holds for them (*HighAneuploidy(p) ⇒ Progress(p)*) lowers the progression hazard (weight -0.005 per SD). The highest-scoring knowledge-graph path runs from **ZHX1** to **Triethyl phosphate**. Progression was recorded at 32.7 months; the model's estimate given progression was 23 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

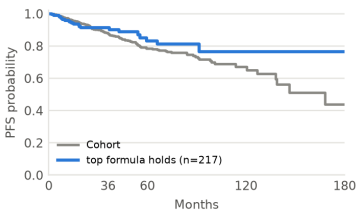


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-B6-A3ZX** amplified → **ZHX1** interacts with → **TP53** targeted by → **Triethyl phosphate**	51	No learned formula fires on this path; its score is neural only. cohort: HR 1.07, n = 283 carriers	**0.31**
2	**TCGA-B6-A3ZX** amplified → **ATAD2** interacts with → **BAG4** in pathway → **Regulation of HSF1-mediated heat shock response**	46	No learned formula fires on this path; its score is neural only. cohort: HR 0.95, n = 220 carriers	**0.27**
3	**TCGA-B6-A3ZX** amplified → **TAF2** in pathway → **RNA Polymerase II Transcription Initiation**	41	No learned formula fires on this path; its score is neural only. cohort: HR 1.06, n = 135 carriers	**0.24**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

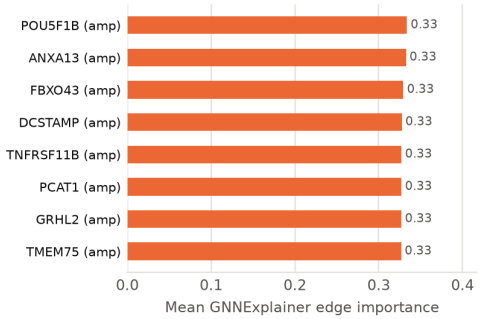
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.005 ↓ risk	HighAneuploidy(p) ⇒ Progress(p)	single atom	214 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MYC · driver	High-level amplification	GISTIC ±2	20	0.32
POU5F1B	High-level amplification	GISTIC ±2	0	0.33
ANXA13	High-level amplification	GISTIC ±2	0	0.33
FBXO43	High-level amplification	GISTIC ±2	0	0.33
DCSTAMP	High-level amplification	GISTIC ±2	1	0.33
TNFRSF11B	High-level amplification	GISTIC ±2	1	0.33
PCAT1	High-level amplification	GISTIC ±2	0	0.33
GRHL2	High-level amplification	GISTIC ±2	0	0.33
TMEM75	High-level amplification	GISTIC ±2	0	0.33
CASC8	High-level amplification	GISTIC ±2	0	0.33
MIR1205	High-level amplification	GISTIC ±2	0	0.33
FAM83A-AS1	High-level amplification	GISTIC ±2	0	0.33



135 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); TRHR → Protirelin (DB09421); NDUFB9 → NADH (DB00157); RRM2B → Cladribine (DB00242); SLC30A8 → Zinc chloride (DB14533); SLC30A8 → Zinc sulfate, unspecified form (DB14548); ATP6V1C1 → Tiludronic acid (DB01133); ATP6V1C1 → D-tartaric acid (DB01694); ATP6V1C1 → Thonzonium (DB09552) … and 8 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A5FL	65	II	Basal	0.69	no progression, followed 0.8 mo
TCGA-C8-A137	34	II	Her2	0.67	progressed at 4.5 mo (group A)
TCGA-EW-A1PD	61	II	missing	0.64	no progression, followed 13.9 mo
TCGA-AO-A1KQ	84	III	missing	0.64	no progression, followed 61.9 mo
TCGA-B6-A0WY	40	III	LumA	0.63	progressed at 66.1 mo (group D)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group C** · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**68**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

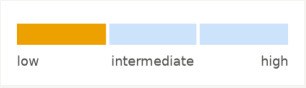
Sample: **TCGA-BH-A1FD-01** · Aneuploidy score: **9** · TMB (nonsyn., /Mb): **0.97** · MSIsensor score: **0.29** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

33rd

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 16%

Model-estimated progression time

≈ 43 months

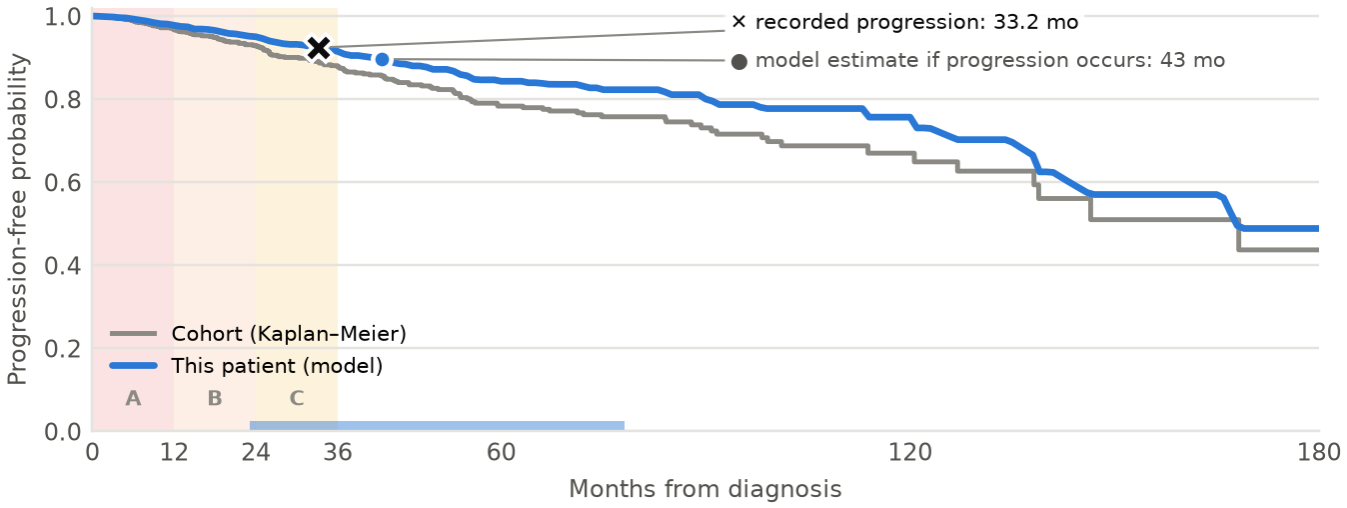
Window > 36 months · IQR 23–78 mo if progression occurs

Recorded progression

33.2 months

■ **Group C** · Progression in year 3

✗ **recorded window differs** · |error| 9 mo



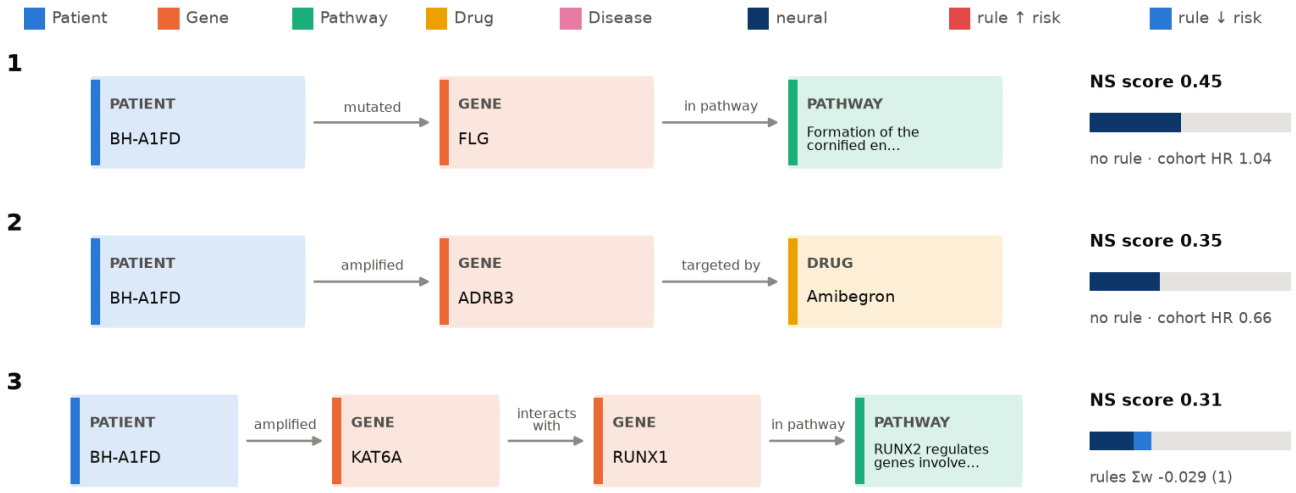
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 68-year-old patient has a LumB tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **33rd risk percentile** (intermediate tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.117 per SD). The highest-scoring knowledge-graph path runs from **FLG** to **Formation of the cornified envelope**. Progression was recorded at 33.2 months; the model's estimate given progression was 43 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

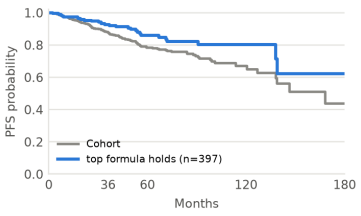


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-BH-A1FD** mutated → **FLG** in pathway → **Formation of the cornified envelope**	75	No learned formula fires on this path; its score is neural only. cohort: HR 1.04, n = 155 carriers	**0.45**
2	**TCGA-BH-A1FD** amplified → **ADRB3** targeted by → **Amibegron**	58	No learned formula fires on this path; its score is neural only. cohort: HR 0.66, n = 105 carriers	**0.35**
3	**TCGA-BH-A1FD** amplified → **KAT6A** interacts with → **RUNX1** in pathway → **RUNX2 regulates genes involved in differentiation of myeloid cells**	37	↓ **slower progression** · Σw -0.029 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'mRNA decay by 5' to 3' exoribonuclease') ∧ Altered(p, g) ∧ InPathway(g, 'Ub-specific processing proteases') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of RUNX1 Expression and Activity') ⇒ Progress(p) cohort: HR 1.09, n = 81 carriers	**0.31**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

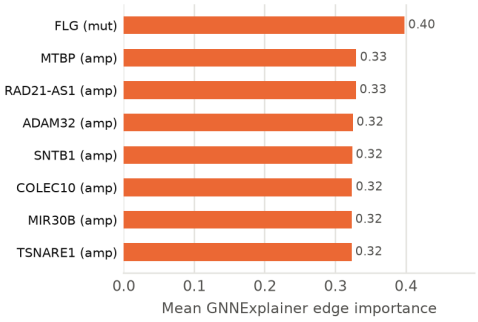
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.117 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	388 pts
-0.031 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Signal transduction by L1') ∧ Altered(p, g) ∧ InPathway(g, 'RNA Polymerase III Abortive And Retractive Initiation') ∧ Altered(p, g) ∧ InPathway(g, 'RAB GEFs exchange GTP for GDP on RABs') ⇒ Progress(p) [KG]	RL-proposed	36 pts
-0.029 ↓ risk	Altered(p, g) ∧ InPathway(g, 'mRNA decay by 5' to 3' exoribonuclease') ∧ Altered(p, g) ∧ InPathway(g, 'Ub-specific processing proteases') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of RUNX1 Expression and Activity') ⇒ Progress(p) [KG]	RL-proposed	36 pts
-0.021 ↓ risk	NStage(p, N0) ∧ Altered(p, g) ∧ InPathway(g, 'Integrin signaling') ⇒ Progress(p) [KG]	mined pair	62 pts
-0.010 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Synthesis of PA') ⇒ Progress(p) [KG]	single atom	187 pts
-0.008 ↓ risk	NStage(p, N0) ∧ TStage(p, T1) ⇒ Progress(p)	mined pair	132 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.V173L	57	0.32
PIK3R1 · driver	Mutation	p.I559T	79	0.32
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.31
ESR1 · driver	High-level amplification	GISTIC ±2	141	0.31
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
FLG	Mutation	p.S1127=	1	0.40
MTBP	High-level amplification	GISTIC ±2	0	0.33
RAD21-AS1	High-level amplification	GISTIC ±2	0	0.33
ADAM32	High-level amplification	GISTIC ±2	0	0.32
SNTB1	High-level amplification	GISTIC ±2	0	0.32
COLEC10	High-level amplification	GISTIC ±2	2	0.32
MIR30B	High-level amplification	GISTIC ±2	0	0.32



110 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); PIK3R1 → Isoprenaline (DB01064); PIK3R1 → SF1126 (DB05210) … and 183 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0CP	60	I	LumA	0.78	no progression, followed 92.5 mo
TCGA-E2-A15C	61	I	LumA	0.76	no progression, followed 22.8 mo
TCGA-AR-A24S	61	I	LumB	0.75	no progression, followed 97.8 mo
TCGA-BH-A0BG	73	I	Basal	0.75	no progression, followed 61.5 mo
TCGA-A2-A0YH	53	III	LumB	0.74	no progression, followed 21.7 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group C** · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**68**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Normal**
Ethnicity	**Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N3B**
Radiation therapy	**Yes**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-EW-A1P1-01** · Aneuploidy score: **0** · TMB (nonsyn., /Mb): **0.07** · MSIsensor score: **1.16** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

88th

High risk tier · fused neuro-symbolic model



Probability of progression

16% by 36 mo

12 mo 4% · 24 mo 10% · 60 mo 31%

Model-estimated progression time

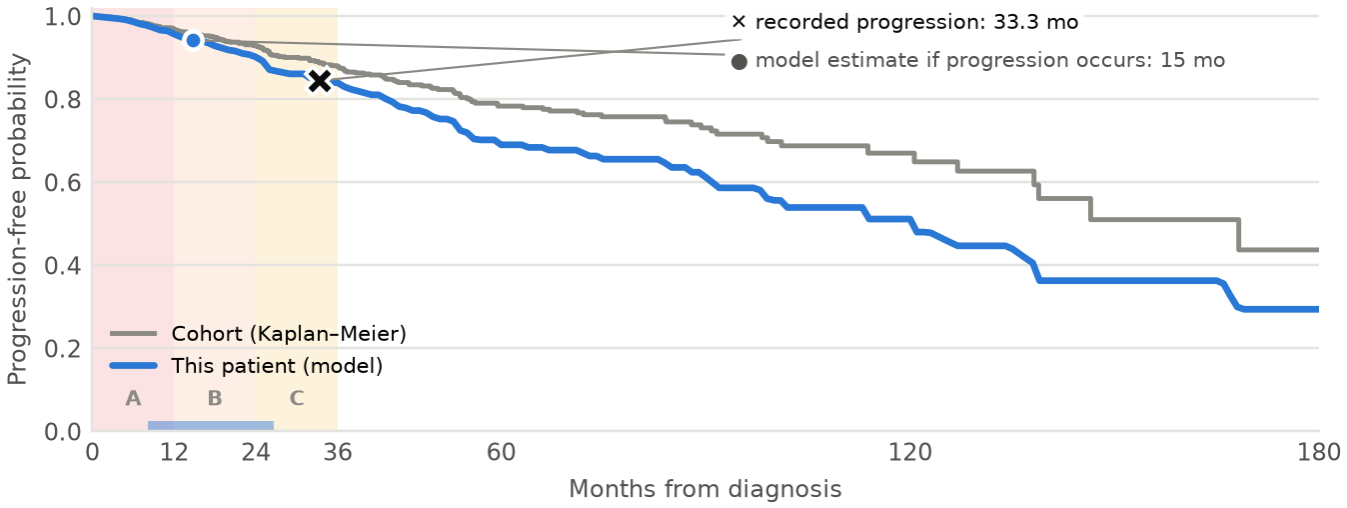
≈ 15 months

Window 12–24 months · IQR 8–27 mo if progression occurs

Recorded progression

33.3 months

■ **Group C** · Progression in year 3
✗ **recorded window differs** · |error| 19 mo



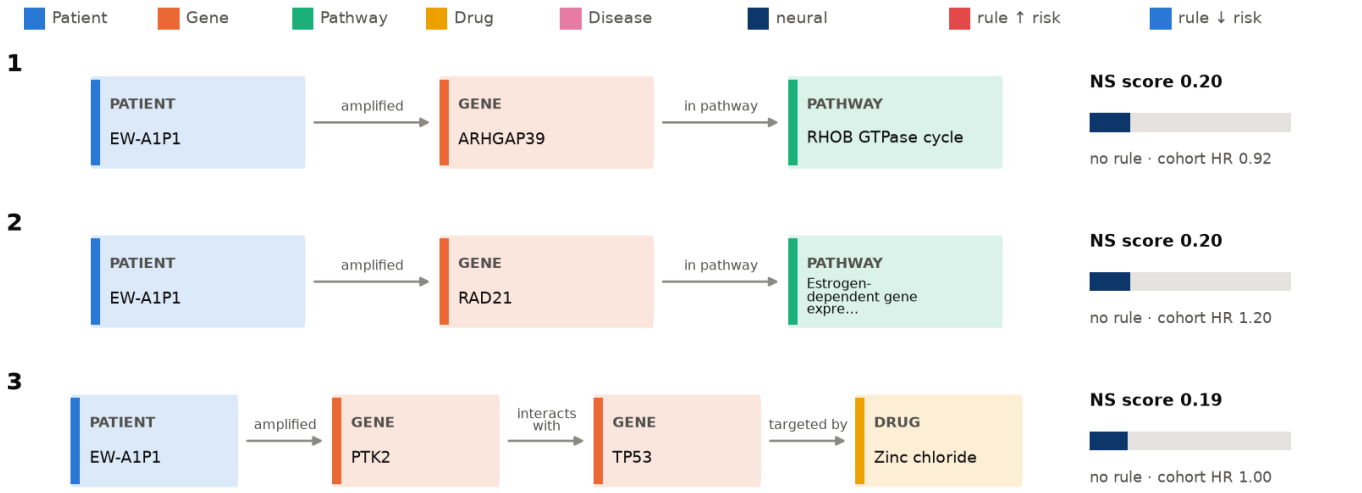
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 68-year-old patient has a Normal tumour, AJCC stage III (T2, N3, MX). The fused neuro-symbolic model places them in the **88th risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **ARHGAP39** to **RHOB GTPase cycle**. Progression was recorded at 33.3 months; the model's estimate given progression was 15 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

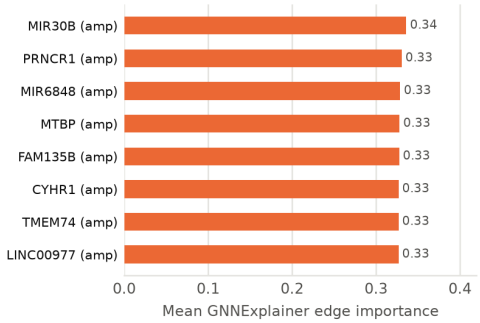
#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-EW-A1P1** amplified → **ARHGAP39** in pathway → **RHOB GTPase cycle**	34	No learned formula fires on this path; its score is neural only. cohort: HR 0.92, n = 135 carriers	**0.20**
2	**TCGA-EW-A1P1** amplified → **RAD21** in pathway → **Estrogen-dependent gene expression**	34	No learned formula fires on this path; its score is neural only. cohort: HR 1.20, n = 455 carriers	**0.20**
3	**TCGA-EW-A1P1** amplified → **PTK2** interacts with → **TP53** targeted by → **Zinc chloride**	31	No learned formula fires on this path; its score is neural only. cohort: HR 1.00, n = 414 carriers	**0.19**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

No learned formula with a non-zero weight fires for this patient; the symbolic risk is the baseline of the logic model.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.R273H	57	0.32
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
MIR30B	High-level amplification	GISTIC ±2	0	0.34
PRNCR1	High-level amplification	GISTIC ±2	0	0.33
MIR6848	High-level amplification	GISTIC ±2	0	0.33
MTBP	High-level amplification	GISTIC ±2	0	0.33
FAM135B	High-level amplification	GISTIC ±2	0	0.33
CYHR1	High-level amplification	GISTIC ±2	0	0.33
TMEM74	High-level amplification	GISTIC ±2	0	0.33
LINC00977	High-level amplification	GISTIC ±2	0	0.33
SPAG1	High-level amplification	GISTIC ±2	0	0.33
COL22A1	High-level amplification	GISTIC ±2	2	0.32



300 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813) … and 90 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A108	64	III	Normal	0.88	no progression, followed 27.5 mo
TCGA-C8-A131	82	III	Basal	0.85	no progression, followed 13.5 mo
TCGA-BH-A0DS	71	III	LumA	0.84	no progression, followed 2.6 mo
TCGA-D8-A1XZ	81	III	LumB	0.84	no progression, followed 15.3 mo
TCGA-A8-A09N	57	III	LumB	0.84	no progression, followed 1.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-BH-A1FH · Group C

Report date 31 Oct 2026

Group C · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**47**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**—**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-BH-A1FH-01** · Aneuploidy score: **—** · TMB (nonsyn., /Mb): **0.8** · MSIsensor score: **0.05** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

99th

High risk tier · fused neuro-symbolic model



Probability of progression

60% by 36 mo

12 mo 21% · 24 mo 43% · 60 mo 83%

Model-estimated progression time

≈ 51 months

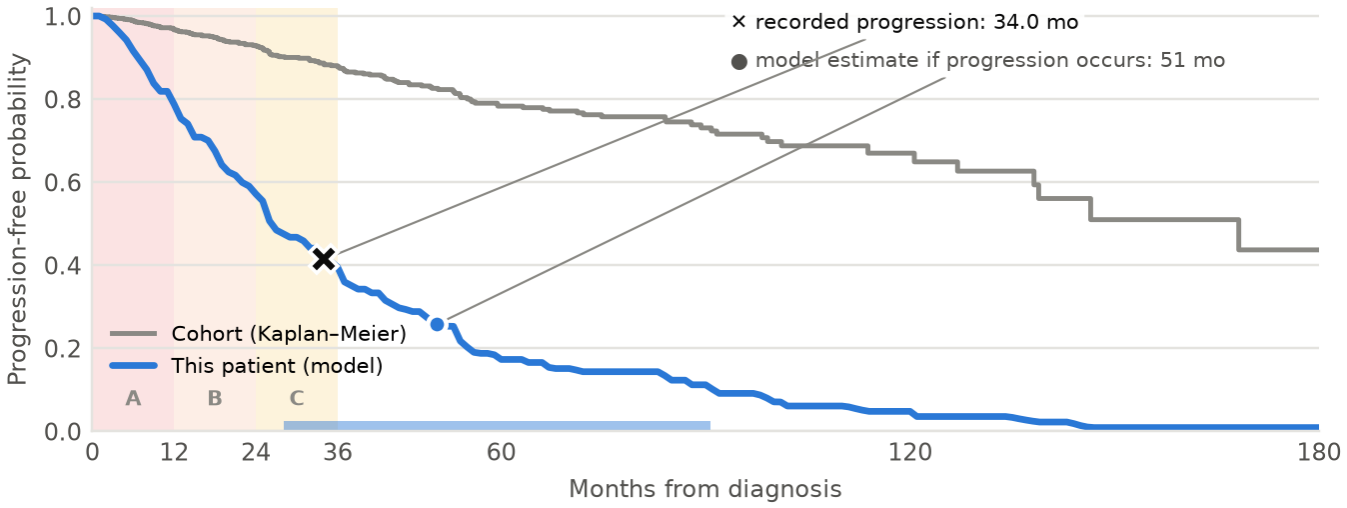
Window > 36 months · IQR 28–91 mo if progression occurs

Recorded progression

34.0 months

Group C · Progression in year 3

✗ recorded window differs · |error| 17 mo



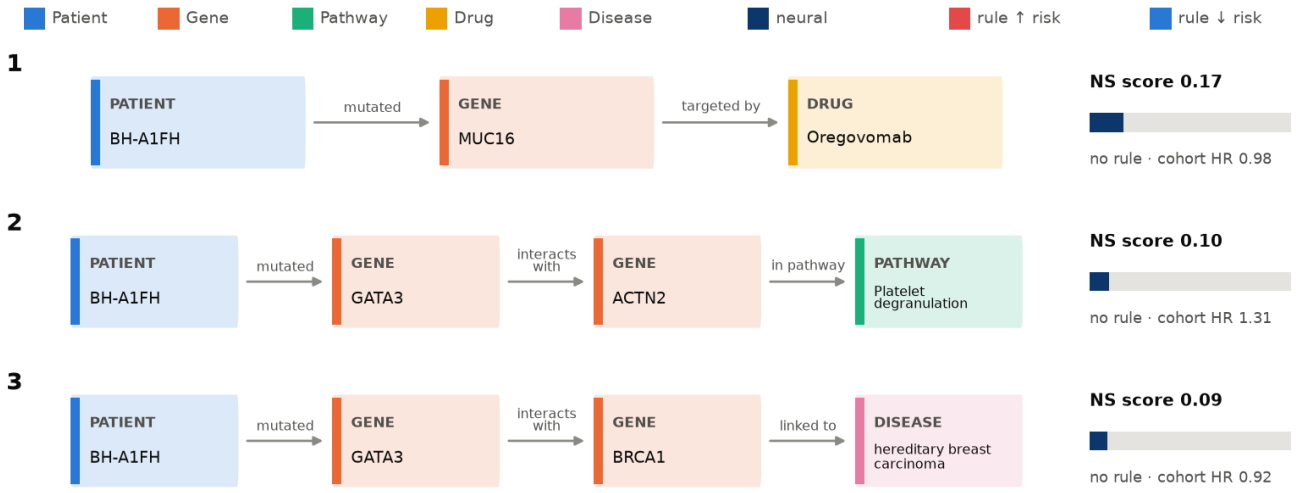
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 47-year-old patient has a LumA tumour, AJCC stage IV (T2, N1, M1). The fused neuro-symbolic model places them in the **99th risk percentile** (high tier). The strongest learned formula that holds for them (*Subtype(p, LumA)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.010 per SD). The highest-scoring knowledge-graph path runs from **MUC16** to **Oregovomab**. Progression was recorded at 34.0 months; the model's estimate given progression was 51 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

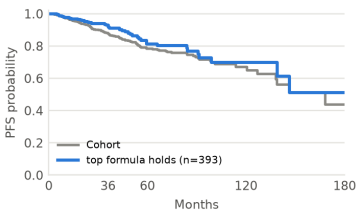


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-BH-A1FH** mutated → **MUC16** targeted by → **Oregovomab**	28	No learned formula fires on this path; its score is neural only. cohort: HR 0.98, n = 122 carriers	**0.17**
2	**TCGA-BH-A1FH** mutated → **GATA3** interacts with → **ACTN2** in pathway → **Platelet degranulation**	16	No learned formula fires on this path; its score is neural only. cohort: HR 1.31, n = 311 carriers	**0.10**
3	**TCGA-BH-A1FH** mutated → **GATA3** interacts with → **BRCA1** linked to → **hereditary breast carcinoma**	15	No learned formula fires on this path; its score is neural only. cohort: HR 0.92, n = 793 carriers	**0.09**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

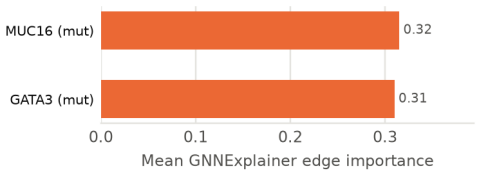
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.010 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	390 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GATA3 · driver	Mutation	p.R398Qfs*109	13	0.31
MUC16	Mutation	p.S8053F	8	0.32



2 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MUC16 → Oregovomab (DB04964). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AC-A62V	58	IV	missing	0.82	progressed at 11.4 mo (group A)
TCGA-C8-A273	29	II	LumA	0.71	no progression, followed 16.9 mo
TCGA-A2-A0CV	41	II	LumA	0.67	no progression, followed 99.0 mo
TCGA-D8-A27K	47	II	LumA	0.66	no progression, followed 48.0 mo
TCGA-B6-A0X4	62	II	LumA	0.65	progressed at 23.5 mo (group B)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-C8-A3M7 · Group C

Report date 31 Oct 2026

Group C · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**60**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**Asian**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIB**
Genetic ancestry	**EAS**	Pathologic T	**T4B**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-C8-A3M7-01** · Aneuploidy score: **8** · TMB (nonsyn., /Mb): **4.33** · MSIsensor score: **0.01** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

75th

High risk tier · fused neuro-symbolic model



Probability of progression

13% by 36 mo

12 mo 3% · 24 mo 8% · 60 mo 25%

Model-estimated progression time

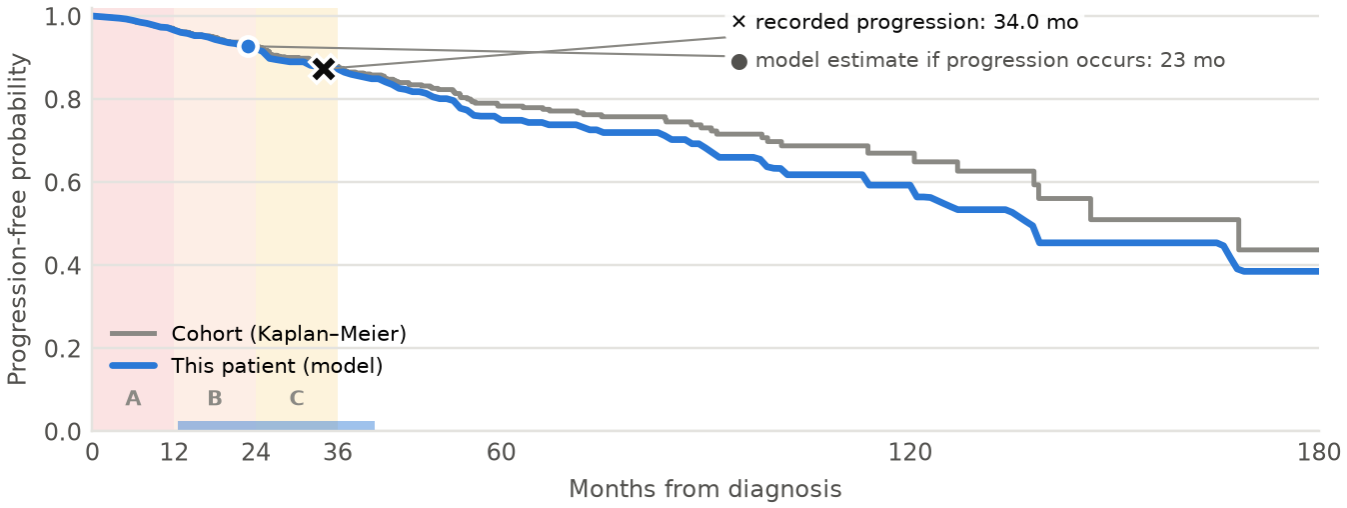
≈ 23 months

Window 12–24 months · IQR 13–41 mo
if progression occurs

Recorded progression

34.0 months

Group C · Progression in year 3
✗ **recorded window differs** · |error| 11 mo



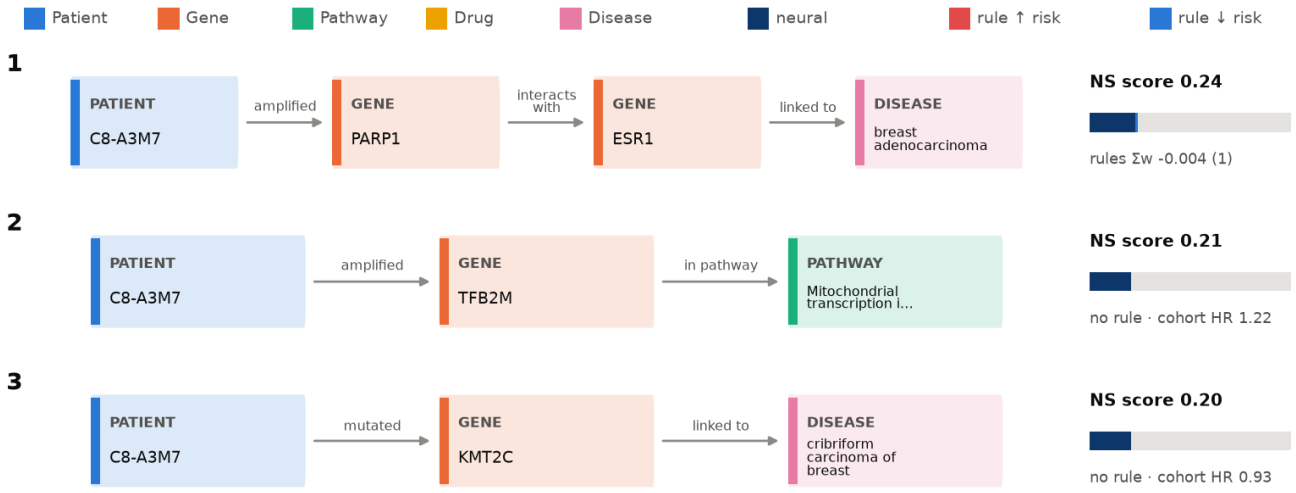
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 60-year-old patient has a LumA tumour, AJCC stage III (T4, N0, M0). The fused neuro-symbolic model places them in the **75th risk percentile** (high tier). The strongest learned formula that holds for them (*NStage(p, N0) ⇒ Progress(p)*) lowers the progression hazard (weight -0.099 per SD). The highest-scoring knowledge-graph path runs from **PARP1** to **breast adenocarcinoma**; the learned formulas it instantiates point to slower progression (Σw -0.004). Progression was recorded at 34.0 months; the model's estimate given progression was 23 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

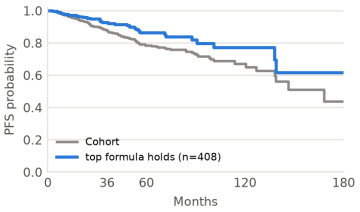


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-C8-A3M7** amplified → **PARP1** interacts with → **ESR1** linked to → **breast adenocarcinoma**	38	↓ **slower progression** · Σw -0.004 over 1 formula(s) NStage(p, N0) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) cohort: HR 0.80, n = 743 carriers	**0.24**
2	**TCGA-C8-A3M7** amplified → **TFB2M** in pathway → **Mitochondrial transcription initiation**	34	No learned formula fires on this path; its score is neural only. cohort: HR 1.22, n = 84 carriers	**0.21**
3	**TCGA-C8-A3M7** mutated → **KMT2C** linked to → **cribriform carcinoma of breast**	34	No learned formula fires on this path; its score is neural only. cohort: HR 0.93, n = 669 carriers	**0.20**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

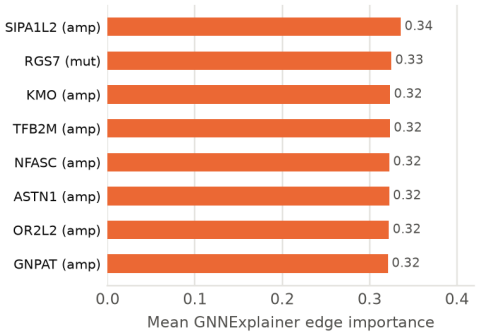
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.099 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	398 pts
+0.096 ↑ risk	TStage(p, T4) ⇒ Progress(p)	single atom	28 pts
+0.040 ↑ risk	Stage(p, III) ∧ Amplified(p, FAM72C) ⇒ Progress(p)	mined pair	33 pts
-0.018 ↓ risk	NStage(p, N0) ∧ Altered(p, g) ∧ TargetedBy(g, 'ATP') ⇒ Progress(p) [KG]	mined pair	162 pts
-0.009 ↓ risk	NStage(p, N0) ∧ Altered(p, g) ∧ InPathway(g, 'Synthesis of PIPs at the plasma membrane') ⇒ Progress(p) [KG]	mined pair	212 pts
-0.004 ↓ risk	NStage(p, N0) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) [KG]	mined pair	252 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
BAP1 · driver	Mutation	p.E566K	9	0.32
KMT2C · driver	Mutation	p.Q3151*	4	0.32
CDH1 · driver	Mutation	p.Y341Vfs*9	17	0.31
NF1 · driver	Mutation	p.S1420*, p.S1443*	6	0.31
ERBB2 · driver	Mutation	p.L377=, p.S305C	55	0.31
PIK3CA · driver	Mutation	p.H1047R	70	0.31
SIPA1L2	High-level amplification	GISTIC ±2	0	0.34
RGS7	High-level amplification	GISTIC ±2	2	0.33
RGS7	Mutation	3'UTR	2	0.33
KMO	High-level amplification	GISTIC ±2	1	0.32
TFB2M	High-level amplification	GISTIC ±2	2	0.32
NFASC	High-level amplification	GISTIC ±2	3	0.32



92 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 62 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-B6-A0RQ	68	II	Normal	0.78	no progression, followed 140.3 mo
TCGA-AN-A0FN	61	I	LumA	0.77	no progression, followed 7.2 mo
TCGA-AR-A5QP	54	II	LumA	0.77	no progression, followed 39.0 mo
TCGA-BH-A0B5	40	III	LumA	0.76	no progression, followed 70.2 mo
TCGA-BH-A0C1	61	III	LumA	0.76	progressed at 36.9 mo (group D)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-AO-A0J9 · Group C

Report date 31 Oct 2026

Group C · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**61**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N3**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-AO-A0J9-01** · Aneuploidy score: **17** · TMB (nonsyn., /Mb): **3.57** · MSIsensor score: **0.05** · AJCC edition: **—**

2 · Progression assessment

Model risk (cohort percentile)

72nd

High risk tier · fused neuro-symbolic model



Probability of progression

12% by 36 mo

12 mo 3% · 24 mo 7% · 60 mo 24%

Model-estimated progression time

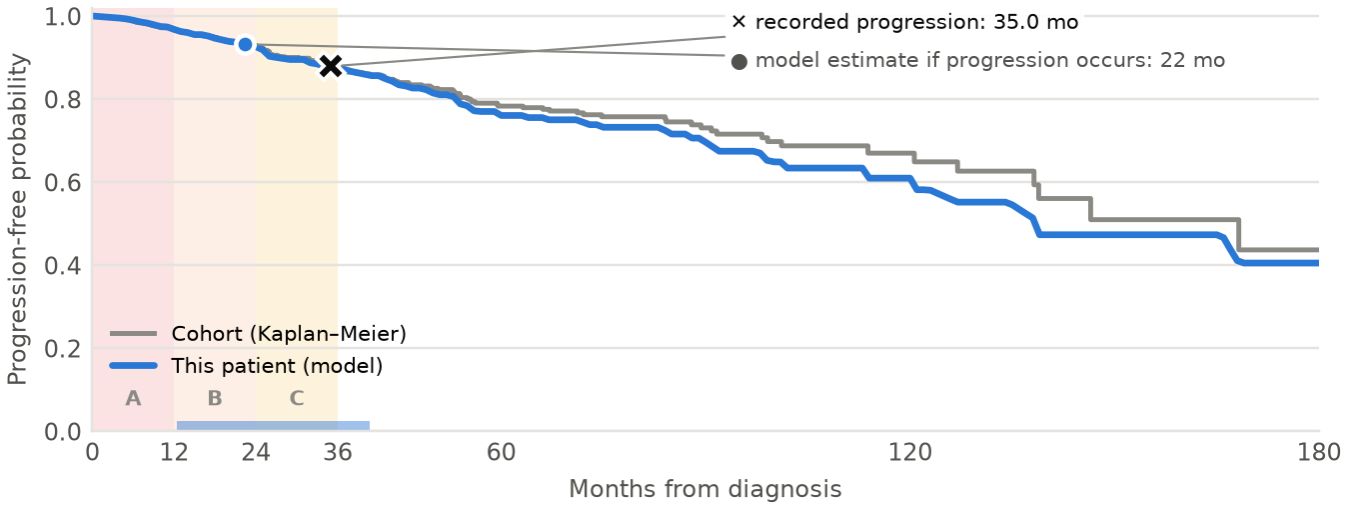
≈ 22 months

Window 12–24 months · IQR 12–41 mo
if progression occurs

Recorded progression

35.0 months

Group C · Progression in year 3
✗ **recorded window differs** · |error| 13 mo



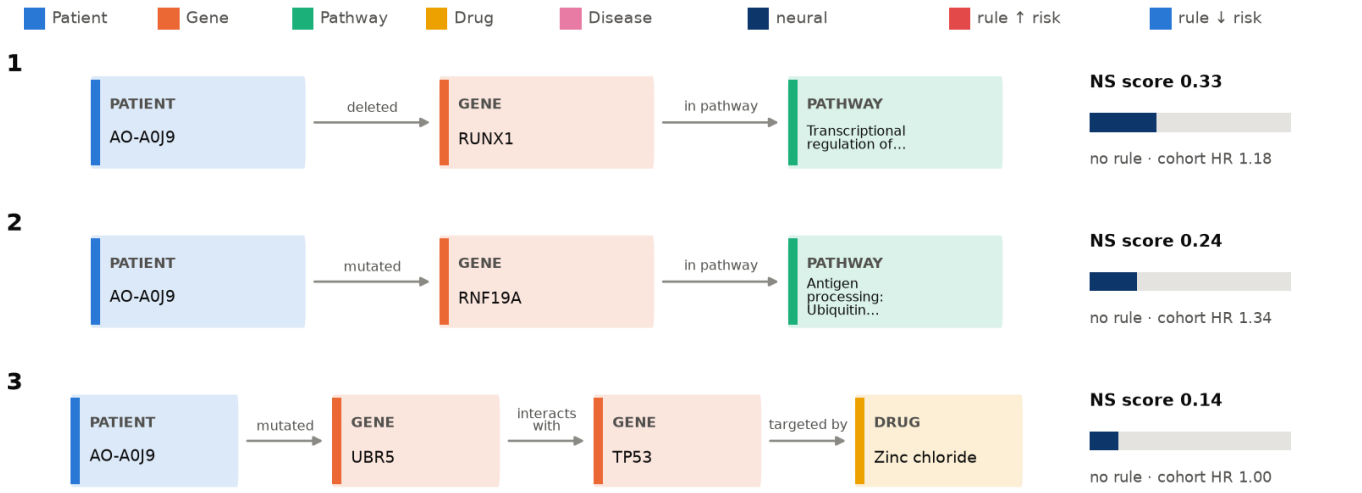
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 61-year-old patient has a LumA tumour, AJCC stage III (T2, N3, M0). The fused neuro-symbolic model places them in the **72nd risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **RUNX1** to **Transcriptional regulation of granulopoiesis**. Progression was recorded at 35.0 months; the model's estimate given progression was 22 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

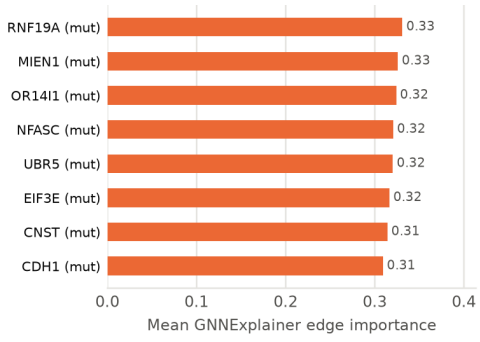
#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-AO-A0J9** deleted → **RUNX1** in pathway → **Transcriptional regulation of granulopoiesis**	55	No learned formula fires on this path; its score is neural only. cohort: HR 1.18, n = 209 carriers	**0.33**
2	**TCGA-AO-A0J9** mutated → **RNF19A** in pathway → **Antigen processing: Ubiquitination & Proteasome degradation**	39	No learned formula fires on this path; its score is neural only. cohort: HR 1.34, n = 139 carriers	**0.24**
3	**TCGA-AO-A0J9** mutated → **UBR5** interacts with → **TP53** targeted by → **Zinc chloride**	24	No learned formula fires on this path; its score is neural only. cohort: HR 1.00, n = 414 carriers	**0.14**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

No learned formula with a non-zero weight fires for this patient; the symbolic risk is the baseline of the logic model.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.E518Nfs*4	17	0.31
RUNX1 · driver	Homozygous deletion	GISTIC ±2	18	0.31
RNF19A	Mutation	p.T586A	1	0.33
MIEN1	Mutation	p.E36K	0	0.33
OR14I1	Mutation	p.G94S	1	0.32
NFASC	Mutation	p.N122K	3	0.32
UBR5	Mutation	p.R2062K	0	0.32
EIF3E	Mutation	p.I10=	6	0.32
CNST	Mutation	p.H639Y	0	0.31



9 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A5FK	60	III	LumA	0.80	no progression, followed 26.7 mo
TCGA-AO-A0J8	61	II	LumA	0.73	no progression, followed 22.4 mo
TCGA-A7-A5ZX	48	III	LumA	0.73	no progression, followed 11.0 mo
TCGA-OL-A5RV	43	III	LumA	0.73	no progression, followed 34.9 mo
TCGA-B6-A0IH	81	III	LumA	0.73	no progression, followed 112.4 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group C · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**55**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**Yes**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8523/3**

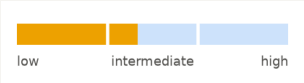
Sample: **TCGA-EW-A1P0-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **0.67** · MSIsensor score: **0.37** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

45th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 17%

Model-estimated progression time

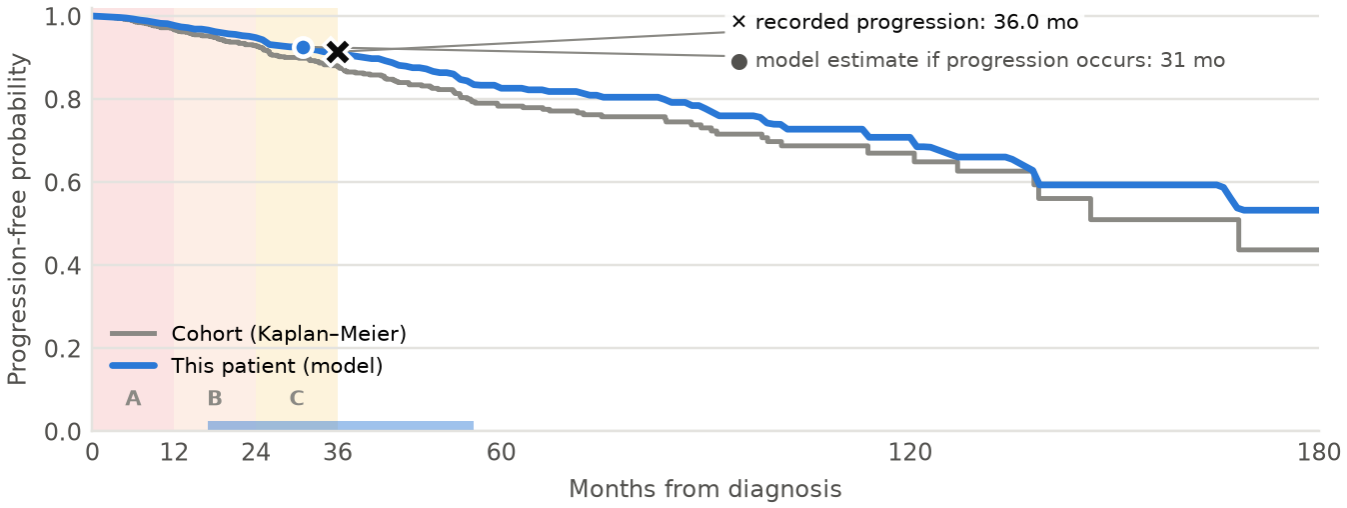
≈ 31 months

Window 24–36 months · IQR 17–56 mo
if progression occurs

Recorded progression

36.0 months

Group C · Progression in year 3
✓ **estimate in recorded window** · |error| 5 mo



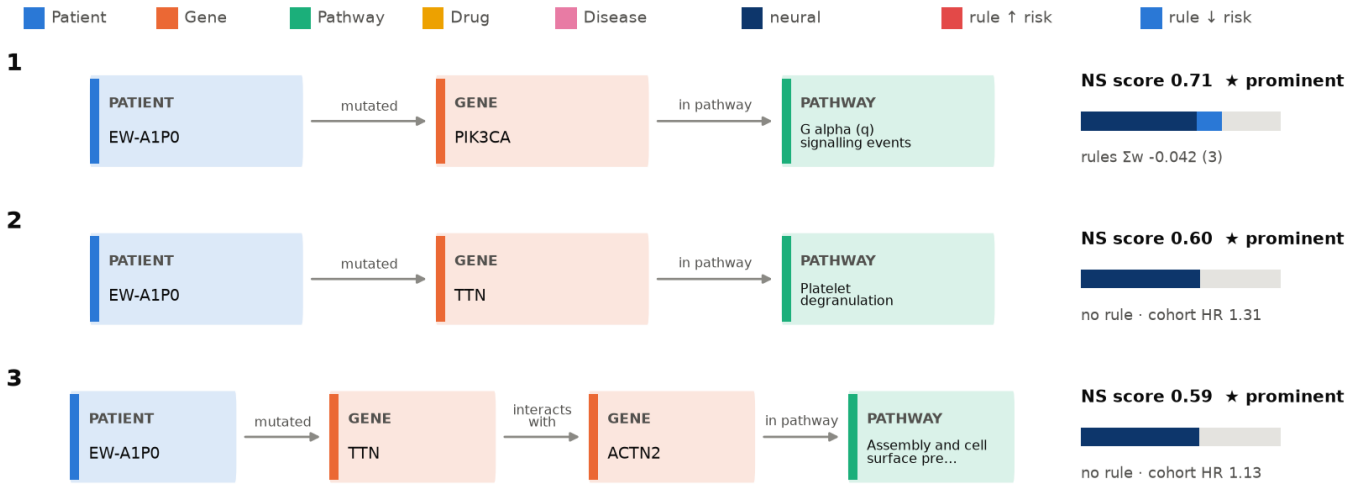
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 55-year-old patient has a LumB tumour, AJCC stage II (T2, N1, MX). The fused neuro-symbolic model places them in the **45th risk percentile** (intermediate tier). The strongest learned formula that holds for them (*Stage(p, II)* ∧ *Altered(p, g)* ∧ *InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling')* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.025 per SD). The highest-scoring knowledge-graph path runs from **PIK3CA** to **G alpha (q) signalling events**; the learned formulas it instantiates point to slower progression (Σw -0.042). Progression was recorded at 36.0 months; the model's estimate given progression was 31 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

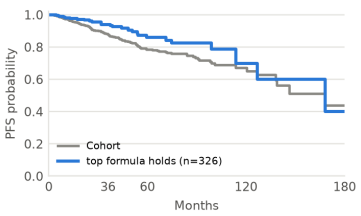


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-EW-A1P0** mutated → **PIK3CA** in pathway → **G alpha (q) signalling events**	97	↓ **slower progression** · Σw -0.042 over 3 formula(s) Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) cohort: HR 0.85, n = 397 carriers	**0.71**
2 ★	**TCGA-EW-A1P0** mutated → **TTN** in pathway → **Platelet degranulation**	99	No learned formula fires on this path; its score is neural only. cohort: HR 1.31, n = 312 carriers	**0.60**
3 ★	**TCGA-EW-A1P0** mutated → **TTN** interacts with → **ACTN2** in pathway → **Assembly and cell surface presentation of NMDA receptors**	99	No learned formula fires on this path; its score is neural only. cohort: HR 1.13, n = 99 carriers	**0.59**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

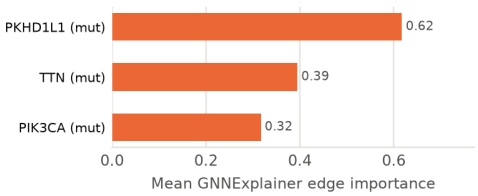
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.025 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) [KG]	mined pair	320 pts
-0.011 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI-3K cascade:FGFR2') ⇒ Progress(p) [KG]	mined pair	233 pts
-0.006 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI-3K cascade:FGFR4') ⇒ Progress(p) [KG]	mined pair	233 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3CA · driver	Mutation	p.D939G	70	0.32
PKHD1L1	Mutation	p.S393R	0	0.62
TTN	Mutation	p.V10463L	2	0.39



3 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AO-A0J7	71	II	LumB	0.80	no progression, followed 20.3 mo
TCGA-A2-A25C	50	II	LumB	0.76	no progression, followed 17.2 mo
TCGA-B6-A0X5	61	II	LumB	0.74	progressed at 44.2 mo (group D)
TCGA-D8-A1X7	40	II	LumB	0.70	no progression, followed 16.7 mo
TCGA-OL-A5S0	66	II	Basal	0.69	no progression, followed 20.4 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.